

Creating a Biomimetic Humanoid Robot
with Pneumatic Artificial Muscles and a Synthetic Neural Network

by

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Abstract

Biomimetic robots, robots whose biomechanics are modeled on human or animal anatomy, offer a platform for studying neuromechanical control through experiments that would be impractical or unethical with living subjects. This dissertation advances a long-running effort at Portland State University to build a humanoid robot, from the trunk down, actuated by pneumatic artificial muscles and controlled by a synthetic neural network. The work addresses the foundational question of that effort: can pneumatic artificial muscle actuators be characterized and arranged so that a biomimetic robot reproduces human joint force capabilities?

First, the isometric force of Festo braided pneumatic actuators (BPAs) with uninflated diameters of 10 mm and 20 mm was characterized over a wide range of resting lengths, contractions, and pressures using a custom test jig. The analysis produced novel equations for the maximum isometric force at 620 kPa as a function of resting length — a previously unreported dependency that the manufacturer's tool predicts with the wrong trend — and a three-coefficient nondimensional force surface that predicts isometric force from relative strain and relative pressure with high fidelity (adjusted $R^2 > 0.99$), correcting the large force over-predictions of existing tools and models at short resting lengths.

Second, the actuator model was carried to the joint level through isometric knee torque experiments on two test stands: a simplified pinned knee and a biomimetic four-bar knee

with a migrating instantaneous center of rotation. A hybrid torque calculation method isolated the sources of prediction error, revealing constant length offsets, series compliance in brackets and artificial tendons, and a loss of usable actuator length when BPAs wrap around the joint. Correction terms for each effect were identified through multiobjective optimization and substantially improved torque prediction across flexor and extensor configurations. The biomimetic knee driven by optimized BPA placements was shown to meet or exceed human isometric knee torque over most of the range of motion.

Third, the dissertation defines the path from this validated biomechanical plant to neuromechanical control: conversion of the lab's modified OpenSim leg model to MuJoCo via MyoConverter, a synthetic nervous system implemented in SNS-Toolbox built around a two-layer central pattern generator with per-leg rhythm generation and pattern formation and left-right coordination, proprioceptive (Ia, Ib, II) and heel/toe contact feedback, and verification by tonic-stimulation tests against the locomotion literature.

Dedication

*

Darian

azīzam, jānam,

nūr-e cheshmam — the light of my eyes

qorbānat beram

Bābā kheyli dūset dāre



*"Los corazones obreros que latén
por nuestra causa, felices serán.
Si entusiasmados y unidos combaten,
de la victoria, la palma obtendrán."*

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1 Introduction

1.1 Motivation

Biomimetic robots provide a platform for investigating how mechanical structure and control interact to generate movement, offering complementary insights to simulation-based studies [Andrews, 1985, Millard et al., 2013, Arnold et al., 2010]. Wheeled robots remain the platform of choice in structured environments, where their speed, efficiency, and payload capacity are difficult to match; in the unstructured terrain of the natural and built world, legged machines hold the advantage, and biomimetic legs that reproduce human force capabilities have the potential to excel where wheels cannot [Frey et al., 2026, Ha et al., 2025]. Musculoskeletal humanoids and limb systems driven by tendon-like actuators have been shown to support biologically meaningful experiments on motion generation, internal force transmission, and neural control hypotheses [Asano et al., 2017, 2019, Hitzmann et al., 2018, Liu et al., 2018, Chen et al., 2019]. The dual perspective of using robots to learn about biology, and using biology to design more adaptive robots, has been emphasized in both biomimetic robotics and biomechanics [Ijspeert, 2020, Asano et al., 2017]. With such robotic systems, experiments can be performed that would not be practical or ethical with human test subjects, such as removing a specific muscle from the system or blocking a sensory feedback pathway. Experiments of this kind allow direct testing of how the components of the neuromechanical system interact, and they offer new opportunities for testing neural control theories that are faster and cheaper than human

observation studies [Shin et al., 2018, Goldsmith et al., 2021, Schilling et al., 2022].

Artificial muscles are a key enabling technology for this class of robot. Among the available actuation technologies, McKibben-style braided pneumatic actuators (BPAs) combine low mass, high force-to-weight ratio, and intrinsic compliance with a force-length curve that qualitatively resembles that of skeletal muscle [Liang et al., 2020, Hunt et al., 2017a]. BPAs have been used in quadrupedal robots [Aschenbeck et al., 2006, Hunt et al., 2017a, Scharzenberger, 2019], musculoskeletal humanoids [Asano et al., 2017, Hitzmann et al., 2018], and dexterous manipulation systems [Chen et al., 2019]. However, BPAs also differ from biological muscle in important ways: their maximum tensile force occurs at resting length rather than at an optimal fiber length, their maximum contraction ratio is substantially lower than that of human muscle, and their force output is a highly nonlinear function of internal pressure, contraction, and loading state [Hunt et al., 2017a, Sárosi et al., 2017, Liang et al., 2020]. Designing a biomimetic robot around BPAs therefore requires quantitative, predictive models of the actuator and of the joint torques it can produce.

1.2 The Biomimetic Humanoid Robot Project

This dissertation is part of a long-running effort at Portland State University to build a biomimetic humanoid robot from the trunk down, actuated by pneumatic artificial muscles and controlled by a synthetic neural network (SNN). The project’s guiding hypothesis is that a robot with human-like biomechanics — human-like muscle numbers, architecture, and routing, human-like joint ranges of motion, and human-like isometric torque profiles — controlled by a biologically grounded neural architecture, will be capable of biomimetic

movement and will serve as a testbed for neuromechanical hypotheses, including loss-of-function experiments that are impossible in living animals. A by-product of this work is expected to be improved prosthetic and orthotic device design.

The present work builds directly on a series of studies by the author and collaborators. In the author's master's project, the kinematics and dynamics of a biomimetic robot leg were matched against the Gait2392 OpenSim model, artificial muscle attachment and wrapping locations were determined for the robot legs, and MATLAB tooling was developed for evaluating muscle lengths, moment arms, and isometric joint torques over the full range of motion [Bolen, 2019, Bolen and Hunt, 2019]. That work showed that simply transplanting human muscle attachment locations onto a BPA-actuated robot does not reproduce human torque profiles, and it motivated a multiobjective optimization of artificial muscle placements that finds attachment locations matching human torque curves and ranges of motion [Morrow et al., 2020]. In parallel, a biomimetic four-bar knee linkage with a migrating instantaneous center of rotation was designed and validated by Steele [Steele et al., 2017, Steele, 2018, Steele et al., 2018], and a canine-inspired quadruped platform (formerly DoggyDeux, current name of Muscle Mutt) with antagonistic BPA pairs and synthetic neural network control was developed by Scharzenberger [2019], extending an earlier demonstration by the Agile and Adaptive Robotics Laboratory (AARL) of central pattern generator (CPG) control of a pneumatically actuated robot [Hunt et al., 2017b].

Two further strands of preliminary work connect this dissertation to its neural control ambitions. First, the Sensory Afferent Database, developed within the AARL's NeuroNex research coordination network, organizes the literature on sensory afferent properties and their role in motor control into a structured, model-oriented resource; the author led two

studies describing the database and evaluating the use of AI-assisted literature triage to expand it [Bolen et al., 2023, Bolen and Hunt, 2024]. Second, an inverted-pendulum balance platform with BPA-driven plantarflexion/dorsiflexion demonstrated pseudo-vestibular feedback with inertial measurement units (IMUs), a first step toward the vestibular and proprioceptive control layers envisioned for the humanoid [Bolen et al., 2024b,a,c]. Finally, companion work by Elzein, Lutz, Bolen, and Hunt developed a state-space, optimization-based model of pressure and force dynamics in BPAs under pulse-modulated control [Elzein et al., 2025], which complements the static force characterization developed here.

1.3 Problem Statement

Within this context, three gaps stand between the existing robot design and a robot that can walk under neural control.

First, the actuator force models available at the start of this work were inadequate for design purposes. The manufacturer’s simulation tool over-predicts the force of short BPAs by large margins, and the published empirical models are fitted to a single resting length each, so they do not accurately predict how force changes when a designer chooses a different actuator length [Sárosi et al., 2017, Martens and Boblan, 2017, Hunt et al., 2017a]. Because muscle resting length is one of the primary degrees of freedom available to the robot designer, this gap blocks rational actuator selection.

Second, actuator force alone does not determine joint torque. The robot knee applies BPA force through brackets, artificial tendons, and wrapping paths. It was unknown how much these effects reduce usable torque relative to the rigid-body prediction, or how

to best correct for them. Additionally, the biomimetic knee’s optimized muscle routing [Morrow et al., 2020] had never been validated experimentally or compared against human isometric torque data.

Third, no simulation pipeline existed for closing the loop between a musculoskeletal robot model of this kind and a biologically realistic neural controller. The Neuromechanical Modeling literature provides the ingredients — two-level spinal CPG architectures [Rybak et al., 2006, McCrea and Rybak, 2008], neuromechanical simulation frameworks [Markin et al., 2010], and synthetic nervous system tooling [Nourse et al., 2023] — but they had not been assembled for a BPA-driven bipedal robot with human-like routing.

1.4 Research Objectives

This dissertation addresses the first two gaps experimentally and lays the groundwork for the third. The objectives are:

1. **Characterize the isometric force of Festo BPAs** with uninflated diameters of 10 mm and 20 mm over a wide range of resting lengths, contractions, and pressures; derive improved maximum-force and normalized force–strain–pressure models; and compare them against existing models (Chapters 3 and 4, Section 4.1).
2. **Validate and improve the prediction of isometric knee torque** produced by BPAs on a pinned joint and on a biomimetic four-bar knee; identify and model the losses due to constant length offsets, series compliance, and wrapping; and compare the resulting torque envelopes with human capabilities (Section 4.2).

3. **Define a path to neuromechanical control** of the resulting robot using a converted MuJoCo model, the SNS-Toolbox synthetic nervous system environment, and a two-layer CPG architecture with proprioceptive and contact feedback (Chapter 6).

1.5 Dissertation Organization

Chapter 2 reviews the relevant background: biomimetic and musculoskeletal robots, artificial muscle technologies and BPA modeling, human lower-limb biomechanics and the OpenSim benchmark models, the neural control of locomotion with an emphasis on two-level CPG architectures and sensory afferent feedback, and the Sensory Afferent Database and simulation tooling. Chapter 3 presents the materials and methods, including the actuator force model, the force characterization experiment, the knee test stands, and the improved torque model. Chapter 4 reports the experimental results, including space for experiments that are in progress at the time of writing. Chapter 5 discusses the findings, their limitations, and their implications for real-time control. Chapter 6 describes the planned neuromechanical modeling and control work, from the OpenSim-to-MuJoCo pipeline through the two-layer CPG with full sensory feedback to the eventual treadmill robot. Chapter 7 concludes.

2 Background

2.1 Biomimetic and Musculoskeletal Robots

Biomimetic robots offer many potential applications, ranging from assisting in factories and environmental exploration to serving as physical models of human movement. By developing high-fidelity biomimetic robots, we can perform tests that help improve our understanding of both biomechanics and the underlying neuromechanical systems that control movement [Shin et al., 2018, Asano et al., 2019, Ijspeert, 2020, Goldsmith et al., 2021, Hunt et al., 2017b, Schilling et al., 2022]. A few such biomimetic robots exist, such as the impressive Kengoro and Kenshiro robots of Asano et al. [2017] and Asano et al. [2019], which have proportionally human bodies, anatomically correct musculotendon arrangements, rolling-sliding joints, and joint ranges and output power comparable with human-measured values; their artificial muscles are implemented with brushless DC motors, wires as tendons, gears, and integrated tension measurement. A dog-like robot developed by Aschenbeck et al. [2006] and actuated with Festo BPAs was used to show how central pattern generators provide robust locomotion patterns that are adaptive to perturbations in ground speed [Hunt et al., 2017b]. Robots built with artificial muscles thus provide opportunities to investigate how muscles are controlled to produce desired motions; however, artificial muscles do not exactly replicate muscle properties, and existing robots do not capture all of the degrees of freedom and joint torques of the animal limbs they mimic.

2.2 Actuation Technologies and Artificial Muscles

A wide range of actuation technologies have been explored in robotics, including electric motors, hydraulic systems, smart-material actuators, and artificial muscles [He and Gao, 2020, Liang et al., 2020]. Electric motors are widely used due to their controllability and maturity, but they are limited by thermal constraints, gearing requirements, and reduced compliance compared to muscle-like actuators [Veale and Xie, 2016, Suzumori and Faudzi, 2018]. Hydraulic actuators offer exceptionally high power density and have enabled highly dynamic legged systems, but they remain relatively heavy, less efficient than biological muscle, and require a dedicated fluid infrastructure [Suzumori and Faudzi, 2018]. Dielectric elastomer actuators (DEAs), a class of electronic electroactive polymers, demonstrate high specific work and good bandwidth in comparative studies [Liang et al., 2020]; however, they are limited by the need for kilovolt-range driving voltages [Madden et al., 2004], electromechanical instability, and the requirement for compliant electrodes [Liang et al., 2020]. Compliant actuator technologies have also been explored for wearable and rehabilitation robotics [Aksöz et al., 2019, Leibach, 2020, Li et al., 2020], where the human-interactive nature of the application makes force control and backdrivability especially important. Recent reviews of pneumatic artificial muscle technology document continued progress in actuator design and in rehabilitation and assistive applications [Zhagiparova et al., 2025].

2.3 Braided Pneumatic Actuators

In contrast to the technologies above, McKibben-style braided pneumatic actuators (BPAs) combine low mass, high force-to-weight ratio, and intrinsic compliance with a force-length curve that qualitatively resembles that of skeletal muscle [Liang et al., 2020, Hunt et al., 2017a]. A BPA consists of an inflatable bladder inside a braided sleeve; pressurizing the bladder expands the bladder radially and contracts the sleeve axially, producing tensile force. As shown by Chou and Hannaford, the force-length relationship of McKibben actuators is governed by geometric constraints imposed by the braided shell, leading to a nonlinear dependence of force on actuator length through the braid angle [Chou and Hannaford, 1996]: actuator force is proportional to pressure and to a nonlinear geometric term $(3 \cos^2 \theta - 1)$, which governs both the maximum contraction limit and the curvature of the force-length relationship.

However, BPAs exhibit several important differences from biological muscle. First, BPA maximum tensile force occurs at resting (i.e., uninflated) length (l_0) rather than at an optimal fiber length [Liang et al., 2020]. Second, their maximum contraction ratio is substantially lower than that of human muscle, presenting challenges for achieving biomimetic joint ranges of motion [Hunt et al., 2017a, Sárosi et al., 2017, Liang et al., 2020]. Third, BPAs have a highly nonlinear dependence on internal pressure (P), contraction (ϵ), and loading state [Hunt et al., 2017a, Martens and Boblan, 2017], which complicates control and motivates high-fidelity force-length-pressure modeling. Despite these differences, their muscle-like properties have made them attractive for biomimetic quadrupeds [Aschenbeck et al., 2006, Scharzenberger, 2019], humanoids [Asano et al., 2017, Hitzmann

et al., 2018], and musculoskeletal arms [Kurumaya et al., 2016, Chen et al., 2019].

2.4 Modeling BPA Force–Length–Pressure Behavior

Previous studies have generated empirical models of the force–length–pressure relationship of BPAs. Sárosi et al. [2017] discuss several high-fidelity BPA force models. They present a static force model with a 21-coefficient polynomial function for a Festo MAS-20-200N (i.e., $\phi 20$ mm, $l_0 = 200$ mm) with an impressive $R^2 = 0.9994$, as well as Sárosi’s static force model for a Festo DMSP-20-400N-RM-RM (i.e., $\phi 20$ mm, $l_0 = 400$ mm), with six coefficients and a $R^2 = 0.9995$ [Sárosi, 2012, Sárosi et al., 2013]. Martens and Boblan present an even more accurate model, in terms of absolute error, using five coefficients for a DMSP-10-250 (i.e., $\phi 10$ mm, $l_0 = 250$ mm) and a DMSP-20-300 (i.e., $\phi 20$ mm, $l_0 = 300$ mm) [Martens and Boblan, 2017]. However, as demonstrated in this work, different resting lengths produce different force–length curves, preventing the use of these single-length models for designing systems with different resting lengths.

Hunt et al. [2017a] looked at six resting lengths of $\phi 10$ mm Festo BPAs, accounting for differences in maximum contractile percentages, and elucidated the force–length–pressure relationship. In particular, for a given contraction percent and force F (in Newtons), the scalar pressure P (in kPa) required to produce this force for a $\phi 10$ mm Festo artificial muscle can be determined by solving the equation:

$$P = 254 \text{ kPa} + F \cdot 1.23 \text{ kPa N}^{-1} + S \cdot 15.6 \text{ kPa} + 192 \text{ kPa} \cdot \tan \left(2.03 \left(\frac{\epsilon}{\epsilon_{\max} - F \cdot 0.331 \times 10^{-3} \text{ N}^{-1}} - 0.46 \right) \right) \quad (2.1)$$

where S is the artificial muscle hysteresis factor such that $S = 1$ indicates the muscle is shortening, $S = -1$ indicates it is lengthening, and $S = 0$ under static conditions. An important note for Equation (2.1) is that the coefficients have been updated with the correct values, as the values reported in Hunt et al. [2017a] contained typographical errors. The contraction ϵ , the maximum free-load contraction at 620 kPa (ϵ_{620} , corresponding to $\epsilon_{\max}$ in Equation (2.1) for these actuators), and the relative contraction ϵ^* used in that model are defined in Section 3.2.

Festo provides a manufacturer tool, MuscleSIM, for predicting force as a function of pressure and contraction [Lang, Robert, 2005], but experimental comparisons indicate that it does not adequately account for variation in l_0 or actuator-specific contraction limits. Comparisons with experimental data indicate that none of the existing models effectively take into account differences in maximum force due to initial actuator length, and the Hunt model was taken with low external forces ($\leq 106 \text{ N}$), so it is unclear how well it captures actuator behavior at higher forces. Chapter 4 compares these models with data collected for this dissertation and develops an improved characterization.

2.5 Human Lower-Limb Biomechanics and the OpenSim Benchmark

Designing a biomimetic leg requires a quantitative benchmark of human muscle–joint behavior. This dissertation uses OpenSim [Delp et al., 2007, Seth et al., 2011, 2018] and the Gait2392 model, which descends from the interactive lower-extremity model of Delp et al. [1990] and includes musculoskeletal geometry from Hoy et al. [1990], planar knee kinematics from Yamaguchi and Zajac [1989], and a lower-limb model with 23 degrees of freedom and 54 musculotendon actuators [Arnold et al., 2010]. Muscle moment arms, force-generating properties, and functional roles in such models have been characterized extensively [Zajac and Gordon, 1989, Brand et al., 1982, Menegaldo et al., 2004, Millard et al., 2013, Sherman et al., 2013, Suderman and Vasavada, 2012], and standard reporting conventions for kinematics and joint mechanics are established by the International Society of Biomechanics [Wu and Cavanagh, 1995, Wu et al., 2002, 2005, Derrick et al., 2020]. The human knee is itself a complex joint: its instantaneous center of rotation migrates posteriorly with flexion, a behavior captured by four-bar linkage descriptions of the joint [Romero et al., 2020, Steele, 2018].

In the author’s master’s work, muscle lengths, moment arms, and isometric joint torques were computed over the range of motion for both the Gait2392 model and a PAM-actuated robot leg model, demonstrating that several robot muscle torque profiles could closely match the human model while many fell far short [Bolen, 2019, Bolen and Hunt, 2019]. Morrow, Bolen, and Hunt subsequently developed a multiobjective optimization that finds artificial muscle attachment locations reproducing human torque profiles and ranges of motion about the trunk and leg joints [Morrow et al., 2020]. Steele’s

biomimetic knee joint, a 1-DoF four-bar linkage whose crossed links produce a migrating instantaneous center of rotation similar to the human knee, provides the physical knee used in this dissertation [Steele et al., 2017, Steele, 2018, Steele et al., 2018].

2.6 Neural Control of Locomotion: CPGs and Sensory Feedback

Locomotion in mammals is driven by spinal central pattern generators (CPGs), spinal neural circuits that can produce alternating rhythmic activity of flexor and extensor motoneurons in the absence of rhythmic input and proprioceptive feedback [Brown, 1911, Rybak et al., 2006]. The modern architectural understanding of the mammalian locomotor CPG comes largely from the modeling work of the Rybak group and collaborators. Rybak et al. [2006] proposed that the locomotor CPG consists of two levels: a half-center rhythm generator (RG) and a pattern formation (PF) network, with reciprocal inhibitory interactions between antagonist neural populations at each level. This two-level organization explains, among other phenomena, spontaneous deletions of motoneuron activity during fictive locomotion in which the locomotor period and phase are maintained, and it permits independent control of the level of motoneuron activity and of step-cycle timing. McCrea and Rybak [2008] reviewed the electrophysiological support for this organization, and Rybak et al. [2015] related the model architecture to genetically identified spinal interneuron classes. Left-right coordination is provided by commissural pathways between the two half-centers [Molkov et al., 2015, Danner et al., 2016], and the rhythmogenic core of the CPG continues to be refined experimentally [Rancic and Gosgnach, 2021]. Neuromechanical models coupling these spinal circuits to musculoskeletal dynamics have shown how

afferent feedback shapes CPG output [Markin et al., 2010, Shevtsova et al., 2016]. Physiologically plausible neuromechanical models of human locomotion continue to dissect the relative roles of reflexes and CPGs [Di Russo et al., 2023], and biomechanics and sensory feedback have been shown to regularize the behavior of different CPG architectures [Deng et al., 2022].

Sensory feedback is not a passive correction layer but an integral part of locomotor pattern generation. Reviews of afferent control of the CPG [Van de Crommert et al., 1998] and of muscle proprioceptive feedback and spinal networks [Windhorst, 2007, 2022] summarize the classical picture: group Ia afferents from muscle spindles signal muscle length and velocity, group Ib afferents from Golgi tendon organs signal force, and group II afferents signal length, with load-related feedback playing a role distinct from length-related feedback during locomotion [Dietz and Duysens, 2000]. Removing proprioceptive feedback degrades locomotor patterns in mice [Akay et al., 2014]; proprioceptive input regulates and adapts locomotor activity [Lam and Pearson, 2002]; and the relative contributions of proprioceptive and vestibular systems to locomotion are the subject of ongoing study [Akay and Murray, 2021, Akay, 2020]. Of particular relevance to the tests planned in Chapter 6, tonic central and sensory stimuli have been shown to facilitate involuntary air-stepping in humans [Selionov et al., 2009], demonstrating that simple graded stimulation of motoneuronal and rhythmic circuits can elicit stepping-like output — an experiment readily reproduced in simulation by tonic stimulation of pattern-formation and rhythm-generator populations.

Two-layer CPG organization has also informed the AARL’s own neuromechanical modeling. In preliminary work, the author and Morrow constructed a planar bipedal

walker in the AnimatLab neuromechanical simulation environment [Cofer et al., 2010], with a nine-segment, sagittal-plane biomechanics after the bipedal model of Li et al. [2017] and a synthetic nervous system whose rhythm-generator, pattern-formation, and sensorimotor networks followed the rat hindlimb two-layer CPG of Deng et al. [2019]. That model produced alternating, stepping-like activity and motion while suspended in the air, and preliminary results were presented by the author at the 2021 NIH BRAIN Initiative Investigators Meeting [Bolen and Hunt, 2021]. Its limitations — hand-tuned neural parameters and untuned afferent feedback — shaped the approach of Chapter 6; the model is described in Section 3.10.

2.7 Synthetic Nervous Systems and the Sensory Afferent Database

For robots, these biological architectures can be implemented as synthetic nervous systems (SNSs): networks of conductance-based spiking and non-spiking neurons that are small enough to run in real time yet biologically grounded enough to be compared against animal data. The AARL has pursued this approach across several platforms: CPG-based neural control of a pneumatically actuated dog robot [Hunt et al., 2017b], analytically designed dynamic neural networks for human balance control [Hilts et al., 2019], sensory-entrained CPG networks for steering [Li et al., 2016], inter-leg coordination analyses in stick-insect-inspired controllers [Nourse et al., 2019], reinforcement-reflex hybrids in insects [Goldsmith et al., 2021], adaptive hindlimb walking controllers [Schilling et al., 2022], and the DoggyDeux quadruped designed for SNN control [Scharzenberger, 2019]. The SNS-Toolbox [Nourse et al., 2023] makes such networks practical: it is an open-source

Python package that simulates hundreds to thousands of spiking and non-spiking neurons in real time or faster, supports multiple software and hardware backends including GPUs and embedded platforms, and couples directly to physics simulators such as MuJoCo for musculoskeletal control. The neuromorphic legged-locomotion field this toolbox serves is growing: Szczecinski et al. [2023] survey its past, present, and future, and physical testbeds in this lineage now extend from insect-scale models to a robotic cat hindlimb [Armstead et al., 2025].

The biological data needed to parameterize such networks are scattered across thousands of papers. To address this, the Sensory Afferent Database (SAD) was developed within the AARL’s NeuroNex Communication, Coordination, and Control in Nervous Systems (C3NS) effort as a structured, model-oriented resource characterizing sensory afferent properties across animals and their role in motor control [Bolen et al., 2023]. A follow-up study evaluated AI-assisted literature triage for expanding the database, finding that current tools can accelerate population of the database but still require substantial human oversight for accuracy [Bolen and Hunt, 2024]. The SAD and the locomotor neurophysiology literature it organizes — including the two-level CPG architecture, Ia/Ib/II feedback pathways, and contact-related reflex modulation — directly inform the sensory feedback design of the neural controller planned in Chapter 6.

2.8 Simulation Tooling: From OpenSim to MuJoCo

Two open-source ecosystems anchor the simulation pipeline planned for this project. OpenSim [Delp et al., 2007, Seth et al., 2018] provides the biomechanical modeling side:

musculoskeletal models with anatomically accurate geometry, muscle routing, and force-generating properties, along with tools for analyzing moment arms and isometric torque. MuJoCo [Todorov et al., 2012] provides the physics-engine side: fast, accurate simulation of rigid-body dynamics with contacts, along with muscle-like actuators that include activation dynamics. MyoConverter bridges the two by converting OpenSim 4.x musculoskeletal models to MuJoCo format with optimized muscle kinematics and kinetics [Caggiano et al., 2022, Wang et al., 2022]. Combined with SNS-Toolbox’s MuJoCo backend [Nourse et al., 2023], this makes it possible to convert the AARL’s modified OpenSim leg models into a MuJoCo robot, wrap it in an SNS network with Ia, Ib, and II proprioceptive channels plus contact sensors, and run the full neuromechanical loop faster than real time. This pipeline, and the CPG architecture at its core, are described in Chapter 6.

2.9 State of the Art in Neuromechanical Simulation

Neuromechanical simulation — the coupling of a musculoskeletal plant to a model of the neural controller that drives it — has advanced along three partially overlapping lines. The first line embeds reflex physiology directly into the controller. The muscle-reflex walking model of Geyer and Herr [2010] demonstrated that positive force, length, and velocity feedback around each muscle, organized with simple interlimb coordination, is sufficient to produce self-stabilizing human-like walking dynamics and realistic muscle activations without any central pattern generator. That model became the reference point for a generation of reflex-based neuromechanical controllers, including the cat hindlimb models of Section 2.6 [Markin et al., 2010, Shevtsova et al., 2016] and their extensions.

Hybrid work makes the complementarity of the two strategies explicit: Haeufle et al. [2018] combined feed-forward muscle activations with neuronal reflex feedback in simulated human walking and found that the robustness benefit of each depends on which muscle carries it.

The second line replaces hand-designed circuitry with machine learning. Song et al. [2021] trained deep reinforcement learning policies on a physiologically realistic musculoskeletal model in OpenSim, reproducing human kinematics and muscle activations across locomotion tasks, with the learned policy re-usable for experiments that would be difficult in the laboratory. This approach inherits the fidelity of the musculoskeletal plant but encodes control in opaque function approximations: the trained policy can reproduce behavior, but the individual “neurons” of the controller do not correspond to identifiable physiological circuits, so the class of scientific questions it can answer — lesions, deletions, pathway stimulation — is narrower than its behavioral repertoire. Recent perspectives on predictive musculoskeletal simulation [De Groote and Falisse, 2021] place both learned and optimal-control approaches in the same frame: the plant models are mature, and the open problems are the cost of solving them and the biological interpretability of the control they find.

The third line, in which this dissertation sits, is biologically grounded synthetic nervous systems: controllers whose neurons and synapses are explicit, conductance-based, and mapped to identified physiological circuits, simulated at speeds sufficient for real-time robotics [Nourse et al., 2023]. Hierarchical CPG control is meanwhile spreading to bipedal-capable musculoskeletal plants — motor-module bipedal walkers [Ichimura and Yamazaki, 2019], two-layer rhythm-generator and pattern-formation control of a musculoskeletal

macaque model [Nishizaki et al., 2026], and two-level CPG control of muscle-driven robotic hind legs [Fukuoka et al., 2021] — though always in simulation with generic musculature or on non-humanoid platforms; a two-layer SNS driving a custom-anatomy, artificial-muscle humanoid leg remains unclaimed. What distinguishes the present program from all three lines is the combination it attempts: an anatomically grounded controller (two-layer CPG with Ia, Ib, II, and contact feedback), a biomechanically validated pneumatic plant (Chapters 3–4), and an eventual physical robot. Simulation speed is the practical enabler: the MuJoCo backend evaluates the musculoskeletal loop faster than real time on commodity hardware [Todorov et al., 2012, Caggiano et al., 2022], and SNS-Toolbox sustains real-time or faster network evaluation from workstations down to embedded boards [Nourse et al., 2023]. A single overnight run can therefore sweep thousands of parameter combinations or execute hundreds of simulated gait cycles per condition — throughput that no human-subject or animal protocol could match, and that transforms lesion and stimulation studies from bespoke demonstrations into statistically powered experiments.

2.10 Testing Neuromechanical Hypotheses on Robots: The Ethical Case

The experiments this research program is designed to enable — deleting a muscle from a limb, blocking a sensory pathway, lesioning part of a rhythm generator, applying tonic stimulation to spinal circuits — cannot be performed on healthy humans. Invasive intervention on the human neuromuscular system is limited to clinical populations, is tightly constrained by ethics review, and permits neither the within-subject control nor the re-

versibility that neuromechanical hypotheses demand. The same experiments in animals carry real moral cost, and their justification is governed by the Three Rs framework: replacement of animal subjects with non-animal methods, reduction of the numbers required, and refinement of the remaining procedures [Russell and Burch, 1959, Tannenbaum and Bennett, 2015]. A robot that implements a validated neuromechanical hypothesis is precisely such a replacement method: every deletion, lesion, and stimulation experiment that runs on the synthetic nervous system is an experiment that does not require a living subject, and the robot population can be duplicated exactly, so that “subjects” are statistically identical across conditions — a form of reduction no breeding colony can offer. Refinement follows as well: the failure modes of these experiments damage hardware, not tissue.

For a robot to serve as a replacement subject, however, it must first earn standing as a scientific model. Webb’s influential analyses of biorobotic methodology [Webb, 2001, 2002] argue that a robot is a good model of biological behavior when the mechanism it implements is tested at the same level of organization as the target phenomenon — matching dynamics, control structure, and sensory ecology rather than surface appearance. The validation chain of this dissertation is constructed to that standard: the actuator force model is validated against bench data (Section 4.1), the joint torque model against isometric measurements and human torque envelopes (Section 4.2), and the neuromechanical controller will be verified against published tonic stimulation and deletion phenomena [Selionov et al., 2009, Rybak et al., 2006] before any locomotion tuning is attempted. Only then can a robot experiment be read as evidence about the hypothesized biological circuit rather than about the robot.

The argument is ultimately bidirectional, completing the loop described in Section 1.2.

Experiments that are unethical, impractical, or statistically unreachable in humans and animals become routine on the robot; the resulting insights into neuromechanical control feed back into biology, into the design of the next robot iteration, and directly into prosthetic and orthotic devices that face the same problem of controlling muscles-like actuators for human users, whose central pattern generator controllers are an active topic in lower-limb exoskeleton research [Duan et al., 2024, Pery et al., 2026]. The ethical case and the scientific case are the same case: the robot is the instrument that makes these experiments possible at all.

3 Materials and Methods

Publication note: Sections 3.2–3.3 and the actuator characterization results reported in Chapter 4 appeared in Bolen et al. [2026] (Bolen, Elzein, Pang, and Hunt, *Actuators*, 2026). Sections 3.4–3.7 and the knee torque results reported in Chapter 4 are also the subject of an unpublished manuscript in preparation by Bolen, Burgess, Morrow, Elzein, Pang, Scharzenberger, and Hunt [Bolen et al., 2025]. This manuscript has not been submitted for publication. For both works, B.B. performed the conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, and original draft preparation; M.E., L.P., L.B., and C.M. contributed methodology, software, investigation, and data curation; and A.H. supervised the work, administered the projects, and acquired funding. All co-authors contributed to writing, review, and editing.

3.1 Overview

The experimental program of this dissertation proceeds from the actuator to the joint. Because the robot joints are driven by braided pneumatic actuators (BPAs), the foundation of any torque prediction is an accurate model of the force produced by a single actuator as a function of its length, pressure, and contraction. To measure artificial muscle force as a function of length and pressure, we built a test jig from extruded aluminum and 3D printed parts. Festo BPAs with 10 mm and 20 mm uninflated diameters were tested with various resting lengths and different amounts of contraction. These data are used to develop an improved model that more accurately predicts maximum muscle force based

on the resting length l_0 , current muscle length l_m , and pressure P (Sections 3.2 and 3.3).

With the actuator model in hand, the focus shifts to the joint. This study begins with building a jig to measure knee torque produced by BPAs at various knee angles (θ_k). With the expectation that measured torque will be less than that predicted by the model, the robot muscle arrangement and knee design are simplified to elucidate the variables that most affect the results. Two distinct joints are used and tested: a simplified pinned knee joint and a biomimetic knee [Steele, 2018]. Each joint is driven by a single flexor and a single extensor muscle. Variations were made to the muscle origin/insertion locations, resting lengths, diameters, and whether they used an artificial tendon. The results are compared with two models, and the new data are used to create an improved torque model that accounts for system compliance and muscle wrapping (Sections 3.4–3.7).

3.2 Actuator Force Model

The actuator force model used throughout this dissertation is described here. First, the amount of contraction *varepsilon* in a BPA is calculated as

$$\varepsilon = \frac{(l_0 - l_m)}{l_0} \quad (3.1)$$

where l_0 is uninflated resting length of the BPA measured from hose clamp to hose clamp. Current muscle length is l_m .

Similarly, ε_{620} is the amount of contraction in a BPA without external load when inflated at 620 kPa, calculated as

$$\varepsilon_{620} = \frac{(l_0 - l_{620})}{l_0} \quad (3.2)$$

where l_{620} is the muscle length measured at 620 kPa and free from external loads.

The amount of relative contraction of a BPA ε^* can then be calculated by dividing equation (3.1) by (3.2), such that

$$\varepsilon^* = \frac{\varepsilon}{\varepsilon_{620}} \quad (3.3)$$

describes the BPA's current contraction as a normalized percentage of it's total.

The amount of normalized isometric force F^* in a Festo BPA is then calculated as

$$F^*(\varepsilon^*, P^*) = \begin{cases} c_0 \cdot (\exp(-c_1 \cdot \varepsilon^*) - 1) + P^* \cdot \exp(-c_2(\varepsilon^*)^2) & \text{for } F^* > 0 \\ 0 & \text{for } F^* \leq 0 \end{cases} \quad (3.4)$$

where P^* is pressure P normalized by 620 kPa. BPA diameter dependent coefficients are determined by the actuator characterization experiments reported in Chapter 4 (Table 4.2).

The maximum force F_{max} in a BPA is calculated as

$$F_{max}(l_0, P_{max}) = a_1 \cdot P \cdot \arctan(a_2 \cdot P \cdot (l_0 - 0.0075)) \quad (3.5)$$

with coefficients determined by the actuator characterization experiments reported in Chapter 4 (Table 4.1). When maximum system pressure is set to $P = 620$ kPa, $F_{max}(l_0, 620) = F_{620}(l_0)$. Multiplying (3.4) by (3.5) yields

$$F(\varepsilon^*, P^*, l_0) = F^*(\varepsilon^*, P^*) \cdot F_{620}(l_0) \quad (3.6)$$

as the equation for isometric force F in a Festo BPA.

3.3 BPA Isometric Force Characterization Experiment

To measure artificial muscle force as a function of length and pressure, we built a test jig from extruded aluminium and 3D printed parts. Festo BPAs with 10 mm and 20 mm diameter were tested with various resting lengths and different amounts of contraction. These data are used to develop an improved model that more accurately predicts maximum muscle force based on l_0 , current muscle length l_m , and pressure P . The results are compared with existing models and the new data are used to create an updated model.

A test jig was made of 80/20 brand 1515 series extruded aluminum (Figure 3.1). Artificial muscles were placed vertically in the jig one at a time. The upper end was attached to a load cell. Force data was collected using a CALT DYLY-103 100 kg S-shaped load cell for the $\phi 10$ mm BPAs and a Loadstar RAS1-2HKS-S for the $\phi 20$ mm BPAs. The was load cell was used in conjunction with a HX711 Load Cell Amplifier. Pressure data was obtained from a Freescale MPX5700 GP 5 V pressure sensor. Building air supply pressure was controlled with two pressure regulators in series. The first was a Parker model 20R113GC 0 – 120 psi pressure regulator. The second was a Husky 3/8 in. High Performance Air Regulator HDA72200. A Festo VTUG-10-MRCR-S1T-26V20-T516LA-UL-532S-8K valve manifold VTUG-G was used to deliver air from the pressure regulator to the actuator. This manifold is comprised of eight two-in-one bidirectional normally closed Festo VUVG-S10-

T32C-AZT-M5-1T1L valves.

Pressure and load cell amplifier data were sent to MATLAB 2021b via an Arduino Uno style Sparkfun BlackBoard C microcontroller. The computers that used MATLAB were running either Windows 10 or Windows 11. During phases when the Arduino was collecting force and pressure data to send to MATLAB, the Arduino would also trigger (via MATLAB) an Onsemi 2N4401 NPN transistor to make the valve manifold open or close the valve. For other data collection, the valve was manually opened and closed.

The inner distance between the hose clamps on each BPA was measured with a FANUC tape measure to determine the muscle's length l and resting length (l_0). This is how Festo AG & Co. KG [2026] defines l_0 , although in Hunt et al. [2017a] it was measured to also include end cap length. We then inflated each BPA to $P_{620} = 620$ kPa, with one end allowed to move freely in the axial Degree of Freedom (DoF), and measured the length l_{620} to calculate maximum contraction at 620 kPa using Equation (3.2). The distance between the crossmembers was then controlled to obtain different amounts of contraction, the muscles were inflated to various pressures, and the contractile force was recorded at the pressure–contraction pairs. This was done for $\phi 10$ mm BPA resting lengths (l_0) of 112 mm–518 mm. For $\phi 20$ mm BPAs, resting lengths (l_0) of 300 mm–509 mm were used.

3.4 Robot Leg Model and Knee Test Stands

The primary components of the robot leg are the knee joint, femur, tibia, BPA assemblies, and artificial tendons (Fig. 3.2A–C). The artificial bone components are 3D printed using a combination of Onyx, carbon fiber, and high-strength high-temperature fiberglass on

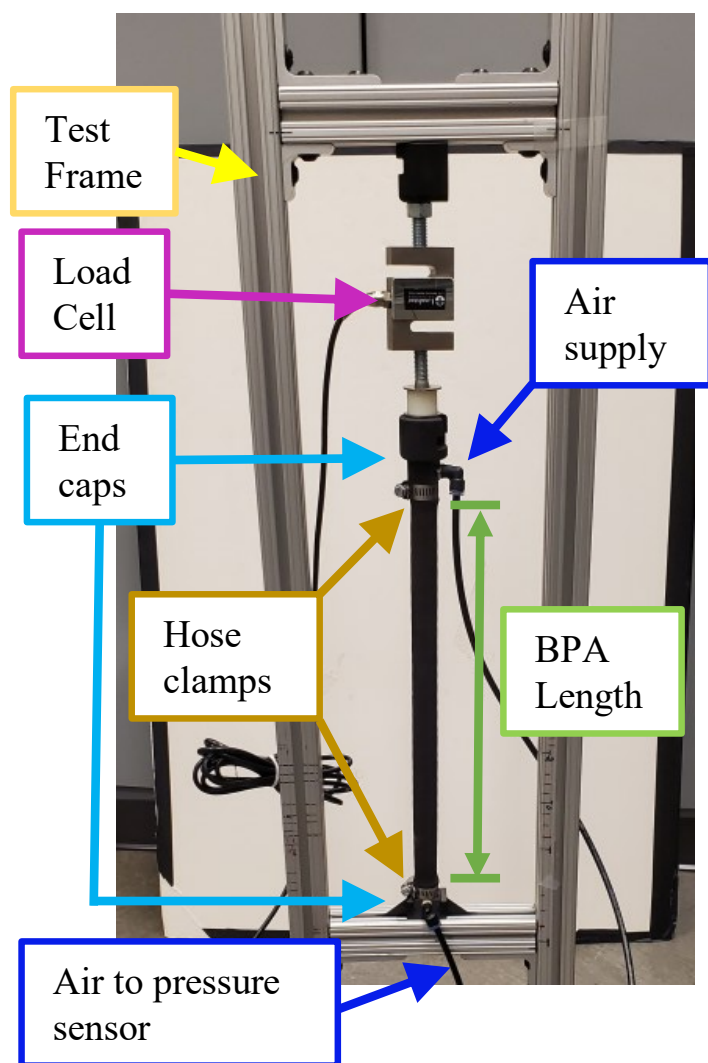


Figure 3.1: A picture of a BPA in the isometric force test stand with components labeled.

Markforged Onyx One and Mark Two printers. Onyx is a proprietary Markforged material that consists of chopped carbon fiber in nylon. Artificial tendons are made with Shimano bicycle brake cable.

The leg assembly geometry and dimensions are based on bones scans of a 6 ft (1.83 m) tall person [Bolen and Hunt, 2019, Seth et al., 2018, Delp et al., 1990, 2007]. Two knee joints were tested, one pure rotational joint, and one biomimetic joint with rotation and translation. Each joint is driven by two antagonistic Festo actuators. Both robot joints have the same center of rotation in the femur frame at $\theta_k = 0$ deg.

Details on the biomimetic robot knee used in this study can be found in Steele [2018], Steele et al. [2017, 2018]. In brief, this 1-DoF joint (Fig. 3.3) uses a four-bar linkage to change the effective moment arm during rotation. The intersection point of the internal crossed links define the instantaneous center of rotation (ICR). This causes the joint ICR location in the femur frame to translate posteriorly in the X direction and distally in the Y direction except for a brief proximal move near full flexion. Reference coordinate frames are given by Wu and Cavanagh [1995], Wu et al. [2002]. This is similar to how the human knee behaves [Bolen and Hunt, 2019, Morrow et al., 2020, Seth et al., 2018, Delp et al., 2007, 1990]. The correct location of the ICR is important for both theoretical and experimental torque calculations.

One uniarticular knee extensor and one uniarticular knee flexor are used to actuate the joint. Each knee joint type was tested using $\phi 10$ mm Festo BPAs. Additionally, the biomimetic knee joint was tested using a $\phi 20$ mm Festo BPA. Each artificial muscle was pinned at the muscle origin location on the femur while the other end was either pinned or attached to the muscle insertion location on the tibia via an artificial tendon.

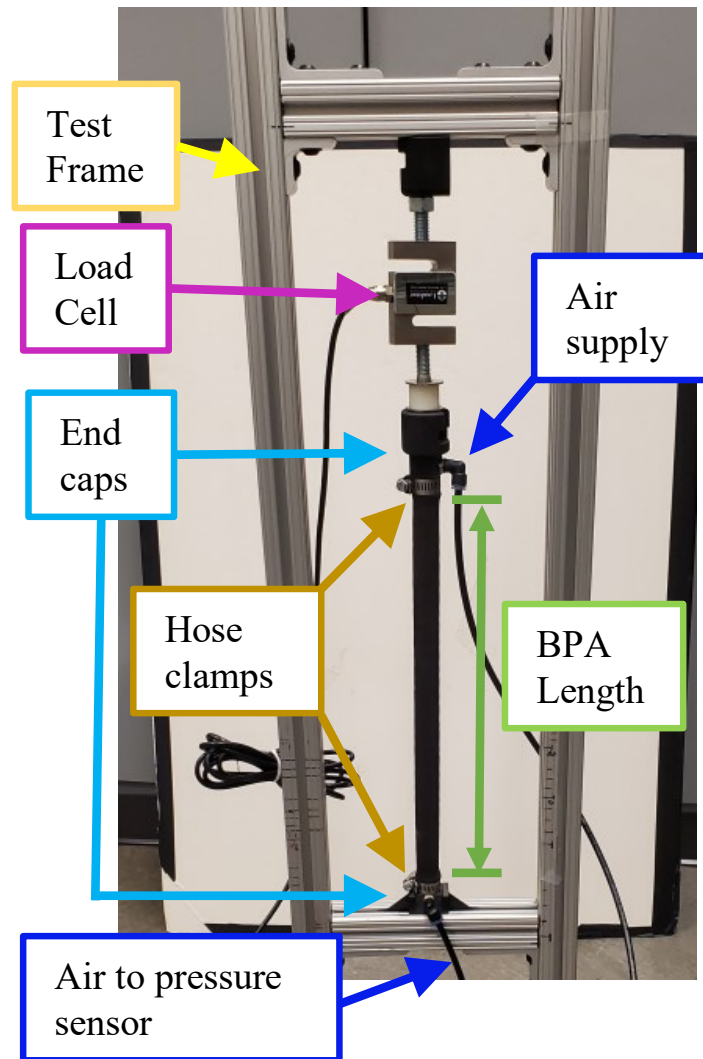


Figure 3.2: **(A)** CAD solid model assembly of the revolute joint knee in the test stand. **(B)** 1-DoF pinned-joint robot knee in the test apparatus with important components labeled. Leg is shown with 30 degrees flexion, i.e. $\theta_k = -30^\circ$. **(C)** Picture from isometric torque testing. Setup shows the revolute knee configured for a test of the extensor BPA. The knee is flexed with $\theta_k = -120^\circ$. Also note the compressed shape of the inflated BPA on the knee during this high degree of flexion.

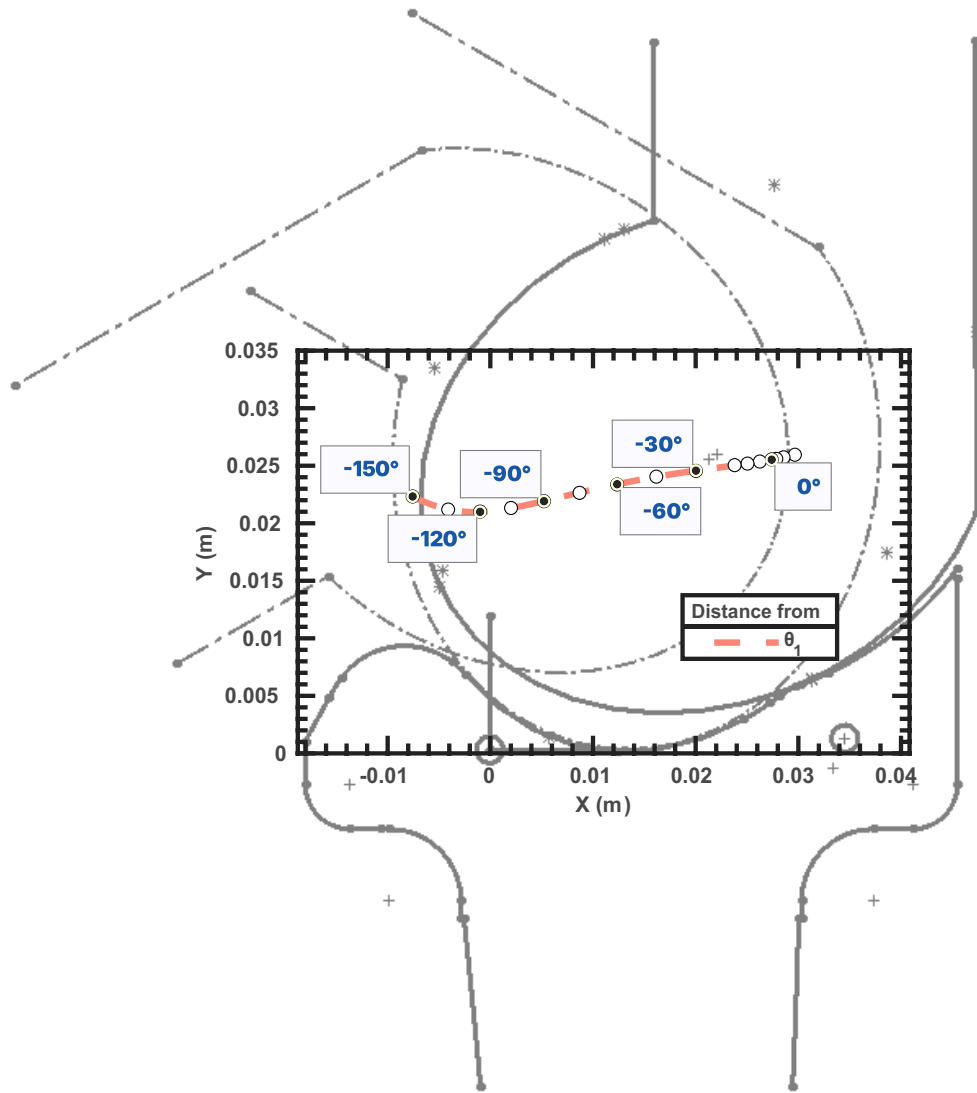


Figure 3.3: ICR for biomimetic robot knee as a function of knee angle. Measured from θ_1 on the tibia. Plot is overlaid on model of the knee at $\theta_k = 0^\circ$ (solid line), and -60° and -120° (dotted dash lines). White circles represent measured points in -15° increments from -150° to -15° , -5° increments from -15° to 0° , and then at 2° , 5° , and 10° .

Moment arms are calculated using vector components between muscle locations and the joint's instantaneous center of rotation (ICR). Given a vector $\vec{d}$ from a joint to the line of action of the force vector $\vec{F}$, the moment arm $\vec{r}$ is the shortest distance to this line of action. In this study, it is calculated using the method specified by Young et al. [2019]. The scalar moment arm, r_k , about the z_k axis (i.e., the Z axis in the knee frame) is calculated as

$$r_k = p_{proj,xy} \cdot \frac{\vec{p}_f \times \hat{z}_k}{\|\vec{p}_f \times \hat{z}_k\|} \quad (3.7)$$

where $p_{proj,xy}$ is the free muscle segment projected onto the XY-plane, which is normal to the joint axis. $\vec{p}_f$ is the projected muscle segment vector. Moment arm is calculated in the tibial body frame.

Details of the existing model and the torque calculation methodology in the modeled robot are given by Bolen and Hunt [2019], Morrow et al. [2020].

3.5 Knee Torque Data Collection and Equipment

A test jig was built to collect knee torque measurements at different knee angles over its range of motion. The test stand frame is made predominantly out of 80/20 1515 series extruded aluminum components. The knee joint is free to rotate the tibia while the femur is fixed to the frame. A force sensor has one end connected to the tibia and the other is connected to a swing arm. The swing arm is tied to a winch with 3/16 inch Kevlar rope from Quality Nylon Rope routed over intermediate idler pulleys (see Fig. 3.2B and 3.2C). At each measured position of θ_k the winch would be locked and the BPA would be inflated and then deflated. This test procedure kept the measured results as isometric as possible.

The knee joint is co-axial with gravity to eliminate gravitational torque (see Fig. 3.2A and Fig. 3.2C).

Force data was collected using one of two different sensors. The first is a MODERN STEP 300 kg digital crane scale. The second force sensor is a CALT DYLY-103 100 kg S shaped load cell. The load cell is used in conjunction with a HX711 Load Cell Amplifier. Pressure is measured with a Freescale MPX5700 GP 5 V pressure sensor. Building air supply pressure is controlled with two pressure regulators in series. The first is a Parker model 20R113GC 0 – 120 psi pressure regulator. The second is a Husky 3/8 inch High Performance Air Regulator HDA72200. A Festo VTUG-10-MRCR-S1T-26V20-T516LA-UL-T532S-8K valve manifold was used to supply air to the actuator. Pressure and load cell amplifier data are sent to MATLAB via an Arduino Uno style Sparkfun BlackBoard C microcontroller.

The length measurements were made using a tape measure. When there was sufficient curvature in the BPA during knee extensor torque tests, a flexible piece of string was used to mark the axial length and then transferred to the tape measure. Angle measurements were taken with a Medigauge digital electronic goniometer or with MALENOO analog goniometers of 6, 8, or 12 inch lengths.

3.6 Hybrid Torque Calculation Method

A Hybrid Torque calculation method was developed to help uncover inconsistencies between the model and recorded torque values. This method measures the actual muscle length and angles of the robot to calculate force and joint torque. With this method, it is

easier to determine which parts of the torque equation did not achieve their predicted value and provide a correction to that part of the equation. The classical mechanics way to calculate the torque $\vec{M}$, about a joint is to take the cross product of distance, $\vec{d}$, and force, $\vec{F}$

$$\vec{M} = \vec{d} \times \vec{F} \quad (3.8)$$

where distance $\vec{d}$ and force $\vec{F}$ are described in the previous section. Knee torque is the scalar torque M about the Z axis, which can simplify Eq. 3.8 to

$$M = r \times F \quad (3.9)$$

where r is the moment arm measured by a tape measure, and F is the scalar force, calculated by measuring the BPA's P , l_0 , l_{620} and l_m values and using the model fit equations determined in the following section.

3.7 Improved Torque Model

The classical kinematic torque model assumed rigid links and inextensible muscle paths. Data collected during the first flexor experiments (Chapter 4) showed that this assumption does not hold for a pneumatic system: measured muscle lengths were consistently shorter than predicted, and measured torque deviated from prediction in a load-dependent manner. This section describes the corrections that were developed to account for a constant length offset, series compliance in the attachment hardware and artificial tendons, and the loss of usable muscle length when a BPA wraps around the joint.

Analysis of measured muscle length data shows that it was consistently shorter than the predicted muscle length at all joint angles. This shorter length results in less muscle force, a shifted RoM, and less torque in the experimental setup. That this shorter length occurs when $\varepsilon = 1$ and $F = 0$ indicates that a constant length offset should be used. Therefore, to produce an improved torque prediction model, l_m was recalculated as

$$l_m = l_{LMT} - 2 l_{fitting} - \chi_0 \quad (3.10)$$

where l_{LMT} is the path length of the musculotendon from origin to insertion point, $l_{fitting}$ is the length of the end caps, and χ_0 is a constant length offset.

Additionally, as the muscles produce more force, compliant components in the limb segments and attachment hardware deform. This deformation changes the effective braided pneumatic actuator (BPA) length, which alters contraction and therefore changes the force predicted by the BPA model. This effect is most apparent at high forces and low contractions. To correct this error, we approximated the combined hardware compliance using Hooke's law in the bracket's principal directions.

A local bracket frame was established at the attachment bracket. The $\hat{x}_{br}$ direction was aligned with the vector from the bracket reference point to the BPA attachment point and represents the stiffest direction of the bracket. The $\hat{y}_{br}$ direction was chosen to point in the posterior direction with respect to the tibia frame, corresponding to the primary cantilever bending direction (see Fig. 3.4), while the $\hat{z}_{br}$ direction completes the right-handed frame. The bracket stiffness matrix is modeled as a diagonal matrix to reflect independent stiffness components along its principal axes:

$$\mathbf{K}_{br} = \begin{bmatrix} \chi_1 & 0 & 0 \\ 0 & \chi_2 & 0 \\ 0 & 0 & \chi_1 \end{bmatrix} \quad (3.11)$$

The corresponding local compliance matrix is

$$\mathbf{C}_{br} = \mathbf{K}_{br}^{-1} \quad (3.12)$$

Force that doesn't account for deformation is $\mathbf{F}$. The force applied to the bracket was expressed in the bracket frame as $\mathbf{F}'$. This force vector can be written in terms of its magnitude and unit direction as

$$\mathbf{F}' = \|\mathbf{F}'\| \hat{\mathbf{u}} \quad (3.13)$$

where $\hat{\mathbf{u}} = [u_x, u_y, u_z]^T$ is the BPA force direction expressed in the bracket frame. The bracket deflection vector, ϵ_{br} , was calculated from Hooke's law as

$$\mathbf{F}' = \mathbf{K}_{br} \epsilon_{br} \quad (3.14)$$

Solving Eq. 3.14 for the bracket deflection gives

$$\epsilon_{br} = \mathbf{K}_{br}^{-1} \mathbf{F}' = \mathbf{C}_{br} \mathbf{F}' \quad (3.15)$$

Substituting Eq. 3.13 into Eq. 3.15 gives

$$\boldsymbol{\epsilon}_{br} = \|\mathbf{F}'\| \mathbf{K}_{br}^{-1} \hat{\mathbf{u}} \quad (3.16)$$

For the diagonal stiffness matrix in Eq. 3.11, Eq. 3.15 is equivalent to the component-wise expression

$$\boldsymbol{\epsilon}_{br} = \begin{bmatrix} F'_{x,br} / \chi_1 \\ F'_{y,br} / \chi_2 \\ F'_{z,br} / \chi_1 \end{bmatrix} \quad (3.17)$$

Although $\boldsymbol{\epsilon}_{br}$ is a three-dimensional deflection vector, the BPA force model depends on the scalar change in effective BPA length (δ). Therefore, only the component of $\boldsymbol{\epsilon}_{br}$ along the BPA line of action directly changes the effective BPA length. This scalar bracket deflection is the projection of $\boldsymbol{\epsilon}_{br}$ onto $\hat{\mathbf{u}}$:

$$\delta_{br} = \hat{\mathbf{u}}^T \boldsymbol{\epsilon}_{br} \quad (3.18)$$

Substituting Eq. 3.15 into Eq. 3.18 gives

$$\delta_{br} = \hat{\mathbf{u}}^T \mathbf{K}_{br}^{-1} \mathbf{F}' \quad (3.19)$$

Using Eq. 3.13, Eq. 3.19 becomes

$$\delta_{br} = \|\mathbf{F}'\| \hat{\mathbf{u}}^T \mathbf{K}_{br}^{-1} \hat{\mathbf{u}} \quad (3.20)$$

The scalar stiffness along the BPA line of action due to the bracket compliance is

therefore

$$k_{\hat{\mathbf{u}},br} = \frac{\|\mathbf{F}'\|}{\delta_{br}} = \left(\hat{\mathbf{u}}^T \mathbf{K}_{br}^{-1} \hat{\mathbf{u}} \right)^{-1} \quad (3.21)$$

This projected-compliance form is used because the force is applied along $\hat{\mathbf{u}}$, while the bracket is allowed to deflect independently along its local $\hat{x}$, $\hat{y}$, and $\hat{z}$ directions. The reduced-order Hooke's law expression for the bracket deflection along the BPA line of action is therefore

$$\|\mathbf{F}'\| = k_{eq} \delta \quad (3.22)$$

where k_{eq} is calculated with (3.21) when there is no artificial tendon. When an artificial tendon was used, it was modeled as an axial spring with stiffness k_{ten} . Since both bracket deflection and tendon stretch reduce the effective BPA length along the actuator line of action, their scalar compliances were added in series:

$$k_{eq} = \left(\frac{1}{k_{br}} + \frac{1}{k_{ten}} \right)^{-1} \quad (3.23)$$

Since δ is the scalar compliance displacement along the BPA line of action, then the compliance-corrected active BPA length used in the force balance is

$$l_m(\delta) = l_m^{(0)} - \delta \quad (3.24)$$

where $l_m^{(0)}$ is calculated from (3.10).

The corresponding relative contraction is $\varepsilon^*(\delta)$. Since the BPA force depends on con-

traction, and contraction depends on the compliance displacement, δ was found using the Newton-Raphson method to solve the nonlinear force-balance equation

$$F_{\max} \cdot F_{\text{BPA}}(\epsilon^*(\delta), P^*) - k_{eq} \delta = 0 \quad (3.25)$$

After solving Eq. 3.25, the corresponding force magnitude was used in Eq. 3.15 to compute the vector bracket deflection. This deflection was transformed back into the limb reference frame and used to update the BPA attachment location. The routed BPA length, force direction, moment arm, and torque were then recalculated using the updated attachment location. In the present implementation, the optimized stiffness terms χ_1 and χ_2 were treated as effective compliance parameters for the attachment system rather than direct measurements of an isolated bracket. This allowed unmodeled compliance in the surrounding hardware to be represented by the reduced-order stiffness model.

Multiobjective optimization using `gamultiobj` in MATLAB was used to determine the length offset (χ_0) and the stiffness terms (χ_1 and χ_2) until the error between the model and the experimental data was minimized for the $l_0 = 48.5$ cm flexor BPA (Fig. 4.4C). Because the bracket forms an arch shape and extends for 75 mm in the Z direction, the extensor configuration required an additional correction. The inner side of a BPA that bends over the joint experiences less tensile stress than the outer side, creating a force imbalance and a loss of part of the arc length of the BPA as usable length. An additional optimization term

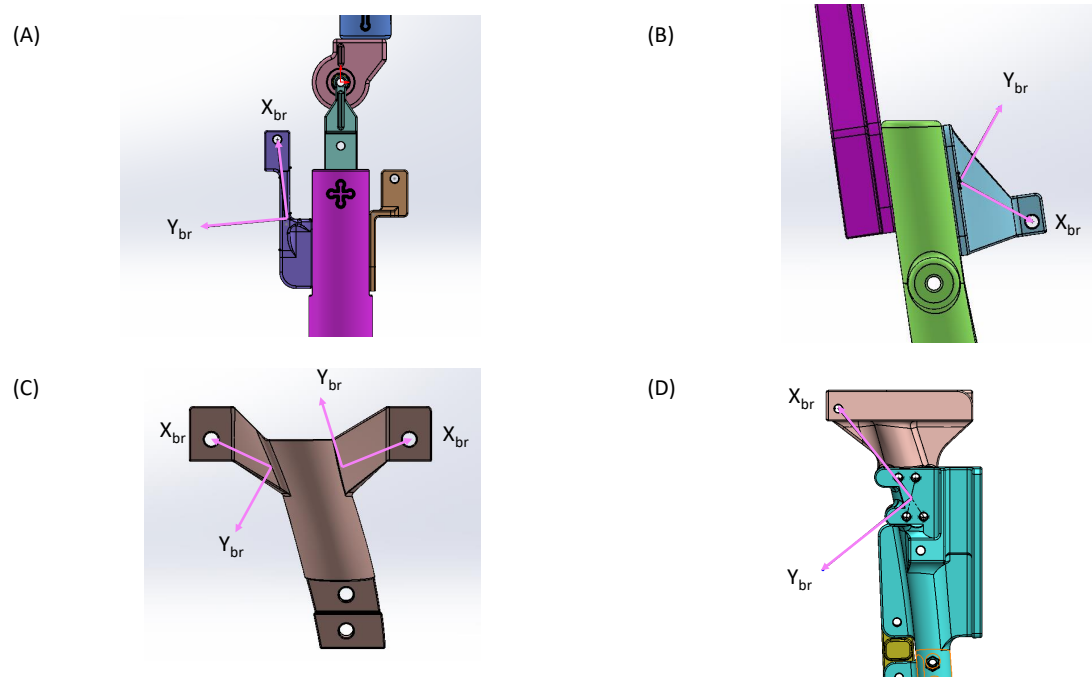


Figure 3.4: Bracket frame orientations for: **(A)** Flexor configuration on the pinned knee, **(B)** extensor configuration on the pinned knee, **(C)** flexor and extensor configurations on the biomimetic knee with $\phi 10$ mm BPAs, and **(D)** flexor configuration on the biomimetic knee with the $\phi 20$ mm BPA.

χ_3 was introduced to capture this wrapping loss, such that

$$\Delta l = \begin{cases} \chi_3 R |\theta_{\text{wrap}} - \theta_k| \zeta^2 & \theta_k \leq \theta_{\text{wrap}} \\ 0 & \theta_k \geq \theta_{\text{wrap}} \end{cases} \quad (3.26)$$

where Δl is the change in usable length of the BPA, R is the wrapping radius. θ_{wrap} is the knee flexion angle when the BPA starts to bend over the knee joint, determined from solid modeling. ζ is the additive complement to relative strain, such that

$$\zeta = 1 - \epsilon^* \quad (3.27)$$

Combining (3.27) into (3.26), and subtracting Δl from (3.10) yields

$$l_m = l_{LMT} - 2 l_{fitting} - \chi_0 - \Delta l \quad (3.28)$$

The optimization algorithm was again `gamultiobj` in MATLAB. This time, the muscle origins bracket was chosen. The bracket frame is shown in Fig. 3.4B. Since the bracket forms an arch shape and extends for 75 mm in the Z direction, for this configuration only we set K_2 in the X_{br} direction, and K_1 in the Y_{br} and Z_{br} directions. The optimization algorithm for this set solves for the best average goodness-of-fit (GoF) measures RMSE, FVU, and maximum absolute residual for two BPAs, while the third one was held out. Three independent rounds of optimization were performed, so that one of each of the BPAs could be held out as validation. Then we picked the solution with the shortest distance between the average normalized GoF and the validation data, while also requiring

improved GoF values for all the individual BPAs, with the exception of maximum absolute residual on the $l_0 = 48.0$ cm BPA.

This initial identification was subsequently extended to the full archived test sets in two stages. First, the flexor model was given a second compliant bracket at the origin-side attachment (a hip-frame cantilever carrying the BPA origin), so that the compliance of both ends of the actuator path is represented; the stiffness pair (χ_1, χ_2) is shared by the two brackets in fixed axis assignments (Appendix C, Table C6), leaving a three-term fit (χ_0, χ_1, χ_2) . The flexor fit was cross-validated over the five pinned-knee flexor tests, holding out the encoder-corrected 47 cm test as validation. Second, the extensor refit locked (χ_1, χ_2) to the flexor front values — enforcing a single stiffness pair across the flexor and extensor configurations — and solved only the extensor-specific terms (χ_0, χ_3) on the six-test extensor validation pool, with the extensor length offset allowed to take negative values. The bounds and the adopted values of both stages are recorded in Appendix C (Tables C7 and C8); the fit outcomes are summarized in Section 4.5.

The goodness-of-fit metrics used to judge the optimization results are the root mean square error (RMSE), the fraction of variance unexplained (FVU), and the maximum absolute residual, each computed between the measured and predicted isometric torque over the tested range of motion.

3.8 Optimization Algorithms and Cost Functions

Two distinct optimization problems underpin the results of this dissertation and the design framework built on them. The first is the identification of the correction terms of

Section 3.7 (χ_0 , χ_1 , χ_2 , and χ_3) from the pinned-knee test data. The second is the redesign of the biomimetic knee route geometry — the BPA origin $\mathbf{p}_1$, insertion $\mathbf{p}_{\text{end}}$, resting length, and tendon length (Appendix C) — so that the flexor and extensor configurations meet or exceed human knee torque capability over the range of motion. Both are carried out in MATLAB [The MathWorks, Inc., 2025].

The correction-term identification is a genuine multiobjective problem: candidate parameter sets are judged simultaneously on RMSE, FVU, and maximum absolute residual (Section 3.7). It is solved with the genetic multiobjective solver `gamultiobj`, an implementation of the elitist nondominated-sorting genetic algorithm NSGA-II [Deb et al., 2002, The MathWorks, Inc., 2025], which returns a Pareto front of parameter sets from which the adopted solution is selected by the holdout-validation procedure described above. The optimizer parameterization (length offsets in centimeters, stiffnesses on a logarithmic scale) and its bounds are recorded in Appendix C.

The route redesign replaces the multiobjective formulation with a single scalar cost assembled from weighted penalty terms, because the route prediction (contact activation, wrapping, and compliance correction) is expensive to evaluate and only moderately smooth. The cost has the form

$$J = w_w J_{\text{worst}} + w_t J_{\text{torque}} + \sum_a J_a + J_{\text{geom}} + J_{\text{len}}, \quad (3.29)$$

where J_{worst} is the largest fraction by which robot torque falls short of the human target at any angle across the range of motion, weighted heavily ($w_w = 10^5$); J_{torque} is the mean-squared shortfall fraction ($w_t = 10^3$); the J_a are configuration-specific auxiliary penalties

described below; and J_{geom} and J_{len} are regularizers that keep the endpoints and actuator lengths near their seed values. Human torque targets are computed from the Gait2392 lower-extremity model [Delp et al., 2007, Seth et al., 2018]: the flexor is graded against the biceps femoris short head and the extensor against vastus medialis. Hard constraints (bone clearance, wrap feasibility, actuator strain and contraction limits, tendon geometry) are enforced alongside the cost, and any candidate whose prediction fails is rejected with a prohibitive penalty.

The solution pipeline is common to both configurations. Each run begins with a full sanity evaluation of the seed design and a small smoke test of the solver setup, then executes `surrogateopt` — MATLAB’s surrogate-assisted global solver for expensive black-box objectives, which builds a spatial interpolation surrogate over sampled candidate points and searches it adaptively [The MathWorks, Inc., 2025] — for 7,000 evaluations in parallel, and finally refines the best feasible point with `patternsearch`, a derivative-free mesh-adaptive direct search method [Audet and Dennis, 2006, The MathWorks, Inc., 2025], for 15,000 further evaluations. The entry scripts and their cost functions are archived with the code (`Opt_run.m`, `Opt_run_Ext.m`).

Although both configurations share this pipeline, their cost functions differ in ways that reflect the mechanics. The flexor BPA routes around a single moving wrap point, so clearance from the tibia and femoral condyles is enforced by continuous signed-distance constraints, and its cost adds an off-axis torque penalty (torques about the joint axes other than the knee axis, relative to the original single-BPA route) and a penalty if the wrap fails to release cleanly as the knee extends. The extensor BPA wraps over the front of the joint through as many as seven sequential contacts, so clearance is enforced instead by the

contact-elimination rule that governs the route, and its cost penalizes torque overshoot above the human curve and excessive actuator contraction, which risks bottoming out at high flexion. The flexor target also carries a required torque margin of 1% above the human curve, whereas the extensor target is the raw human curve. With two BPAs per joint, the flexor route is mirrored about the mid-sagittal plane and the two actuator forces are summed, while the extensor model scales the force of a single shared route by the actuator count.

3.9 Balance Platform and Inverted-Pendulum Testbeds

In parallel with the knee test stands, the author, together with McNeal and Williams, developed a family of hardware testbeds for the balance problem that a walking humanoid must eventually solve: a robotic inverted pendulum actuated by BPAs at the ankle and standing on a motorized oscillating balance platform [Bolen et al., 2024b,a,c]. The testbeds deliberately reduce the robot to a single actuated joint and a single sensory question — which measurement of body orientation, fed through which feedback law, will hold the leg upright on a moving support — so that vestibular and proprioceptive feedback strategies could be exercised in hardware before their adoption in the full humanoid, whose lateral stabilization is future work (Chapter 6). Three generations were built (Figure 3.5).

All three generations share one control template. A microcontroller reads body orientation, compares it against an upright set point, and commands the antagonist BPA pair with a correction proportional to the measured angle error: the command to one muscle of the pair is increased while the command to its antagonist is decreased by the same amount,

about an equilibrium at which both valves sit half open, so that one muscle pressurizes as the other vents. Because the valves are on/off solenoid valves supplied from a manifold, the proportional command is realized by time-proportioning — pulse-width modulation of the valve duty cycle — with the switched pressure routed to the BPA chambers through a transistor-array driver stage, and with chamber pressures monitored by pneumatic pressure transducers. The pressure–force dynamics that this pulse-modulated scheme excites are modeled in companion work [Elzein et al., 2025].

The first generation used a hinge ankle driven by an antagonist pair of 20 mm BPAs acting as plantarflexor and dorsiflexor. Its ankle angle could be measured by a shaft encoder mounted on the hinge axis, estimated from an inertial measurement unit (IMU) mounted on the tibia, or both, selectable as the feedback source for the proportional law; the IMU angle was obtained by integrating the gyro’s angular rate after a static bias correction, with the accelerometer available as a gravity reference. The distinctive feature of this generation is its transmission. The two BPAs operated in parallel on a four-wheel block-and-tackle — a “reverse pulley” system — mounted on the foot, with the artificial tendon reeved through the pulleys and inserting on the tibia. Reeling the tendon through the pulley train multiplies the tendon travel delivered at the joint at a proportional cost in transmitted force and an increased reaction force carried by the foot; the design consequences are examined in Section 5.7.

The second generation, IP 2.0, replaced the hinge with a ball-and-socket ankle, so that the foot could combine plantarflexion and dorsiflexion with inversion and eversion. Three muscles were attached: one soleus-type plantarflexor, and two dorsiflexors — one routed to act as an inverter and one as an everter — arranged so that the muscles can

act antagonistically against one another as the demanded ankle orientation changes; in the controller, the two dorsiflexors shared a single complementary command against the plantarflexor. A ball-and-socket joint presents no shaft on which a rotary encoder can be carried, so angle feedback in this generation was IMU-only.

The final balance platform stands the inverted pendulum, its ankle driven by an antagonist BPA pair, on a motorized platform that oscillates the support surface beneath the foot. An IMU on the tibia serves as a pseudo-vestibular sensor, feeding the same proportional law. This is the configuration in which balance was demonstrated both on flat ground and under the oscillating support (Section 4.3).

3.10 Prior Neuromechanical Simulation: The AnimatLab Bipedal Walker

Before the simulation pipeline of Chapter 6 was defined, the author and Morrow built a neuromechanical model of a bipedal walker in AnimatLab, a 3D graphics environment for constructing biomechanical systems actuated by synthetic neural systems [Cofer et al., 2010]. The course report describing that model [Bolen and Morrow, 2020] is summarized here because its biomechanics and network architecture are the direct ancestors of both the current AnimatLab walker model, whose neuron and synapse parameters are documented in Appendix B, and the two-layer CPG controller planned in Chapter 6.

The biomechanical plant consisted of nine rigid bodies — pelvis, thigh, shank, foot, and toe segments — connected by sagittal-plane hinge joints with human-like rotational capabilities and actuated by antagonistic pairs of Hill-type muscles at the hip, knee, and ankle. The toe joint, the author’s addition to the Li design, carried a passive spring instead

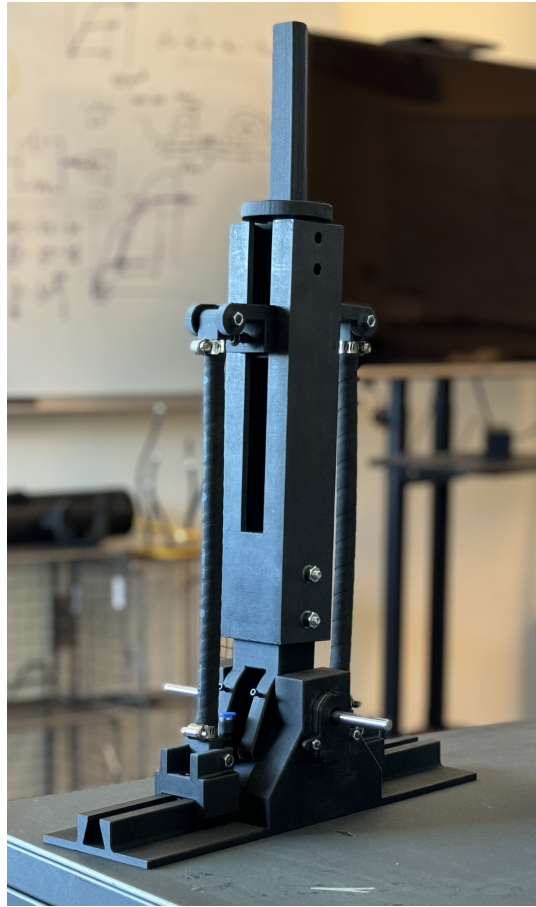


Figure 3.5: The BPA-actuated inverted-pendulum testbed. The pendulum shank rides on a hinge ankle above the foot assembly, and braided pneumatic actuators route down both sides of the shank to the foot, forming the antagonist ankle pair. Photograph from the AARL Inverted Pendulum project repository.

of a muscle pair, increasing foot compliance and conserving energy; its contribution to push-off would make an interesting study in its own right once the model walks on the ground. The mechanics followed the suspended bipedal walking model of Li et al. [2017], whose CPGs were entrained by heel and toe mechanosensory receptors and hip-angle position sensors standing in for the group-I loading, cutaneous, and hip-joint afferents of the biological scheme — muscle afferents were not modeled — and it is worth noting that the Li model required support against falling: a system with only planar joints cannot balance side to side, so lateral stabilization falls to the vestibular layer and hardware design of Chapter 6.

The synthetic nervous network was built from non-spiking, leaky-integrator neurons and followed the two-layer CPG organization of Deng et al. [2019]. A rhythm-generator layer set the timing of the system; a pattern-formation layer reshaped the timing signals into phase-appropriate activation of antagonist muscle groups; and a sensorimotor layer containing Renshaw-cell-like recurrent inhibition drove the motor neurons. The sensorimotor layer received Ia (length and velocity) and Ib (tension) afferent feedback from the muscles, but the feedback gains were never systematically tuned, and no ground-contact signals were modeled. Every applied current, synaptic weight, and muscle-tension parameter was set by hand: each was adjusted by trial and error, one at a time, until the three layers oscillated in phase and the motor neurons produced stepping-like activation of the muscle pairs. That hand-tuning consumed the bulk of the project's effort, left no record of the sensitivity of the behavior to individual parameters, and did not generalize when new feedback pathways were added — limitations that directly motivated the systematic, verification-first tuning approach of Chapter 6.

3.11 Preliminary Neuromechanical Simulation Workflows

Preliminary simulation workflows were assembled to support future integration of the BPA mechanics and synthetic nervous system (SNS). The figures in this section document the biomechanical representations, controller organization, and software implementation. Their purpose is to identify what was constructed; quantitative validation remains separate.

3.11.1 AnimatLab Body and Controller Organization

The walker representation and controller hierarchy are shown in Figures 3.6 and 3.8. The hierarchy separates rhythm generation, hip control, and knee–ankle pattern formation within left- and right-side neural subsystems. The current AnimatLab working model was instrumented for a new 10 s phase-1 run recording joint, neural, contact-sensory, and body-motion channels. This run was used to inspect recorded signals and simulation behavior, without treating it as evidence of stable walking or effective ground-contact gating. It is distinct from the earlier walker study described in Section 3.10. Preliminary joint-motion results are reported in Section 4.4.

Since that run, the working model has been reorganized into a complete bilateral network. The right-side rhythm-generator, hip, and knee–ankle subsystems were mirrored from the left, inhibitory commissural pathways were added between the two rhythm generators, and each leg gained four biarticular muscle pathways (gastrocnemius, biceps femoris short head, semimembranosus, and rectus femoris) with human maximum isometric forces from the Gait2392 data set. The reorganized project opens in the AnimatLab

environment without errors and exports a standalone simulation that runs headless. Its rhythm-generator half-centers hold a static latch under the tonic drive levels tested so far rather than oscillating, so conductance tuning toward robust oscillation is the active step; no locomotor claim is made for the reorganized model beyond its verified assembly.

Figure 3.7 illustrates the rhythm-generator, pattern-formation, and sensorimotor organization at the neuron and synapse level. These reference schematics complement the subsystem view by making the antagonist pathways and sensory connections visible.

3.11.2 MuJoCo Conversion and Synthetic BPA Knee

A reproducible OpenSim-to-MuJoCo conversion workflow was assembled using scripted model repair, MyoConverter, and a documented software environment. Separately, a custom quasi-static BPA model was ported from MATLAB to Python and coupled to MuJoCo tendon kinematics to compute actuator forces and their joint contributions. A minimal SNS with reciprocal inhibition drove a synthetic knee with antagonistic flexor and extensor BPAs. Figure 3.9 distinguishes the converted robot from this separate knee demonstration. The demonstration establishes a controller–actuator–joint interface; it does not establish SNS control of the converted Gait2392 robot or validation of correction-factor-based MuJoCo torque predictions.

At the time of writing the conversion stage is complete and verified: the converted model carries all 92 muscle-tendon actuators with attachment geometry audited against the OpenSim source, and the joint sign conventions of the robot’s modified paths were audited dynamically (from simulation torques rather than static model tables, because

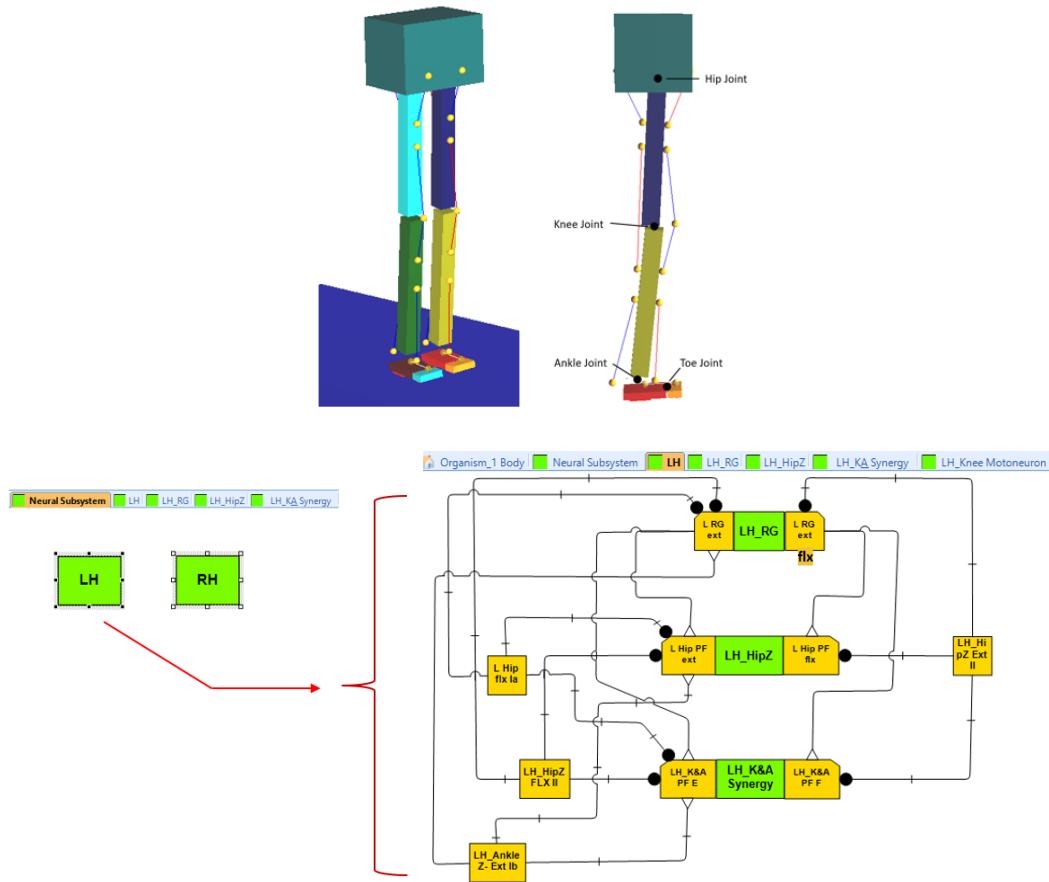


Figure 3.6: AnimatLab model and architecture illustrations. Top: the segmented walker and labeled hip, knee, ankle, and toe joints. Bottom: the left/right subsystem organization and an expanded view of the left-side rhythm-generator, hip, and knee–ankle modules. These architectural views describe model organization and are not snapshots of the phase-1 trajectory.

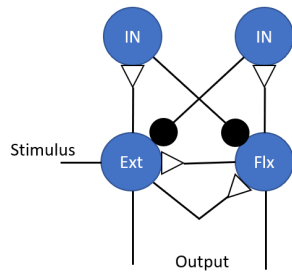
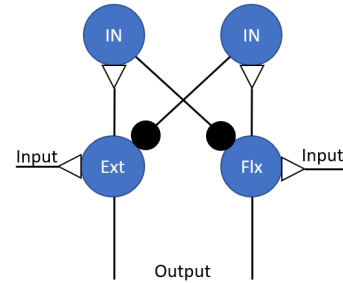
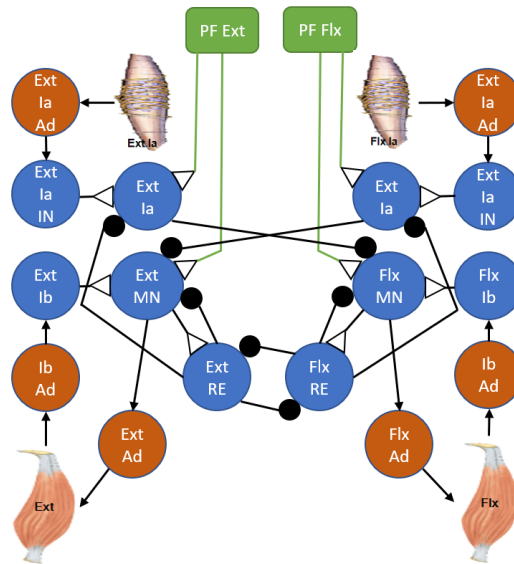
(A) Rhythm generator**(B) Pattern formation****(C) Sensorimotor organization**

Figure 3.7: Reference schematics of **(A)** rhythm generation, **(B)** pattern formation, and **(C)** sensorimotor organization. Ext and Flx identify antagonist extensor and flexor pathways; IN denotes interneurons, MN motor neurons, and Ia and Ib muscle afferent pathways. Open triangles and filled circles distinguish excitatory and inhibitory synaptic connections. These schematics illustrate controller architecture and are not readouts from the phase-1 run.

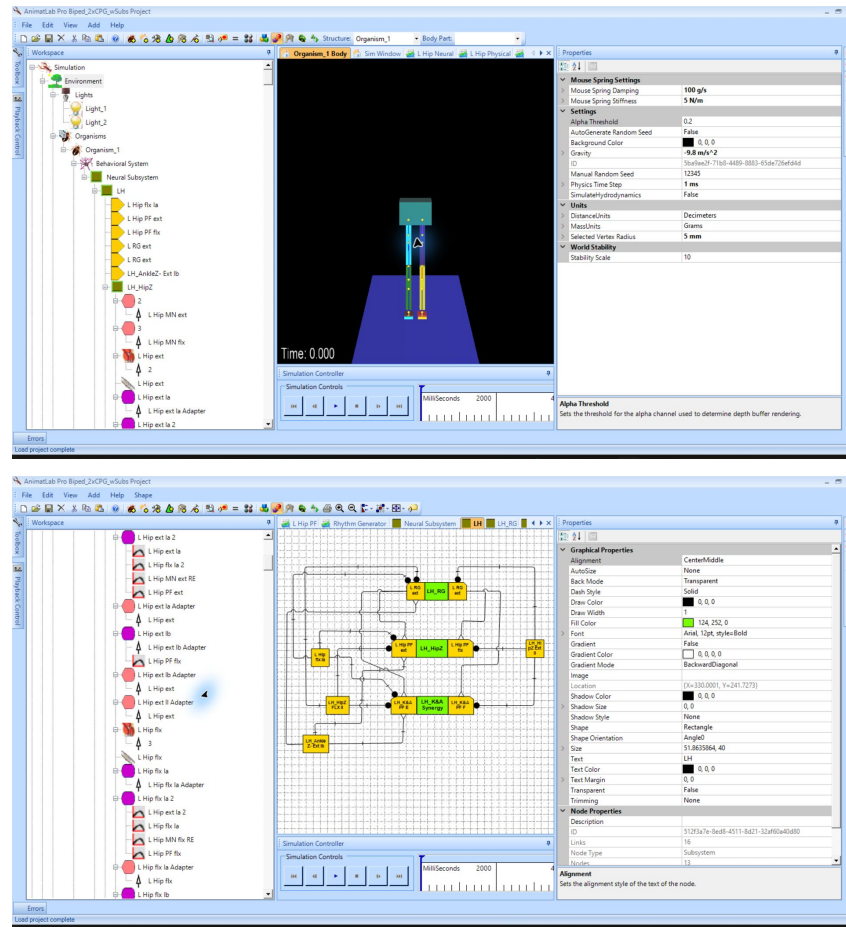


Figure 3.8: The saved AnimatLab working project opened in the application: the biomechanical body at initialization (top) and the expanded left-side controller subsystem (bottom). The workspace tree and property inspector expose the relationship between the body, sensory pathways, and neural modules. These interface captures were acquired without running a new simulation.

coupled path points move). The BPA port reproduces the MATLAB force calculation, including the correction-factor force balance. An SNS-Toolbox network couples to MuJoCo at a 0.1 ms neural step, with muscle force feedback entering as Ib afferent input. On this coupling, a 406-neuron two-layer spinal network has been assembled for the converted model — per-leg rhythm-generator half-centers, four phase-shifted pattern-formation groups per leg, motoneuron pools for all actuators, and Ia, Ib, and II afferent populations — and its rhythm generator has been verified in isolation to oscillate at an approximately 0.9 s cycle with the two legs in antiphase under tonic drive. Full-body closed-loop co-simulation is not yet running: leg-joint states develop non-finite values during longer simulations, with the foot-contact meshes the primary suspect under investigation.

3.11.3 Simulink SNS Library and Reflex Topology

A seven-block Simulink SNS library was also created, comprising a non-spiking neuron, a graded chemical synapse, a spiking leaky integrate-and-fire neuron, Ia and Ib afferents, activation dynamics, and a placeholder BPA force block (Figure 3.10). These components were connected in a reduced-order knee-reflex topology with antagonist actuation, stretch feedback, autogenic inhibition, and reciprocal inhibition (Figure 3.11). The library and the reflex demonstration run in base Simulink without additional toolboxes; the block icons derive their excitatory/inhibitory symbols automatically from the synapse sign, in a color-blind-safe palette, and the journal figures regenerate from scripts. The route for coupling the plant to CAD was settled on URDF export from SolidWorks followed by Simscape Multibody import, and the import path has been verified on a trial knee assembly; the

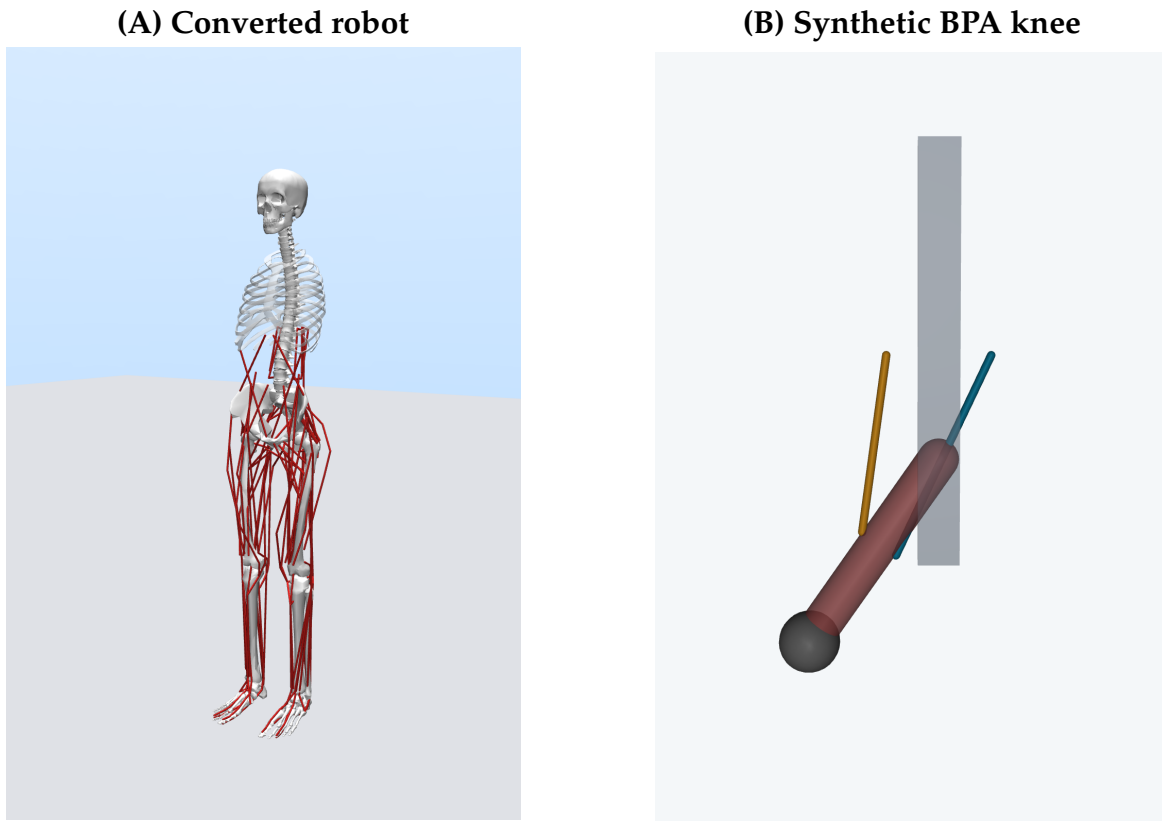


Figure 3.9: Native MuJoCo renders of **(A)** the converted Gait2392 robot at its saved reference pose, with muscle paths visible, and **(B)** the independent synthetic antagonist-BPA knee at an illustrative static angle of -35° . Cyan and gold identify its named flexor and extensor tendon routes; transparency exposes the routes through the view. The models are not shown to the same scale. These views document model construction, not SNS control of the converted robot or validated torque predictions.

tendon hardware is modeled in CAD but not yet inserted into the assembly. The plant retains provisional mechanical parameters, the BPA force block remains a placeholder, and the saved runs are not used to establish quantitative knee-regulation performance.

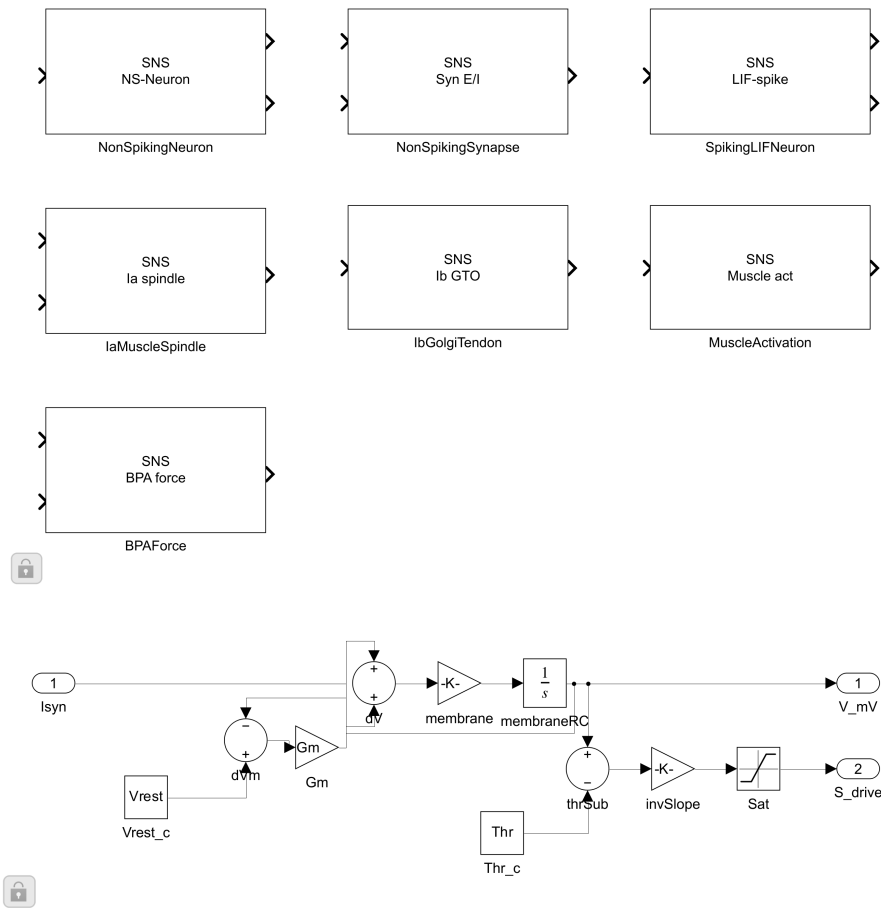


Figure 3.10: Native Simulink diagram exports showing the seven masked library blocks (top) and the internal implementation of the non-spiking neuron (bottom). The library blocks are arranged in a compact display grid for legibility. The BPA force block remains a placeholder. These implementation views do not establish quantitative knee regulation or a completed CAD-coupled Simscape Multibody plant.

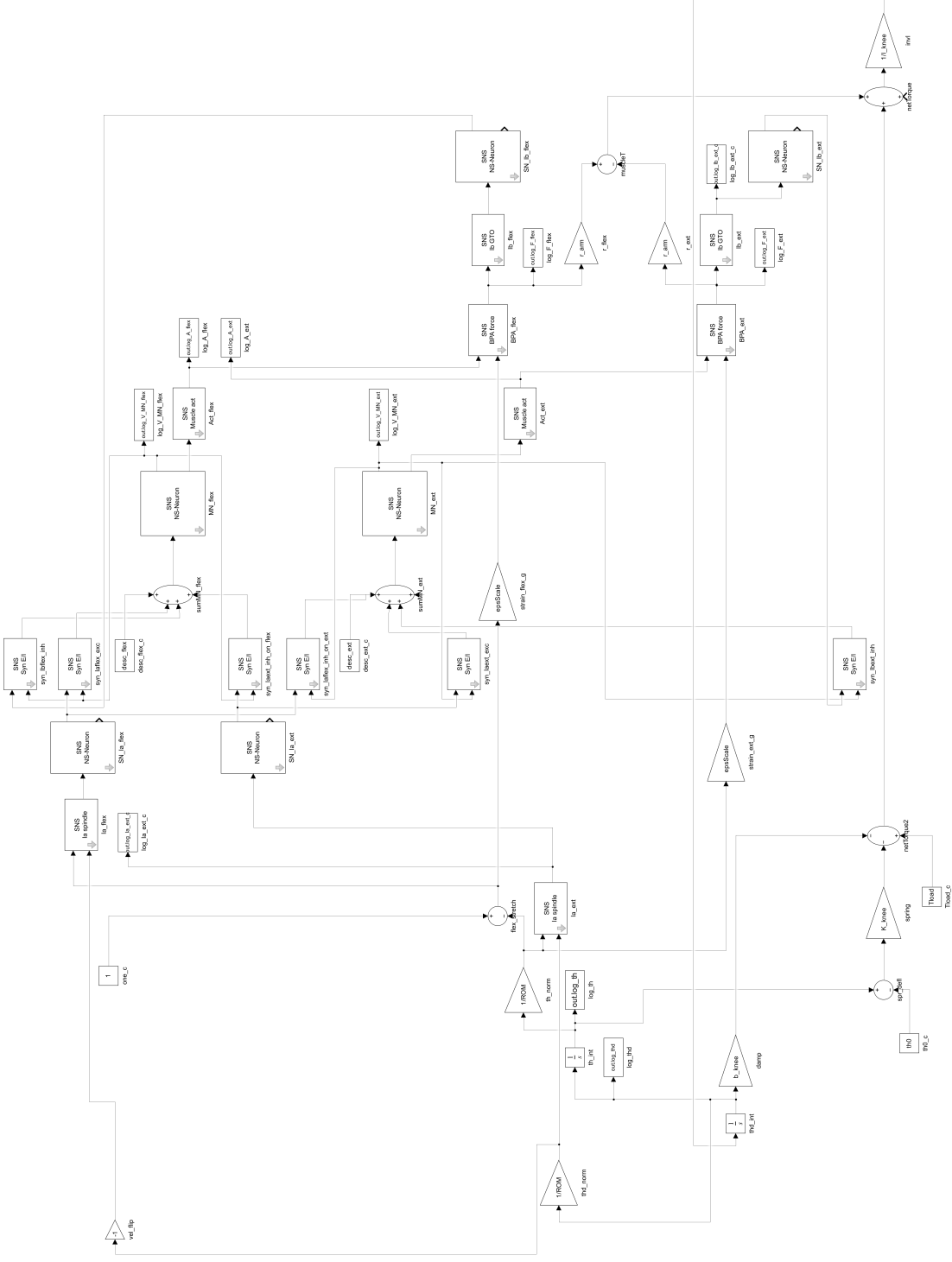


Figure 3.11: The complete reduced-order knee-reflex model exported from Simulink. The diagram connects Ia and Ib sensory pathways, excitatory and inhibitory synapses, antagonist motor neurons, activation and BPA force blocks, and the joint dynamics. Blocks and wires were arranged for display; source and destination ports were checked to confirm unchanged connectivity. Mechanical parameters and the BPA force block remain provisional.

4 Results

Publication note: Section 4.1 and the associated figures and tables appeared in Bolen et al. [2026] (Bolen, Elzein, Pang, and Hunt, *Actuators*, 2026). Section 4.2 and the associated figures and tables are also the subject of an unpublished manuscript in preparation by Bolen, Burgess, Morrow, Elzein, Pang, Scharzenberger, and Hunt [Bolen et al., 2025]. This manuscript has not been submitted for publication. Author contributions for both works are stated in Chapter 3.

4.1 BPA Force Characterization

This section reports the results of the isometric force characterization experiments described in Section 3.3. First, maximum isometric force at 620 kPa is modeled as a function of resting length. The remaining data are then normalized to build a predictive force surface, and the resulting model is compared with the manufacturer's tool and with other published BPA models.

4.1.1 Maximum Force at 620 kPa

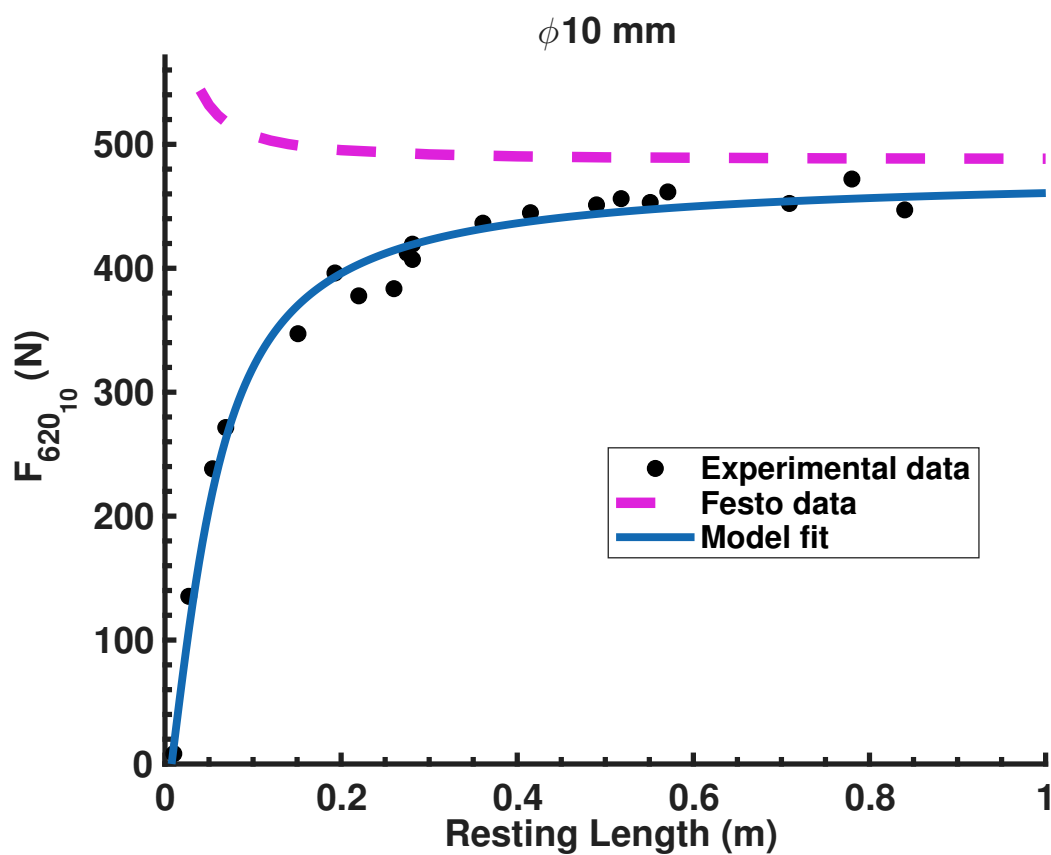
Data of the maximum force from BPA characterization tests of the 10 mm and 20 mm diameters at 620 kPa show a dependency on resting length (Figure 4.1). This is a previously unreported characteristic of these artificial muscles. Detailed analysis of the Festo tool [Festo AG & Co. KG, 2026] does in fact predict a change in maximum force with the

resting length; however, the Festo tool predicts increased force with shorter lengths, while our collected data indicate decreasing force with shorter lengths. The data show a force response resembling an arctan curve along the l_0 dimension. Using the Nonlinear Least Squares method and a Least Absolute Residual robustness, we fit an arctan curve to the data to get the maximum force at 620 kPa given a resting length as:

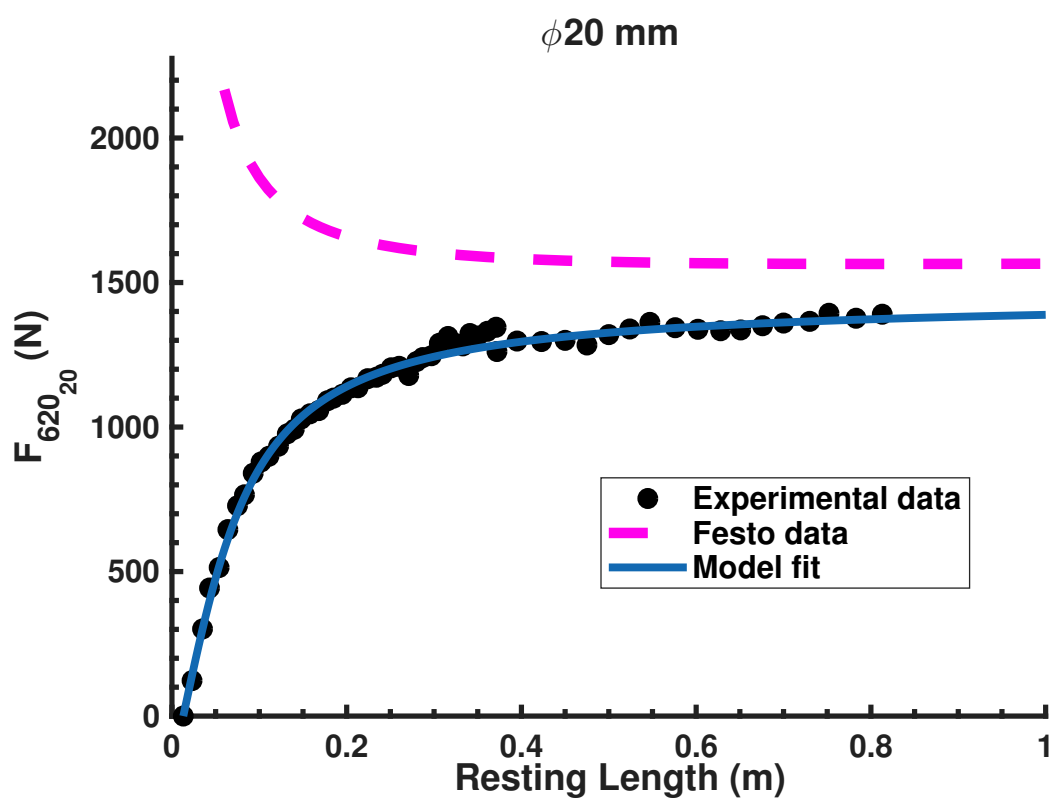
$$F_{620_{l_0}} = 303.5 \text{ N} \cdot \arctan (19.03 \text{ m}^{-1} \cdot (l_0 - 0.0075)) \quad (4.1)$$

$$F_{620_{20}} = 922.4 \text{ N} \cdot \arctan (15.37 \text{ m}^{-1} \cdot (l_0 - 0.013)) \quad (4.2)$$

The length is offset by 0.0075 m and 0.013 m because solid modeling showed that the end caps contact each other at these lengths. At these lengths, the actuator would not be able to contract to produce force.



(a)



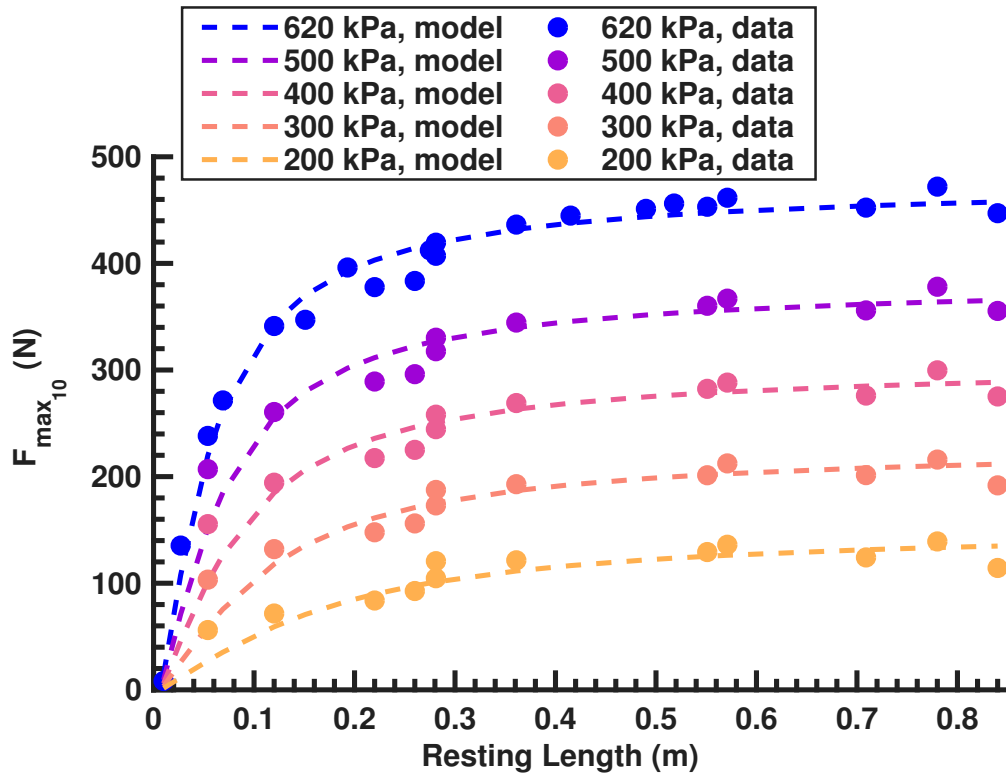
(b)

4.1.3 Maximum Contraction at 620 kPa

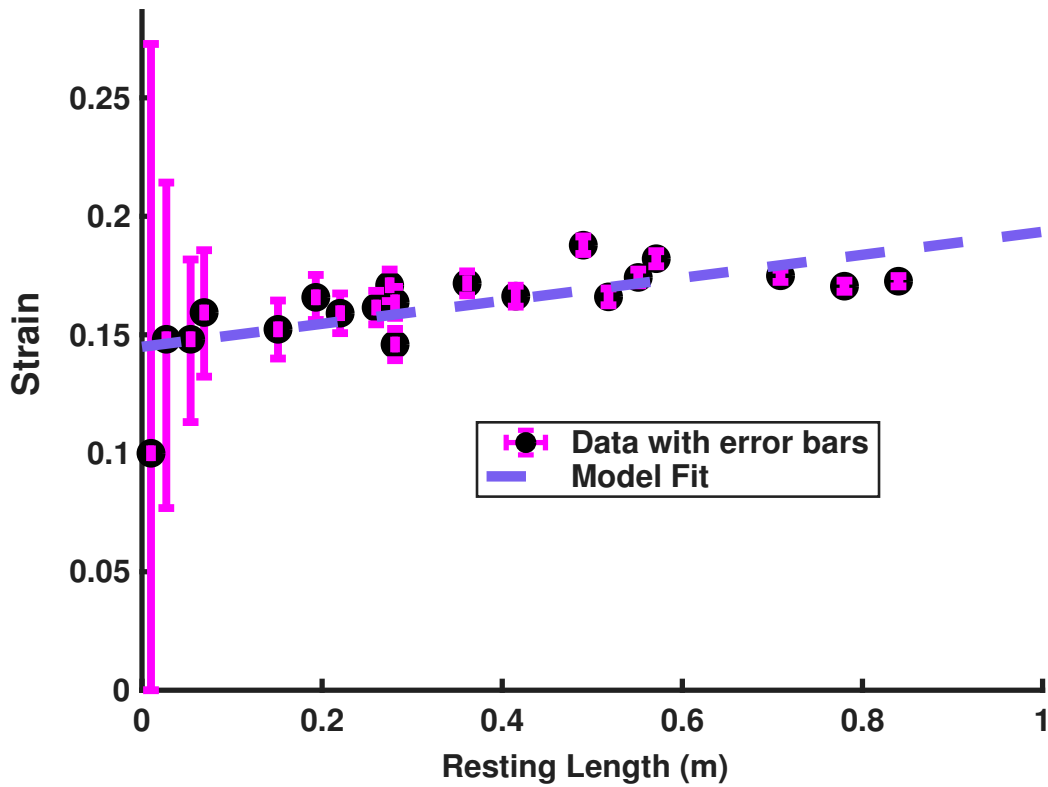
Figure 4.2(b) shows an attempt at a linear fit for maximum contraction at 620 kPa (ϵ_{620}) as a function of resting length (l_0). Error bars have been included to show the effect of ± 1 mm accuracy with measurements of l_0 and l_{620} . There was a large amount of variance in the data, with the linear fit giving an adjusted $R^2 = 0.4124$ and an RMSE = 0.0083. Since there is no direct, predictable relationship between maximum contraction and l_0 , this value should be recorded in each muscle used on a robot to best predict the force it may produce at different pressures and contraction.

4.1.4 Force as a Function of Pressure and Contraction

In addition to being a function of the pressure and resting length, the force produced by the actuator is also a function of the amount of actuator contraction, with less force being applied as the actuator contracts. To build this model, the collected data of force, pressure, and contraction are normalized by dividing by F_{620} , P_{620} , and ϵ_{620} , respectively. This had the effect of compressing the data into a 3D surface ranging from 0 to 1 on all axes. Normalized data are compared with pressure isoclines of force predicted by the Festo tool divided by the maximum force equation described in the previous section in Figure 4.3. With this comparison, it is clear that the Festo tool over-predicts the expected force, especially at lower pressures, contraction, and resting lengths.

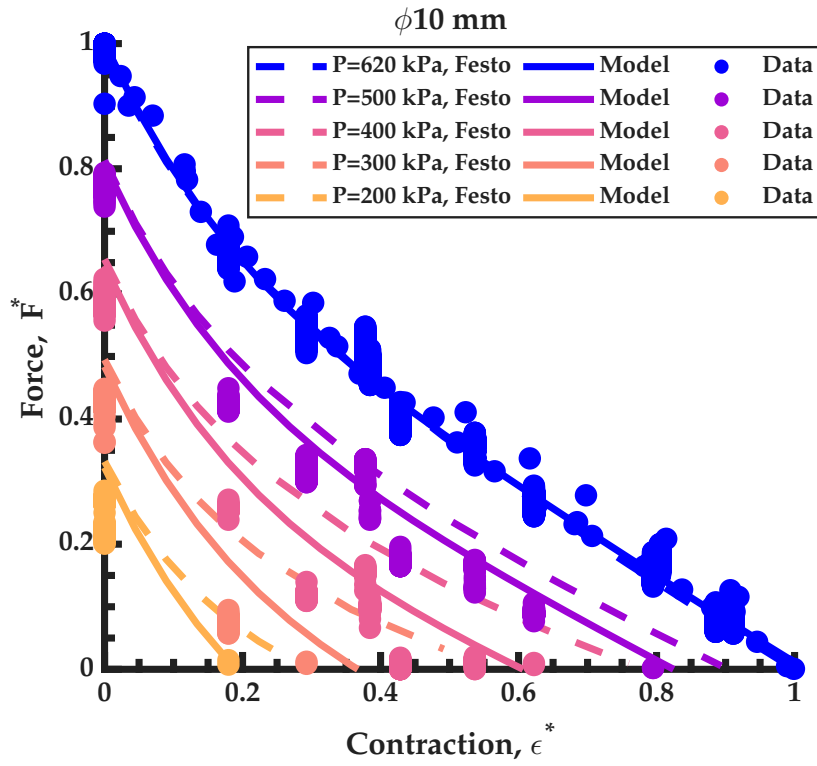


(a)

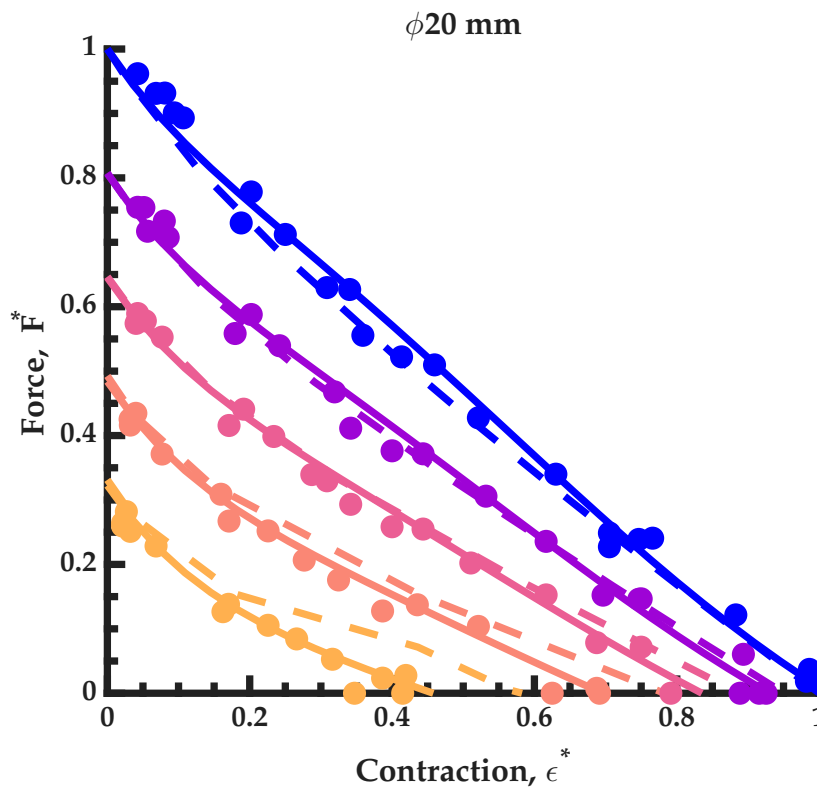


(b)

Figure 4.2: Results for finding the relationship between F_{620} , l_0 , P_{620} , and $\epsilon_{620,\infty}$. (a) Isoclines



(a)



(b)

Figure 4.3: Surface fit for $F^*(\epsilon^*, P^*)$. Solid lines are from our model. Dashed lines from

We therefore derived our own equation for isometric force in the BPA as a function of pressure and contraction. Visual analysis of the experimental data shows an exponential relationship between ϵ^* and F^* , and a linear relationship between P^* and F . We fit a surface to the data using Nonlinear Least Squares and a Least Absolute Residual robustness such that it is described by Equation (3.4). For the $\phi 10$ mm BPAs, the result of the improved fit can be seen in Figure 4.3a. The adjusted $R^2 = 0.9998$, a RMSE = 0.004537, and a maximum absolute residual of 10.6%. Solving (3.4) for the $\phi 20$ mm BPA, yields different coefficients, and results are seen in Figure 4.3b. Coefficient values and goodness-of-fit statistics are found in Table 4.2.

Table 4.2: Normalized isometric BPA force equation (Equation (3.4)) coefficient values and goodness-of-fit measures.

BPA	Coefficient	CI (95%)	Adj. R^2	Model RMSE	Max. Error	Adj. R^2	Validation RMSE	Max. Error
$\phi 10$ mm	$c_0 = 0.5682$	(0.5584, 0.578)	0.9998	0.005118	10.3%	0.9994	0.0245409	10.6%
	$c_1 = 4.254$	(4.126, 4.383)						
	$c_2 = 0.5597$	(0.5429, 0.5766)						
$\phi 20$ mm	$c_0 = 0.2579$	(0.2401, 0.2756)	0.992	0.02294	6.8%	0.9943	0.0231303	5.7%
	$c_1 = 6.477$	(5.558, 7.396)						
	$c_2 = 1.321$	(1.239, 1.403)						

4.2 Isometric Knee Torque

With the actuator force model validated at the actuator level, the same model was used to predict isometric torque about the knee joint for the test stands described in Sections 3.4 and 3.6. Measured torque is compared against three predictions: the original rigid-body torque model, a prediction using the improved BPA force characterization alone, and the improved torque model of Section 3.7 after optimization. The hybrid torque method

(Section 3.6) is used to isolate the sources of error.

4.2.1 Flexor on Pin Joint

Experimentally measured torque is compared with various prediction methods while using flexor BPAs on the pinned knee joint (Fig. 4.4). These results show that the original model and the improved BPA force model overpredict the measured torque at the knee for all joint angles, although the improved model performs slightly better (Fig. 4.4A and 4.4B). However, when the measured parameters (l_m and r_k) are used in the hybrid model, then predictions closely match the experimental data. This indicates that the kinematics of the torque model are incorrectly predicting l_m and/or r_k .

The optimization yielded $\chi_0 = 11.9 \text{ mm}$, $\chi_1 = 102 \text{ kN m}^{-1}$, and $\chi_2 = 10.6 \text{ kN m}^{-1}$ (Table 4.3). It can be seen that the model with these correction factors greatly improves the ability to predict the joint torque, even capturing the reduction in force at small joint angles that was completely missing from the previous model. The $l_0 = 45.7 \text{ cm}$ BPA was used for validation (Fig. 4.4D). It can be seen that the improved model does a good job of predicting the torque with a different muscle length configured to operate over a different joint range of motion. The optimized torque prediction for the $l_0 = 48.5 \text{ cm}$ BPA has an $\text{RMSE} = 1.616$, $\text{FVU} = 0.0958$, and a maximum absolute residual of 3.2527; validation on the $l_0 = 45.7 \text{ cm}$ BPA gives $\text{RMSE} = 2.1214$, $\text{FVU} = 0.0878$, and a maximum absolute residual of 5.1690 (Table 4.4).

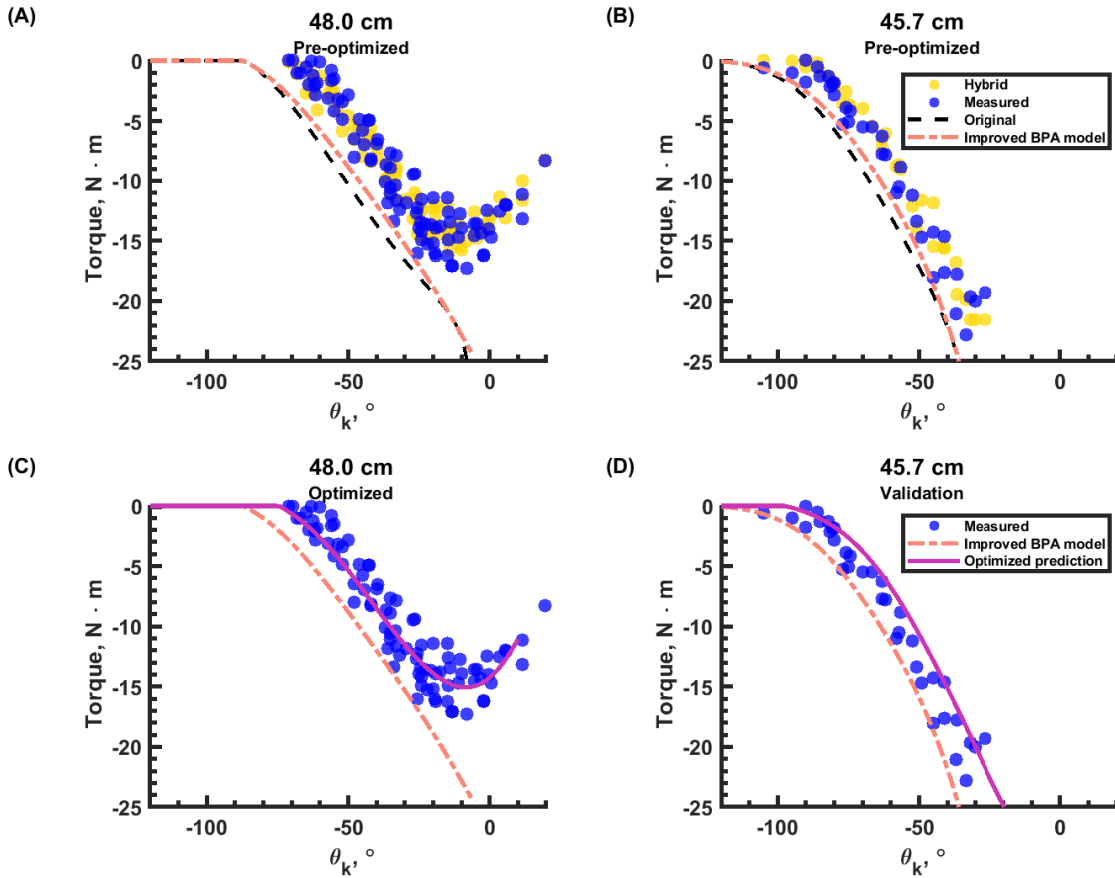


Figure 4.4: Results for the pinned knee joint using flexor BPAs. Dashed black line shows the torque predicted with our original model. Blue dots are the experimentally measured torque. Yellow dots represent the hybrid method of calculating torque. The magenta line represents the predicted torque. **(A)** Measured and pre-optimized torque prediction for the 48.5 cm BPA. **(B)** Measured and pre-optimized torque prediction for the $l_0 = 45.7$ cm BPA. **(C)** Theoretical and measured torque for the $l_0 = 48.5$ cm BPA using optimization results. **(D)** Predicted measured torque for the $l_0 = 45.7$ cm BPA using the optimization results

Table 4.3: Optimized coefficient values for flexor and extensor configurations. X_0 is a constant length offset (in meters), X_1 and X_2 are stiffness values (in N/m) in the bracket's X and Y directions, respectively, and X_3 is the unitless wrapping factor.

	X_0	X_1	X_2	X_3
Flexor	0.0119	1.02E+05	1.06E+04	–
Extensor	-0.0109	1.02E+05	1.06E+04	0.123

4.2.2 Extensor on Pin Joint

When the new stiffness and length offset model was used to predict torque for the knee extensor, measured torque was slightly higher than the original model calculates for the 48 cm resting length BPA, while it was slightly lower than expected in the 46 cm BPA, and it was much lower in the 42 cm. Only the K_1 and K_2 terms from the previous optimization results were used with the extensor configuration. Firstly, by inspection, the zero point for the measured torque was shifted to the right of the original model, indicating an offset that was at least opposite in sign from the χ_0 term in the previous section. Secondly, if we look at the BPA $l_0 = 46$ cm as an example, we see good correlation between the hybrid-method calculated torque and the originally predicted torque, but a divergence in the actual measured torque.

Results are shown in Fig. 4.5. The solution was found while holding the $l_0 = 41.5$ cm BPA for validation. The optimization resulted in $\chi_0 = -10.9$ mm and $\chi_3 = 0.123$. It should be noted here that the length offset χ_0 is almost equal in magnitude, but opposite sign, as the flexor configuration. The goodness of fit values for the torque prediction is found in table 4.4.

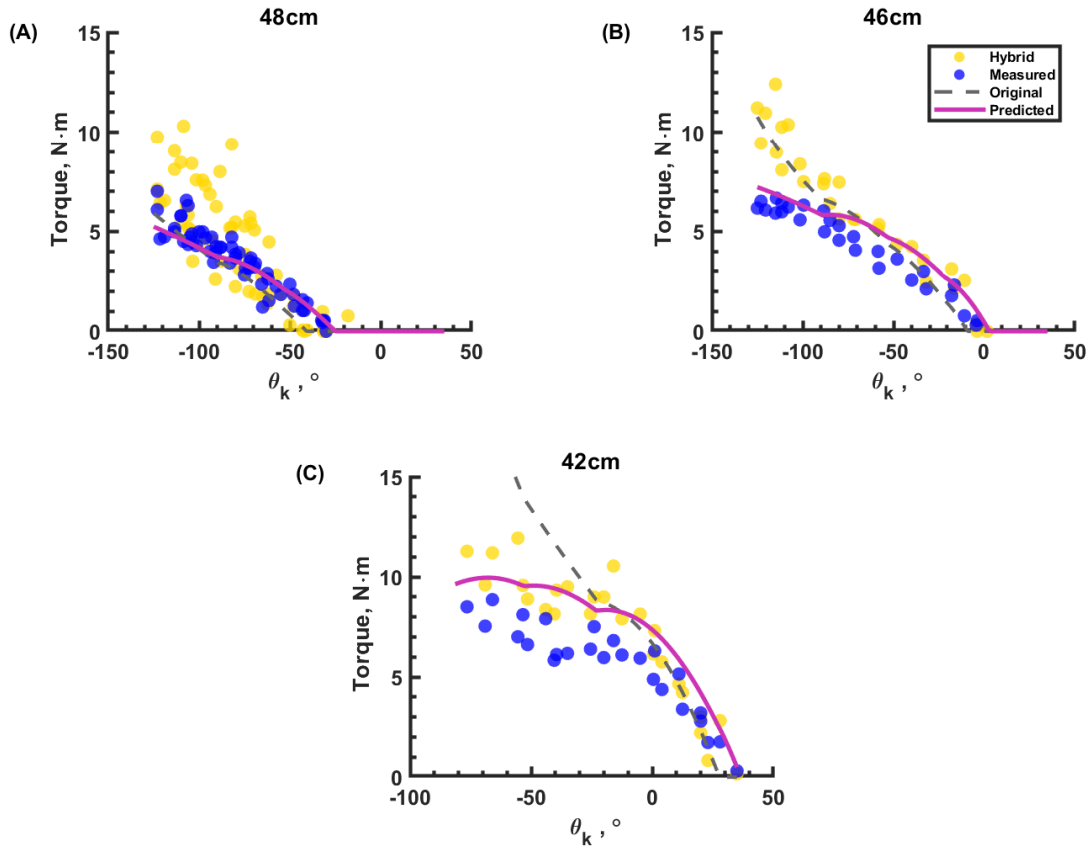


Figure 4.5: Pinned knee torque versus knee angle with the extensor BPAs. Blue dots are experimentally measured data, yellow dots are hybrid calculated torque, black dashed line is the original predicted values, and magenta line is the optimized predicted torque. For resting lengths, (A) 48.0 cm, (B) 45.7 cm, (C) and 41.5 cm.

4.2.3 Extensor on Biomimetic Knee

We obtained results using the biomimetic knee with a $\phi 10$ mm BPA in the extensor configuration. The bracket frame origin is the centroid of the cantilevered segment that holds the muscle origin bracket to the proximal femur (see Fig. 3.4C, right side).

The torque results for a $l_0 = 51.8$ cm BPA are shown in Fig. 4.6. The torque values measured are significantly smaller than both the original prediction and back-calculated torque using the hybrid method. The results from the previous extensor configuration lead to a better torque prediction. GoF values can be found in table 4.4. Fig. 4.7 shows the difficulty of modeling a BPA path that passes through the femoral condyles. Using the optimization results for extensor BPAs on the pinned knee joint, the $l_0 = 51.8$ cm BPA has an RMSE = 0.8920, FVU = 0.3396, and a maximum absolute residual of 1.4188.

4.2.4 Flexor on Biomimetic Knee

Fig. 4.8 shows the isometric torque values of the biomimetic knee from measurements, the Hunt model [Hunt et al., 2017a], and the predictions of the updated model. The values for χ_0 , χ_1 , and χ_2 derived from the flexors on the pinned knee were used. The bracket frame for the $\phi 10$ mm BPA is shown on the left hand side of Fig. 3.4C, and for the $\phi 20$ mm it is 3.4D. Fig. 4.8A represents a $\phi 10$ mm $l_0 = 41.5$ cm BPA with a 21 mm artificial tendon made out of $\phi 1.5$ mm Shimano bicycle brake cable. The brake cable was wrapped four times. The BPA was pressurized to 604 kPa. The predicted values have an RMSE = 0.3007, FVU = 0.0102, and a maximum absolute residual of 0.4647 $N \cdot m$. Fig. 4.8B represents a $\phi 20$ mm $l_0 = 42.0$ cm BPA with a 15 mm artificial tendon made out of Shimano bicycle

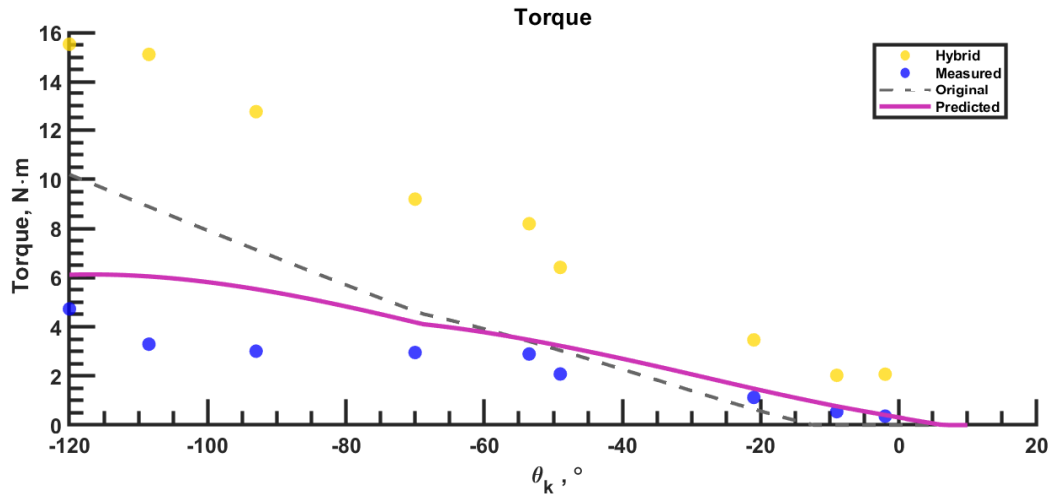


Figure 4.6: Measured versus expected isometric torque results using a $\phi 10$ mm, $l_0 = 51.8$ cm extensor BPA on the biomimetic knee. Measured results are blue dots. Original prediction is a dashed black line. Optimized prediction is a solid magenta line. Hybrid calculation shown in yellow dots.

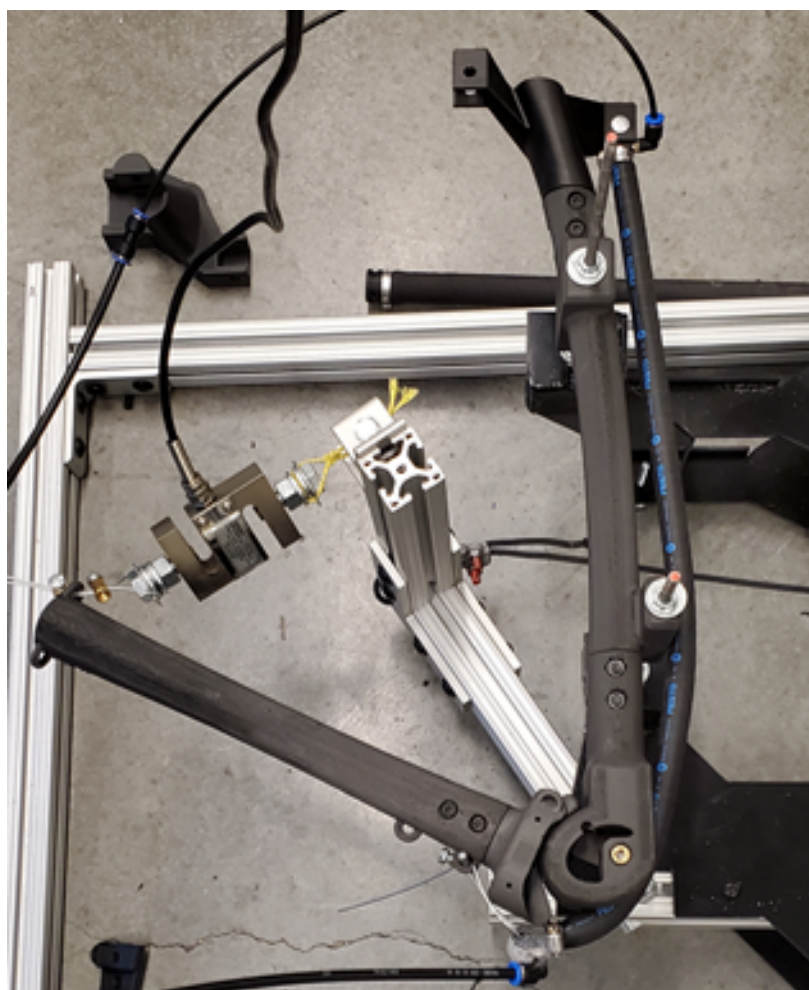


Figure 4.7: A configuration that is particularly hard to model. The BPA passes between the femoral condyles close to ICR.

brake cable (wrapped six times). Maximum human isometric torque values are shown as the black dotted line. The experiment was performed at 620 kPa and 325 kPa. At 620 kPa, the predicted values have an $RMSE = 1.8024$, $FVU = 0.0499$, and a maximum absolute residual of $3.5678 \text{ N} \cdot \text{m}$. The BPA torque does not exceed the magnitude of the human torque from ca. 60 – 100 degrees knee flexion. Allowing for a $\pm 20\%$ deviation from the human mean, the BPA does not produce human levels of torque from 78 – 103 degrees knee flexion.

Table 4.4: Goodness of fit for the improved torque model across test configurations. RMSE and maximum absolute residual are in $\text{N} \cdot \text{m}$.

Configuration	RMSE	FVU	Max. residual
Flexor, pin joint, $l_0 = 48.5 \text{ cm}$ (fit)	1.616	0.0958	3.2527
Flexor, pin joint, $l_0 = 45.7 \text{ cm}$ (validation)	2.1214	0.0878	5.1690
Extensor, biomimetic knee, $l_0 = 51.8 \text{ cm}$	0.8920	0.3396	1.4188
Flexor, biomimetic knee, $\phi 10 \text{ mm}$, $l_0 = 41.5 \text{ cm}$	0.3007	0.0102	0.4647
Flexor, biomimetic knee, $\phi 20 \text{ mm}$, $l_0 = 41.5 \text{ cm}$ (620 kPa)	1.8024	0.0499	3.5678

4.3 Balance Platform and Inverted-Pendulum Demonstrations

The testbeds of Section 3.9 were demonstrated by the author and presented in three conference posters [Bolen et al., 2024b,a,c]; this section summarizes what they showed.

The first-generation pendulum — hinge ankle, selectable encoder/IMU angle feedback, and reverse-pulley transmission — demonstrated self-balancing: under proportional control of the antagonist 20 mm BPA pair with IMU pseudo-vestibular feedback, the leg held itself upright on flat ground, and balance was maintained when the same controller operated on the oscillating balance platform beneath the foot. The second generation, IP

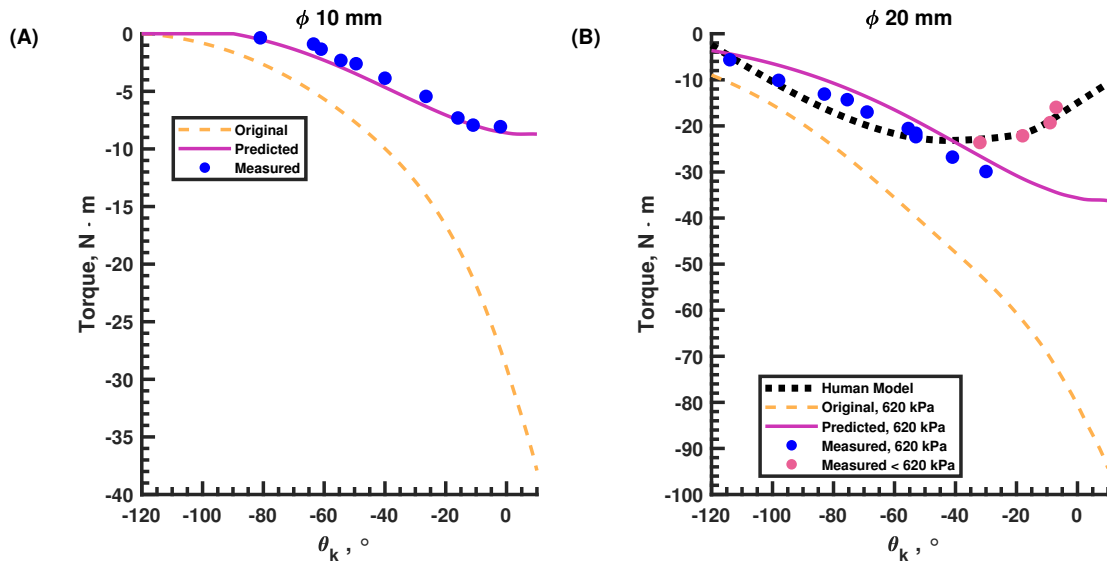


Figure 4.8: Biomimetic knee torque for **(A)** ϕ 10 mm $l_0 = 41.5$ cm flexor BPA with a 21 mm artificial tendon. Blue dots show the measured torque values. Magenta line is the predicted torque using the improved BPA characterization and optimization from the previous sections. The dashed light orange line shows the torque predicted using only the improved BPA characterization. **(B)** ϕ 20 mm $l_0 = 42.0$ cm flexor BPA with a 15 mm artificial tendon uses the same colors for 620 kPa results. Human values are given as a black dotted line. Due to the BPA exceeding the ultimate strength of the femur, torque measurements were taken at $P < P_{620}$ to show the desired RoM is achievable. Results for torque measurements at 560 kPa, 421 kPa, 325 kPa and 281 kPa (from left to right, respectively), are shown as pink dots.

2.0 — ball-and-socket ankle, the three-muscle plantarflexor/inverter/evertor arrangement, and IMU-only feedback — also maintained balance, on flat terrain. Both demonstrations used the same family of Festo BPAs characterized in Section 4.1.

These demonstrations establish that balance of a BPA-actuated leg under pseudo-vestibular inertial feedback is achievable in hardware, on both a fixed and an oscillating support. They are reported qualitatively, however: the poster materials quote no balance durations, platform oscillation amplitudes or frequencies, ankle-angle envelopes, or valve and pressure levels, so no quantitative values are claimed here. Quantifying the controllers — disturbance rejection, sensor drift, and sustained operation under continuous platform oscillation — remains open work, on the architecture that Chapter 6 plans to extend to the full humanoid.

4.4 Preliminary Neuromechanical Simulation Recordings

The AnimatLab phase-1 run described in Section 3.11 produced synchronized joint, neural, contact-sensory, and body-motion recordings over 10 s. Basic checks confirmed finite values and a uniform 0.2 ms sampling interval in both exported charts. Each chart contained 50,001 samples from 0 through 10 s; nine trailing zero-filled rows beyond the configured chart end were excluded. Figure 4.9 shows bilateral hip, knee, and ankle angles. Neural and contact-sensory recordings remain archived with the run; the contact-sensory channels record membrane voltage rather than contact force. These recordings demonstrate signal acquisition and alternating activity in this run, rather than validated locomotor performance. The model included an externally applied horizontal force during the first

second, and neither stable walking nor effective ground-contact gating was established.

4.5 Follow-Up Identification and Route Redesign

The identification work outlined at the close of the experimental chapters has since been completed, and its outcomes are summarized here; the coefficient values shown with the figures above (Table 4.3) belong to the initial single-bracket identification of those experiments.

The correction terms were re-identified over the full archived test sets with the two-stage procedure of Section 3.7: a two-bracket flexor fit cross-validated over the five pinned-flexor tests (the encoder-corrected 47 cm test held out), followed by an extensor refit with the stiffness pair locked to the flexor front. The adopted values are $\chi_0 = 8.9 \text{ mm}$, $\chi_1 = 5.62 \times 10^4 \text{ N m}^{-1}$, and $\chi_2 = 1.85 \times 10^4 \text{ N m}^{-1}$ for the flexor, and $\chi_0 = -10.1 \text{ mm}$ with $\chi_3 = 0.621$ for the extensor on the locked pair (4.354×10^4 , $1.701 \times 10^4 \text{ N/m}$); full provenance is given in Appendix C, Table C8. At these values, the corrected flexor model predicts the five pinned-flexor tests with per-test RMSE of 1.94/1.51/2.29/1.63/1.21 N m (48, 46, 47, 40-tendon, 41 cm; FVU at most 0.17), the 47 cm test being pure held-out validation. On the six-test extensor validation pool, per-test RMSE spans 0.56–1.88 N m against 0.95–3.62 N m for the uncorrected rigid model — an improvement on every pool test. On the biomimetic extensor test ($l_0 = 51.8 \text{ cm}$) the corrected prediction improves substantially on the baseline (RMSE 2.20 vs. 3.03 N m) but leaves FVU above unity, so the shared-pair model still underpredicts that configuration’s measured torque variation; this remains the open gap in the cross-configuration validation. The solver question raised during the campaign was settled

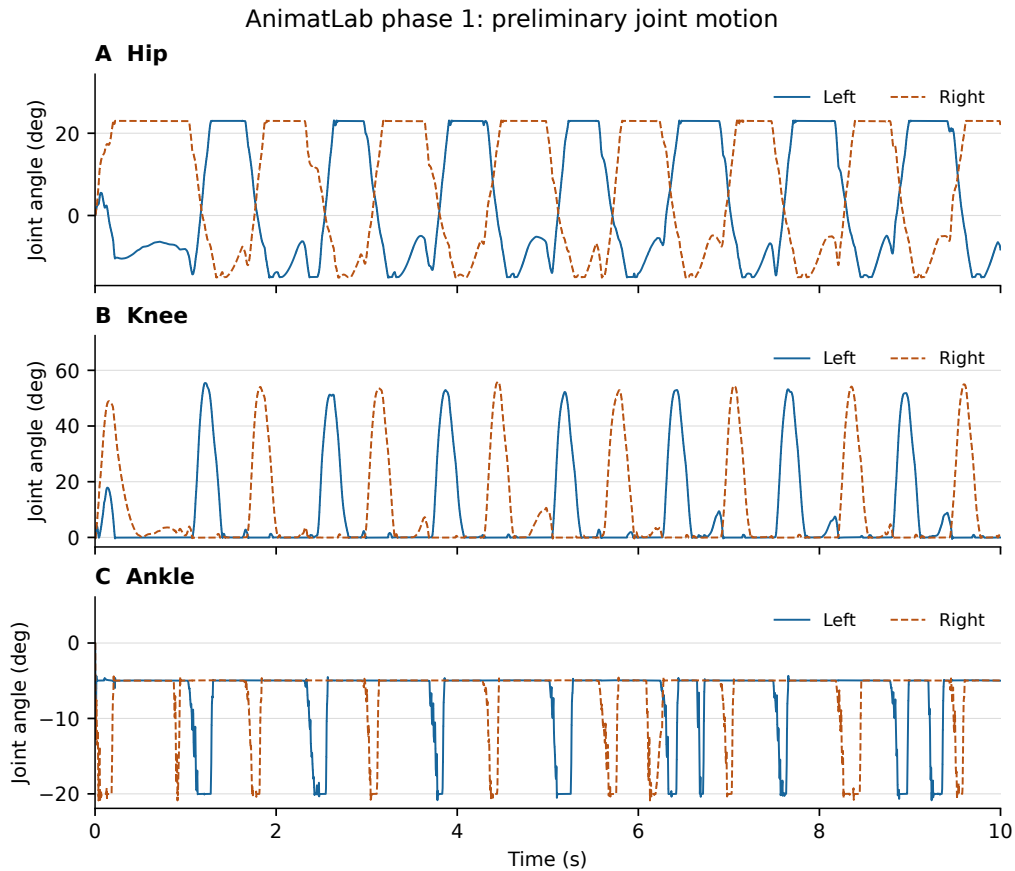


Figure 4.9: Preliminary AnimatLab phase-1 joint motion: **(A)** hip, **(B)** knee, and **(C)** ankle angles in the model's recorded joint conventions. Blue solid and orange dashed traces denote left and right sides. All 50,001 samples within the configured 0–10 s chart window are retained without filtering. The model includes an external horizontal force during the first second. These traces document joint motion in an instrumented run; they do not establish stable walking or effective ground-contact gating.

empirically: the genetic multiobjective solver was retained for the identification, where the Pareto front is required for holdout validation, and the surrogate-plus-pattern-search pipeline was reserved for the route redesign where it is already the method of record (Section 3.8).

The route redesign built on these values is likewise far advanced. The two-BPA flexor arrangement — motivated by the single-muscle shortfall at high flexion angles documented above — reached a feasible solution meeting the flexor torque target under the modeling assumption that both actuators share one attachment route rather than a physically mirrored pair; making the mirrored pair explicit is the next design refinement, and the flexor re-run under the newly adopted correction terms was still in progress at the time of writing. The extensor route was redesigned directly under the adopted correction terms: it meets or exceeds its human-scaled target at every angle in the range of motion, with the thinnest torque margin below 0.1% at a single angle, while respecting the wrap-release schedule and contraction limits of the route-elimination rule (Appendix C). These redesign solutions are the plant definitions carried into the neuromechanical simulation work of Sections 3.11–4.4 and Chapter 6.

5 Discussion

Publication note: Sections 5.1–5.5 and the associated figures and tables appeared in Bolen et al. [2026] (Bolen, Elzein, Pang, and Hunt, *Actuators*, 2026). Section 5.6 is also the subject of an unpublished manuscript in preparation by Bolen, Burgess, Morrow, Elzein, Pang, Scharzenberger, and Hunt [Bolen et al., 2025]. This manuscript has not been submitted for publication. Author contributions for both works are stated in Chapter 3.

5.1 Effects of Resting Length

We have developed new F_{620} equations for 10 mm and 20 mm diameter Festo BPAs that capture the change in maximum force produced by the actuators at 620 kPa as a function of their resting length. When examining Equation (4.1) and measured data in Figure 4.1b, it can be seen that as l_0 goes to infinity, $F_{620_{10}}$ goes to 470 N. When the Festo tool [Lang, Robert, 2005] was queried at $l_0 = 1$ m as an approximation of infinity, it predicts the $\phi 10$ mm BPA can produce $F = 490$ N at P_{620} . This is within the 10% variability that Festo states may occur in manufacturing tolerances. Similarly, the Festo data predict a $\phi 20$ mm BPA will produce $F = 1570$ N. Our results for Equation (4.2) show that $F_{620_{20}}$ goes to 1460 N as l_0 approaches infinity. Although the difference is much larger, it is also with the 10% manufacturing tolerance.

Where our measured data differs significantly from the Festo model, is that as l_0 goes to zero. For the measured data, as the length gets smaller, the force does as well, whereas the Festo tool predicts exponential force increase (see Figure 4.1). As a specific example,

when specifying $l_0 = 112$ mm for the 10 mm diameter BPAs, the Festo predicted F_{620} is 498.6 N. However, using Equation (4.1), F_{620} is calculated to be 335.3 N. Actual force in the $l_0 = 112$ mm was measured at 350.9 N. Therefore, the error in predicted F_{620} for the $\phi 10$ mm BPA is 4.5% for our model and 42.1% using the Festo tool. Researchers using BPA resting lengths under 300 mm should take note of these results.

Festo AG & Co. KG [2026] (p. 12) states that force will be reduced by approximately 10% for a minimum nominal length. In Festo AG & Co. KG [2018] (p. 14) they say the force can deviate by 10% based on factors such as: production variation, material variation, and deviation from nominal length (l_0). Festo AG & Co. KG [2013] (p. 13) notes that testing was done with BPAs that are ten times the uninflated diameter, and that theoretical force can increase by up to 10%. In each datasheet, the characteristic curves appear to be the same. As demonstrated above, the 100 mm $\phi 10$ mm BPA should have much less force than what is shown in the curve. This is our experience and the reason for the testing. It is unclear why the MuscleSIM tool [Lang, Robert, 2005] then predicts higher force as resting length decreases. It could be a reversed sign coefficient in their internal curve-fit model. We have shown here that as the nominal length goes to zero, so does force.

The reason for BPA models predicting higher force than what is measured experimentally is a recurrent theme across many models: they do not account for the end effects of the pressure vessel where end effects dominate. Nominally, the clamped ends fix the inner and outer diameter while allowing axial and radial displacement. In the isometric case, there is only radial displacement. The boundary conditions force steep gradients in stress and strain near the ends. Near the boundary conditions are where axial, radial, and circumferential strains deviate from analytic solutions to a cylindrical model. For a

constant pressure, the end effect region appears to have the same length. Therefore, in shorter BPAs, the end effects dominate. The axial length scale of the boundary section is comparable to l_0 , and the active force differs substantially from that of an assumed perfect cylinder.

Manufacturing variations seem to play an important factor in the behavior of these actuators. It remains to be seen how ϵ_{620} can be known a priori. We still suspect that it is a function of l_0 , but it might also be a function of other factors such as product batch number or UV light exposure [Festo AG & Co. KG, 2026] (p. 15); [Festo AG & Co. KG, 2018] (p. 6); [Festo AG & Co. KG, 2013] (p. 13). We found there to be larger variations in the 10 mm BPAs than in the 20 mm BPAs. Modeling work of ideal McKibben actuators describe how the relationship between contraction and initial length is a function of the total number of twists and the angle of twist in the usable muscle area [Chou and Hannaford, 1996]. It is unclear if there are variations in the angle and number of twists in the Festo BPAs due to manufacturing. Uncovering the relationship between ϵ_{620} and l_0 for the Festo DMSP artificial muscles, if it exists in a meaningful way, will require additional controlled tests.

Similarly, it is not clear how the maximum force might change with manufacturing variations. This was reasonably corrected for by normalizing the equations to each individual's maximum contraction, but more nuanced details might emerge with more data collection. Not knowing these relationships a priori means there is still a need for robots and other systems to go through an iterative design stage if it is discovered that the system does not produce the torque or have the RoM that the design team expects. However, the data and models provided here should enable for faster redesigns with fewer additional measurements.

5.2 Nondimensionalized Force Prediction

We have also introduced the concept of nondimensionalized isometric force that is a function of relative strain and relative pressure (Equation (3.4)). This elegant equation has only three coefficients with low error (see Table 4.2). In the work presented here, we introduce the relative pressure term, $P^* = P/P_{620}$. Our data show that by normalizing force, contraction, and pressure, we are able to create a simplified force equation as a function of contraction and pressure that scales well with initial actuator length, as expressed by Equation (3.6) in Section 3.2.

For example, given $l_0 = 54$ mm and $P = 300$ kPa, the measured force was $F_{actual} = 103$ N. Force prediction using only the Festo tool predicts the force to be $F_{predict} = 270$ N. This is 161% greater than the measured force. Using Equations (4.1) and (3.4) with Equation (3.6) gives $F_{predict} = F^*(0, 0.48) \cdot F_{620}(0.054) = 0.49 \cdot 230 \text{ N} = 112 \text{ N}$. This is an error of 8%, which is much more accurate than using the Festo tool only.

The presented work is capable of predicting force output for 10 mm and 20 mm muscles with different initial muscle lengths, internal pressure, and contraction amount with minimal initial data collection. However, it remains to be seen if these relationships are maintained over the full lifecycle of the muscles. There are not yet any peer reviewed studies investigating the long-term behavior of these artificial muscles, though the Festo datasheet provides some indication of what can be expected [Festo AG & Co. KG, 2026] (p.19). They report that the 40 mm BPA can complete 1 million cycles at 6000 N (maximum rated load), or 10 million cycles at 4000 N. The service life of the other size BPAs is not reported; however, it is mentioned that thermal stressing may reduce the lifespan and they

should be inspected every 500,000 cycles for signs of aging [Festo AG & Co. KG, 2018] (pp. 10–11). Future research should look at the behavior of these muscles over time.

In our work, $P_{620} = 620$ kPa is the maximum supply pressure for our system, and other users of this actuator may use a different supply pressure. However, this supply pressure is not required, and normalizing by this value was chosen as a method for improving the optimization techniques by reducing the sensitivity of the term associated with pressure. This equation will work for any pressure supplies below 620 kPa. We also anticipate the equation will also work for pressures up to 800 kPa (the maximum rated pressure for Festo $\phi 10$ mm BPAs); however, we do not have the data to confirm this.

5.3 Comparison with Other Published BPA Models

The proposed model was compared with the formulations by Sárosi [2012], Sárosi and Fabulya [2012], and Martens and Boblan [2017], to evaluate their predictive performance across actuator sizes and operating conditions (see Table 5.1 and Figure 5.1). For the $\phi 20$ mm BPA, parameters from Sárosi [2012] were used, corresponding to actuators with resting lengths of 200 mm and 400 mm. These were compared against the experimental data at $l_0 = 300$ mm and $l_0 = 450$ mm, respectively. For the $\phi 10$ mm BPA with resting lengths of 257 mm and 233 mm, parameters from Sárosi and Fabulya [2012] were applied. In each case, the maximum contraction parameter (k_{max}) was determined numerically by solving $F(6.2 \text{ bar}, k_{max}) = 0$, ensuring consistency with the experimental maximum pressure condition.

Table 5.1: Goodness-of-fit comparison for the Bolen, Sárosi, and Martens and Boblan models at different resting lengths (in mm) for 10 mm and 20 mm BPAs. RMSE and maximum error are reported in newtons, and FVU denotes the fraction of variance unexplained.

Diameter	Length	Bolen			Sárosi			Martens and Boblan		
		RMSE	FVU	Max. Error	RMSE	FVU	Max Error	RMSE	FVU	Max. Error
10 mm	257	22.9	0.0353	46.0	143.8	1.3903	235.8	73.8	0.3668	122.9
	233	16.8	0.0187	30.3	132.7	1.1655	222.6	62.6	0.2590	101.2
20 mm	300	35.6	0.0144	114.0	132.1	0.1992	313.4	72.8	0.0605	137.8
	450	67.6	0.0258	166.6	177.7	0.1785	294.8	862.1	4.2022	1321.1

While the Sárosi [2012] and Sárosi and Fabulya [2012] models capture general trends in force–contraction behavior, discrepancies were observed when applying parameters derived from different actuator lengths. In particular, the model tends to over-predict force at higher pressures and low amounts of contraction, suggesting sensitivity to geometric scaling effects not explicitly accounted for in the formulation.

The Martens and Boblan [2017] model is derived from a geometric and material-based framework and provides a more physically grounded representation of actuator behavior. However, its accuracy depends strongly on precise geometric parameters, including braid angle, wall thickness, and fiber length. Small deviations in these parameters can lead to significant differences in predicted force, particularly at higher pressures.

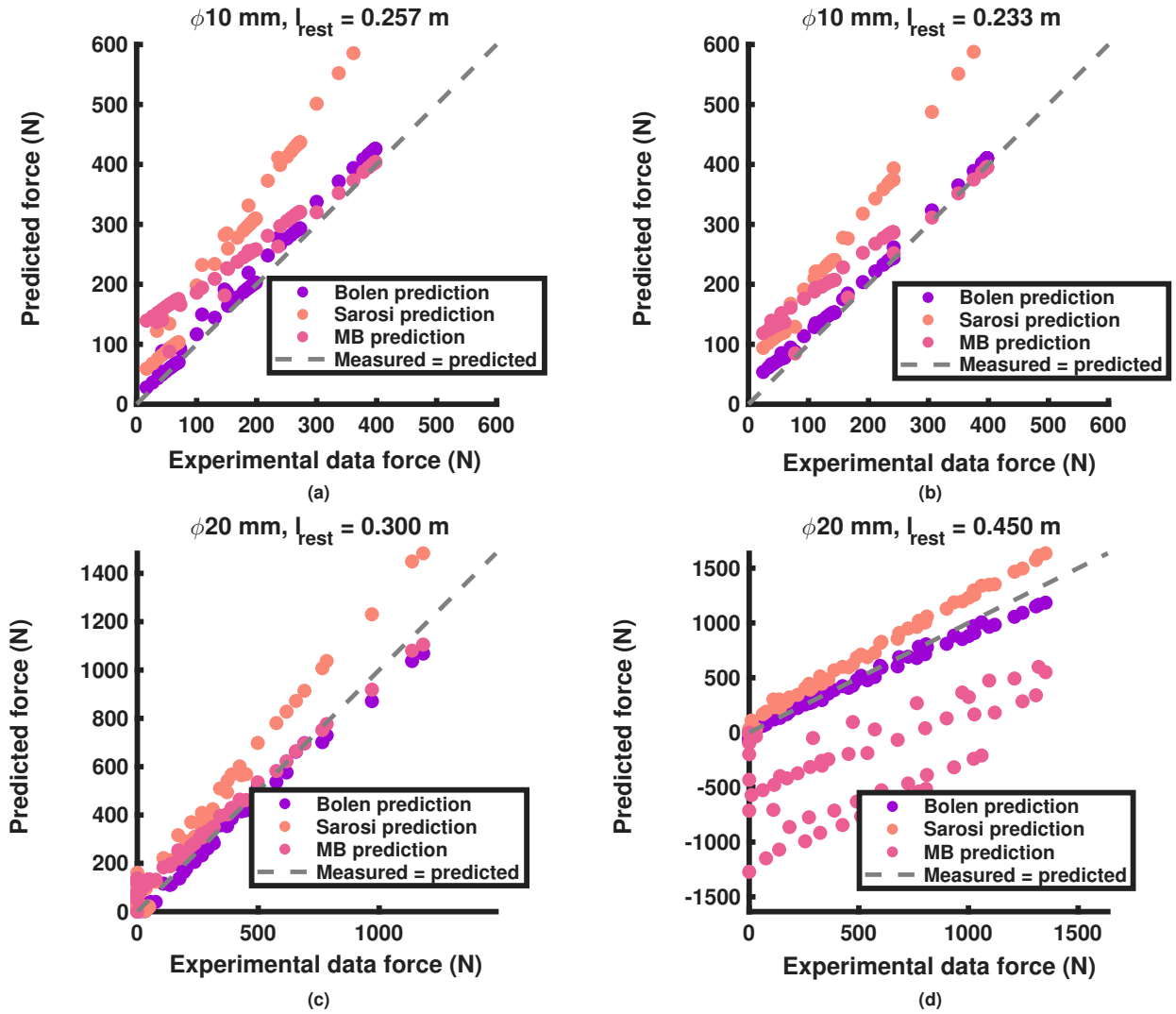


Figure 5.1: Comparison between measured and predicted BPA force at varying amounts of contraction and pressure. A $y = x$ perfect fit line is added for reference. The BPAs compared are: (a) $\phi 10 \text{ mm}$, $l_0 = 0.257 \text{ mm}$, (b) $\phi 10 \text{ mm}$, $l_0 = 0.233 \text{ mm}$, (c) $\phi 20 \text{ mm}$, $l_0 = 0.300 \text{ mm}$, and (d) $\phi 20 \text{ mm}$, $l_0 = 0.450 \text{ mm}$.

The Martens and Boblan model was first evaluated at the same pressure–contraction points as in their experimental data [Martens and Boblan, 2017] (Tables A1 and A2),

indicating that it reproduces the general published force map shape but is sensitive to the assumed initial geometry. Examining their $\phi 20$ mm BPA, for example, using the explicitly stated $l_0 = 300$ mm and initial diameter $D_0 = 20$ mm significantly under-predicted the results compared to using the implied $l_0 = 296$ mm ([Martens and Boblan, 2017] (Table A1)) and $D_0 + H_0 = 21.8$ mm, where $H_0 = 1.8$ mm is the material thickness. The diameter adjustment can be implied from [Chou and Hannaford, 1996]. This adjusted diameter was also included for the $\phi 10$ mm BPA to give a better fit. As can be seen in Figure 5.1 and Table 5.1, the highest accuracy is observed when $\phi 20$ mm and $l_0 = 300$ mm, which is at or very near their experimental measurements. For the $\phi 10$ mm BPAs the accuracy decreases even though their experimentally measured $l_0 = 250$ mm is close to our $l_0 = 257$ mm and $l_0 = 233$ mm. The largest error occurs at $\phi 20$ mm and $l_0 = 450$ mm and the model starts to predict negative force. We have calculated that the $F_{PE} \cdot \frac{dD}{DL}$ term becomes very large compared to the others [Martens and Boblan, 2017] (p. 5). This suggests that the model is sensitive to geometric parameter selection and interpretation, particularly with respect to initial length and diameter definitions.

In comparison, the present normalized model demonstrates improved consistency across actuator lengths by incorporating experimentally derived scaling through F_{620} and ϵ_{620} . This normalization reduces sensitivity to geometric variation and enables a unified representation of actuator behavior, while maintaining predictive accuracy over the tested range of pressures and contractions.

5.4 Implications for Real-Time Control and Simulation

Although pneumatic muscle actuators exhibit hysteresis, thermodynamic effects, and other dynamic behavior, accurate and high-performance controller designs can still be achieved using a static force map approach when the underlying static force model is accurate. Martens and Boblan explicitly note that BPA dynamics are often dominated by the air dynamics inside the actuator and the mass flow behavior of the valve, such that precise modeling of the static force characteristic remains crucial for torque control accuracy [Martens and Boblan, 2017]. This view is consistent with more recent BPA control work, including a sensor-less torque control interface for BPA-driven joints [Martens et al., 2018], stable force/position/stiffness control of antagonistic pneumatic-muscle joints [Ugurlu et al., 2019, Hunt et al., 2017b], and sensor-less angle stiffness control of antagonistic BPA systems [Shin and Kogiso, 2022].

In this context, the present model is well suited for real-time use because it reduces the static isometric force prediction to a low-order normalized surface, $F^*(\epsilon^*, P^*)$, scaled by the resting-length-dependent term $F_{620}(l_0)$. This structure is computationally inexpensive to evaluate online and is also suitable for lookup-table implementation. While the present work does not model the full actuator–valve dynamics, it provides the static force relationship needed as a feedforward or inner-model component in real-time torque control, simulation, and preliminary design workflows. It is anticipated that improved real-time control for high-speed and high-precision systems will benefit more from improved modeling of airflow dynamics than dynamics involving actuator hysteresis, damping, and other dynamic behavior. Characterizing the actuators under dynamic operating conditions

— isokinetic (constant-velocity), isobaric, and isotonic (constant-load) contractions — is a natural next step beyond the isometric characterization reported here, for which the pressure–force dynamics model of Elzein et al. [2025] provides the modeling foundation.

5.5 Biomimetic Artificial Muscle Analogy

BPA artificial muscles are often said to be analogous to biological muscles because they have force–length curves and can produce force only in tension. This analogy is worth closer inspection. For example, the describes the passive force–length curve has been described as an exponential function arising from connective tissue elements, while the active force–length relationship is typically modeled as a bell-shaped curve reflecting actin–myosin overlap [Winters, 1990]. These formulations emphasize that muscle force generation is governed by nonlinear geometric and material interactions rather than simple linear elasticity. McKibben-type pneumatic artificial muscles (of which Festo is an example) exhibit analogous nonlinear behavior, though arising from fundamentally different physical mechanisms. As shown by Chou and Hannaford, the force–length relationship of McKibben actuators is derived from geometric constraints imposed by the braided shell, leading to a nonlinear dependence of force on actuator length through the braid angle Chou and Hannaford [1996]. In their formulation, actuator force is proportional to pressure and a nonlinear geometric term $(3 \cos^2 \theta - 1)$, which governs both the maximum contraction limit and the curvature of the force–length relationship. The work presented here demonstrates that this nonlinear curve can be well approximated by an exponential as well.

However, there are some distinct differences between how these artificial muscles behave when compared to biological muscles. For example, the biological muscle optimum fiber length l_{OFL} is not equivalent to l_0 in the muscles. The maximum active force that the artificial muscles can produce is at the resting length, also the maximum length, of the artificial muscles. In contrast, the maximum active force a biological muscle length can produce is somewhere in the middle of the total contractile length of the muscle.

Furthermore, biological muscles can adapt the optimal operating length through structural remodeling (e.g., sarcomere addition or tendon compliance) while BPAs have a fixed geometric configuration. This allows biological systems to self-optimize for the particular tasks the animal is subjected to. In order to re-optimize a robot for new tasks, new actuators would have to be swapped into the system. This distinction highlights a key limitation of BPAs in biomimetic applications, while also reinforcing the importance of accurate nonlinear modeling for control and design.

5.6 Sources of Error in Isometric Torque Generation

Our improved BPA characterization predicts torque more accurately than the original BPA model (see Fig. 4.4A and 4.4B). This only accounts for part of the error. There are certainly many factors that affect the torque results. Compliance in the testing system, position location tolerance, flexibility of brackets, imperfect path modeling, and variation in individual BPAs are other factors.

Simplifying the model and testing it allowed us to see how we were deficient in our previous analysis. The isometric system is not rigid. There is compliance in the winch

cable, artificial tendons, and brackets that bend and stretch. We introduced a constant length offset term χ_0 , axial stiffness χ_1 , and bending stiffness χ_2 . The bracket frame of reference was placed where it starts to cantilever and is rotated about Z. Optimization found results for a $\phi 10$ mm BPA with $l_0 = 48.5$ cm in the simple pinned knee configuration. These results were validated with second $\phi 10$ mm BPA with $l_0 = 45.7$ cm.

The modeling of extensor BPA paths that wrap around the knee joint is an approximation with inherent error (Fig. 4.7). When the moment arm is short, small changes to it can have a large effect on the resultant torque [Young et al., 2019]. It was assumed that wrapping the BPA around the joint would be easier to account for than adding a patella, deterministically designing its location as a function of knee angle, and accounting for friction. During greater magnitude knee flexion angles, shorter l_0 BPAs were noted as being stretched and compressed (see Fig. 3.2C). The same force that caused the stretch and compression would also sometimes cause the BPA to slip off of the anterior bolt head of the bolt that holds the femur to the knee. This would result in a ± 20 mm displacement in the Z direction, which means a change the force vector and a change in torque. The system designer, too, must be mindful. Fig. 4.5 shows a deviation between measured and originally predicted results that widens as expected torque increases. This indicates system compliance. However, the hybrid calculated torque matching or exceeding predicted torque, except for higher degrees of flexion in the $l_0 = 41.5$ cm BPA, indicates a loss of functional l_m ; this is why the χ_3 term was created in (3.26). Fig. 4.5 shows the improved fit from running the optimization that includes this term.

The biomimetic knee joint with an extensor BPA was used to validate the extensor optimization results. The predicted line more closely matches the torque values. However,

the predicted torque is still higher than experimentally measured torque.

In the presented nonlinear biomimetic robotic system, rigid body mechanics simplifications, often encountered in other robotic and biomechanical systems, are removed. Instead, working from the simple to the complex elucidated mechanisms of system compliance that will lead to much more accurate first iteration designs. Technologies have been developed to predict system bending and axial compliance, a constant length offset, and a loss of usable length that occurs as artificial muscle bends around a joint.

In equation $m\ddot{\epsilon}^* + c(\dot{\epsilon}^*)\dot{\epsilon}^* + k(\epsilon^*)\epsilon^* = F$ given $k(\epsilon^*, P^*) = \left. \frac{\partial F}{\partial \epsilon^*} \right|_{P^*} = -c_0 c_1 e^{-c_1 \epsilon^*} - 2c_2 \epsilon^* P^* e^{-c_2 (\epsilon^*)^2}$ then if we set mass m and external force F to zero we get $c(\dot{\epsilon}^*) = (k(\epsilon^*)\epsilon^*)/\dot{\epsilon}^*$

5.7 Tendon Transmission Trade-offs in the Reverse-Pulley Arrangement

The reverse-pulley transmission of the inverted-pendulum testbeds (Section 3.9) [Bolen et al., 2024b,a,c] illustrates a design trade that recurs wherever a BPA drives a joint through a tendon. The four-wheel block-and-tackle buys joint travel with muscle force: reeving the paired BPAs' tendon through the pulley train mounted on the foot multiplies the tendon travel delivered at the joint at a proportional cost in force transmitted to the tibia. For an actuator whose limited contraction forces long routings — the burden visible in the extensor redesign (Section 4.5) and taken up for the humanoid in Chapter 6 — that exchange can be exactly what a design needs. The arrangement also collects its price structurally: because the pulley block mounts on the foot, the tendon tension reacts into the structure that carries the pulleys rather than passing cleanly along the tendon line, and this reaction force deformed the printed foot significantly at the pulley mounting location.

Transmission hardware, in other words, is not kinematically free; it loads and deflects the segments it rides on.

The arrangement also complicates the geometry on which the produced torque depends. With a fixed insertion, the force unit direction used in the moment-arm calculation follows from static attachment geometry; with the tendon wrapping over pulleys that move with the joint, the wrapping point that fixes that direction migrates as the joint angle changes. This migration can increase the effective moment arm over part of the travel — part of the arrangement’s appeal — but it makes the kinematics considerably more daunting: the moment arm becomes a function of joint angle that must itself be modeled, and errors in the assumed wrap geometry propagate into torque just as the moment-arm sensitivity discussed in Section 5.6 predicts.

Two solutions present themselves. The first is to track the moving wrap point in the model, accepting the more demanding kinematics in exchange for the pulley train’s mechanical simplicity and its moment-arm advantage. The second is to eliminate the moving wrap with a Bowden cable — the tendon running in a sheath anchored to both bodies — which fixes the force direction on each body but demands packaging space for a bracket and a barrel adjuster on both bodies, trading kinematic simplicity for routing hardware. Either solution can serve; the choice turns on the goals of the researchers, on whether model fidelity or transmission simplicity is worth more for the joint being designed. Tracking a moving wrap point in the route model is not new ground for this dissertation’s optimization work — the biomimetic flexor route already carries one explicitly (Section 3.8; Appendix C) — but no pulley-aware model of this arrangement has been built, and the arrangement remains a candidate for the humanoid joints whose

optimized routes demand impractically long actuators (Chapter 6).

5.8 Synthesis: From Actuator Characterization to Neuromechanical Control

Taken together, the actuator and joint level results of this dissertation form a coherent design chain for biomimetic legged robots. At the actuator level, the maximum isometric force of Festo BPAs was shown to depend strongly on resting length, correcting a systematic over-prediction by the manufacturer's tool at short resting lengths, and a nondimensional force surface was derived that predicts isometric force from relative strain and relative pressure with only three diameter-dependent coefficients. At the joint level, the improved actuator model was combined with a compliance-corrected musculotendon kinematics model to predict isometric knee torque across the full range of motion for pinned and biomimetic knee joints, and the biomimetic knee driven by optimized BPAs was shown to meet or exceed human isometric torque over most of the range of motion.

This chain was developed deliberately so that each stage can feed the next. The force model supplies the actuator terms for the torque model; the torque model supplies the plant model for designing joint controllers; and the validated biomimetic knee demonstrates that the actuator architecture can reproduce human joint-level force capabilities, which is a prerequisite for the neuromechanical control studies planned in Chapter 6. What remains missing is the neural layer itself: a controller that schedules the measured muscle forces the way the nervous system schedules motor unit activity. The AARL's prior AnimatLab walker (Section 3.10) offers an early demonstration of both halves of that challenge: acting as a feedforward controller, its two-layer CPG drove gait-like stepping of a suspended

biomechanical model — confirming that rhythm generation does not require sensory feedback — but untuned afferent feedback could not support those patterns against the ground, which is why feedback tuning is the critical path from feedforward stepping to weight-bearing walking. Closing that loop in simulation, and finally on hardware, is the subject of the next chapter.

6 Future Work

The experimental work of Chapters 3–4 characterizes the plant: the actuators and the joints they drive. The remaining task of the dissertation project is the controller — a synthetic nervous system that schedules those actuators the way the nervous system schedules muscles — and eventually the physical robot that the controller runs on. This chapter lays out the planned work in four stages: a neuromechanical simulation pipeline (Section 6.1), a two-layer CPG with full proprioceptive and contact feedback (Section 6.2), extension of the sensorium with vestibular, ocular, and cerebellar contributions (Section 6.3), and physical realization on a treadmill robot with real-time neural control (Section 6.4). Some of the work described here is planned for the remainder of the dissertation; the later stages extend naturally into postdoctoral research.

6.1 A Neuromechanical Simulation Pipeline

The first stage converts the validated biomechanical design into a physics simulation that a neural controller can drive. The pipeline has three parts. Much of this stage is now in place (status as of this writing, detailed in Section 3.11): the OpenSim-to-MuJoCo conversion is complete and audited, the SNS-Toolbox coupling runs at a 0.1 ms neural step with force feedback, and the full two-layer spinal network has been assembled and its rhythm generator verified in isolation. What remains of this stage is the blocker described below — non-finite leg-joint states in full-body co-simulation, with the foot-contact meshes the primary suspect — after which the plan shifts from assembly to tuning and validation.

6.1.1 From OpenSim to MuJoCo

The starting point is the author’s modified OpenSim model: the Gait2392-derived robot variant in which the actuator set has been reduced and adapted to the BPA hardware and the muscle origin, insertion, and wrapping points have been updated to the optimized robot routing [Bolen and Hunt, 2019, Morrow et al., 2020]. The model will be kept in OpenSim 4.x format and converted to MuJoCo [Todorov et al., 2012] using MyoConverter [Caggiano et al., 2022, Wang et al., 2022], which converts OpenSim musculoskeletal models to MuJoCo models with optimized muscle kinematics and kinetics. The conversion will be sanity-checked by inspecting the generated model, verifying joint naming and ordering for later sensor and actuator mapping, confirming that the expected number of muscle-tendon actuators exists with plausible attachment geometry, and fixing any conversion gaps by revising the OpenSim model or accepting manual edits in the MuJoCo model.

The MuJoCo model will represent actuation with muscle-like actuators that carry an internal activation state, so that SNS outputs can drive them through activation dynamics. Each timestep, MuJoCo updates the kinematics and the muscle variables (e.g., muscle-tendon lengths and velocities) needed to compute proprioceptive signals.

6.1.2 Coupling the Synthetic Nervous System

The neural controller will be implemented with SNS-Toolbox [Nourse et al., 2023], which supports large networks of spiking and non-spiking neurons and couples directly to MuJoCo. At each timestep, SNS receives Ia, Ib, and II proprioceptive channels plus binary heel and toe contact signals from the MuJoCo model, processes them through the neural

core described below, and returns per-muscle activation commands that are fed into the MuJoCo muscle activation states.

The coupling will be brought up incrementally. Before any gains are tuned against human gait, the mechanics-only loop will be verified with open-loop activation patterns, confirming the sign and magnitude behavior of torques and forces. The SNS will then be added with a minimal sensor set — contact signals plus a single proprioceptive channel — and only later expanded to the full Ia/Ib/II set. Once stable, the activation strengths and neuron-to-muscle mapping parameters will be tuned across a few walking speeds. A Simscape-based mechanical plant could substitute for MuJoCo for validation of specific mechanical subsystems, but Simscape does not natively host the spiking and non-spiking neuron models of an SNS, so MuJoCo remains the primary platform. The AARL’s quadruped has already traveled part of this road: a MuJoCo model of the BPA quadruped was coupled to SNS-Toolbox, produced closed-loop coordinated stepping, and was transferred to hardware with potentiometer and pressure feedback on a treadmill [Lutz, 2025]; the present pipeline extends that approach from the quadruped to a leg built on human anatomy. Comparable neuromechanical simulation frameworks, such as FARMS, are emerging in parallel [Arreguit et al., 2023], underscoring the timeliness of the approach.

6.2 A Two-Layer CPG with Sensory Feedback

The author’s prior AnimatLab walker (Section 3.10) demonstrated both the promise and the limitation of this architecture, and its outcome belongs on record here because it defines

the requirements for the work below. With the rhythm-generator, pattern-formation, and sensorimotor layers oscillating in phase, the suspended model produced alternating stepping-like motion at approximately 4 Hz — faster than the ~ 1 Hz of typical human gait, but adjustable through the sodium-channel activation parameters, and, once ground contact exists, subject to further slowing from ground-reaction forces and load-dependent feedback. When the earlier demonstrated configuration was lowered onto a simulated ground plane, however, the same patterns could not support the body or generate forward motion: afferent pathways had been wired but not tuned, and functional ground-contact gating of stance and swing had not been implemented. Preliminary results from that model were presented at the 2021 NIH BRAIN Initiative Investigators Meeting [Bolen and Hunt, 2021]. The current instrumented phase-1 model records left and right foot and toe contact channels, but their use for closed-loop phase control has not yet been validated. The suspended walker is the CPG acting as a feedforward controller: rhythm generation alone, without functional ground-contact feedback, produced gait-like stepping, exactly as the deafferented-CPG physiology of Section 2.6 predicts. What feedforward stepping cannot do is support the body against the ground, so the critical path from stepping-like motion to weight-bearing walking is the proprioceptive and contact feedback channels described below, and the verification tests that tune them. That premise is itself well supported: biomechanics and sensory feedback have been shown to regularize the behavior of different locomotor CPG architectures [Deng et al., 2022], so the contribution of the present work lies in applying this principle to the suspended walker rather than in the principle. The AnimatLab walker model, the direct descendant of that earlier work, continues to serve as the prototyping environment for the network while the MuJoCo

pipeline is brought up.

6.2.1 Architecture

The neural core follows the two-level organization of the mammalian locomotor CPG established by the Rybak group [Rybak et al., 2006, McCrea and Rybak, 2008, Rybak et al., 2015]: each leg has a rhythm generator (RG) circuit, which produces the basic locomotor rhythm and controls its period, and a pattern formation (PF) circuit, which shapes the rhythm into phase-appropriate activation of the flexor and extensor motoneuron populations. A left–right rhythm generator circuit coordinates the two legs through commissural inhibition and excitation, following the models of left–right coordination [Molkov et al., 2015, Danner et al., 2016]. The per-leg PF output drives the motoneuron pools — here, the BPA activation commands for the hip, knee, and ankle muscle groups of each leg.

6.2.2 Sensory Feedback Channels

Three classes of proprioceptive feedback will close the loop between the MuJoCo plant and the CPG, mirroring the biological afferents organized in the Sensory Afferent Database [Bolen et al., 2023]:

- **Ia afferents** (muscle spindle primary endings) signal muscle length and velocity, providing stretch reflex pathways and phase-dependent modulation of the PF network.
- **Ib afferents** (Golgi tendon organs) signal muscle force, providing load-dependent

regulation of stance-phase extensor activity.

- **II afferents** (muscle spindle secondary endings) signal length, contributing to position sensing and interjoint coordination.
- **Contact receptors** on the heel and toe provide discrete stance and swing boundary signals, driving phase transitions and weight-support switching.

6.2.3 Verification Tests

Before attempting locomotion, the network will be verified with simple simulation tests in which tonic stimulus is applied to the muscle motoneurons, to the pattern-formation layer, and to the rhythm-generator layer, and the resulting outputs are compared with the corresponding experimental literature. The first of these tests is complete at the rhythm-generator level: under tonic drive, the isolated SNS-Toolbox rhythm generators of the assembled spinal network oscillate at an approximately 0.9 s period with the two legs in antiphase (Section 3.11); in the AnimatLab instance of the network, conductance tuning toward robust rhythm-generator oscillation is still under way. The closest biological analogue is the tonic-stimulation air-stepping protocol of Selionov et al. [2009], in which tonic central and sensory stimuli facilitate involuntary stepping in humans; analogous deletion and stimulation phenomena are reproduced by the two-level CPG model itself [Rybak et al., 2006] and by neuromechanical implementations [Markin et al., 2010, Shevtsova et al., 2016]. Comparing the simulated tonic-stimulation responses with these studies will validate the sign, gain, and phase structure of the implemented circuits before any walking-speed tuning is attempted.

6.3 Extending the Sensorium: Vestibular, Ocular, and Cerebellar Contributions

Once stepping is stable, three supraspinal contributions will be added. First, vestibular feedback will be simulated with IMU-like sensing of body orientation and angular velocity, extending the AARL’s balance-platform work, in which an inverted pendulum with pseudo-vestibular IMU feedback maintained balance on an oscillating platform [Bolen et al., 2024b,a,c] and analytic neural networks reproduced human balance dynamics [Hilts et al., 2019]. Second, ocular feedback will be modeled to investigate how visual stabilization cues can help reset IMU drift — a well-known practical problem for inertial sensing on robots — by treating visual measurements of the horizon and surrounding as a low-frequency correction to the vestibular estimate. Third, a cerebellar layer will be explored as an adaptive gateway between higher-level locomotion commands and the CPG: a structure that can gate between different gaits or movement patterns (e.g., walking, running, turning) and tune the phase relationships among them, in line with the cerebellum’s established role in adaptation and coordination. Multilevel spinal-plus-cerebellar synthetic nervous systems have already been demonstrated for insect-like legs [Grummer et al., 2024], and hierarchical cortico-cerebellar-spinal models have been proposed for adaptive step-length control in bipedal locomotion [Chen et al., 2025]. The proprioceptive–vestibular interaction literature [Akay and Murray, 2021] will guide how these signals are weighted.

6.4 Physical Realization: A Treadmill Robot with Real-Time Neural Control

The final stage returns to hardware. The robot will walk on a treadmill with real-time feedback and control implemented on an embedded computing stack: Teensy microcontrollers

for the low-latency per-joint loops (valve control, pressure sensing, contact sensing), and an Nvidia Jetson-class companion computer for the SNS network evaluation and activation computation. A serial connection to a host computer will be used only for data logging and high-level commands (e.g., run, walk, stimulate a named neural pathway), keeping the control loop fully embedded. Walking control of a physical musculoskeletal biped with McKibben muscles has recently been demonstrated with spring-loaded inverted pendulum controllers [Li et al., 2024], supporting the feasibility of embedded control on anthropomorphic pneumatic legs. An actuation-level design lever also remains open: the BPA’s limited contraction can force long musculotendon routings — a burden visible in the extensor redesign bounds of Appendix C and encountered in the author’s master’s routing work [Bolen and Hunt, 2019]. A block-and-tackle “reverse pulley” arrangement, demonstrated on the AARL’s self-balancing inverted-pendulum platform, addresses this by trading transmitted force for travel: paired BPAs pulled through compound pulleys on a tendon coupled to the limb, increasing range of motion at the cost of reduced transmitted force and increased reaction force on the structure [Bolen et al., 2024b,a,c]. Because joint torque is computed in the body frame, configurations that turn that reaction force to advantage may exist, making the arrangement a candidate for joints whose optimized routes demand impractically long actuators. This division of labor mirrors the SNS-Toolbox backend philosophy, which targets embedded computing platforms for real-time neural simulation [Nourse et al., 2023].

Because the static force and torque models of Chapters 3–4 are computationally inexpensive lookup-table candidates, they will serve as the feedforward plant model inside the real-time controller, while the dynamic pressure–force behavior under pulse-modulated

control will draw on the state-space model of Elzein et al. [2025]. Embedded execution of the complete CPG-plus-reflex network on this stack is the intended demonstration of the dissertation project: a robot that walks with human-like biomechanics under a biologically grounded neural controller, and whose nervous system — unlike a living animal’s — can be instrumented, stimulated, and lesioned at will.

6.5 Toward Dynamic Whole-Body Maneuvers

Beyond steady walking, the long-term goal of this line of research is dynamic, whole-body maneuvering: transitions between gaits, reactive balancing, and ultimately ballistic maneuvers that require coordinating the full actuator set through brief, precisely timed activation bursts. The author’s aspiration for this program is a robot that can perform a kickflip — a maneuver that, like its skateboard namesake, demands explosive coordinated extension, precise timing, aerial body orientation, and controlled landing. It is exactly the kind of behavior that exercises every layer of the proposed architecture, from contact receptors and spinal pattern generation to vestibular orientation sensing and cerebellar coordination, and it would demonstrate that biomimetic actuation and neuromechanical control scale from steady locomotion to dynamic whole-body movement.

7 Conclusion

This dissertation moved the biomimetic humanoid robot project from an unvalidated design concept to a quantitatively characterized actuator set and a validated knee-level biomechanical design, and it defined the path from those experimental foundations to neuromechanical control.

At the actuator level, the analysis in Chapters 3 and 4 produced novel equations for calculating isometric force in Festo 10 mm and 20 mm diameter BPAs. The work elucidated the relationship between maximum BPA force at 620 kPa and resting length, $F_{620}(l_0)$, a previously unreported characteristic of these actuators that the manufacturer's tool predicts with the wrong trend, and it produced a nondimensional force surface, $F^*(\epsilon^*, P^*)$, with only three diameter-dependent coefficients. Taken together, these fits predict isometric BPA force across resting lengths from a small number of measurements, correct the large over-predictions of the manufacturer's tool at short resting lengths, and outperform published single-length models. In a design workflow, these equations make it possible to select actuator diameter and resting length, and to evaluate whether a proposed muscle routing can meet force and range-of-motion requirements, before committing to a prototype.

At the joint level, the improved actuator model was combined with a compliance- and wrapping-corrected musculotendon kinematics model and validated on two knee designs. The hybrid torque calculation method isolated the sources of error between prediction and measurement, and the resulting correction terms — a constant length

offset, series compliance represented by an equivalent stiffness, and a wrapping loss term — substantially improved torque prediction on the pinned knee and transferred to the biomimetic four-bar knee. The biomimetic knee driven by optimized BPAs was shown to meet or exceed human isometric knee torque over most of the range of motion, confirming that the optimized muscle routing of the author’s earlier work is realizable in hardware, and identifying the high-flexion regime where additional flexor capacity is needed.

Beyond the specific models, the dissertation’s methodological contribution is the actuator-to-joint design chain itself: characterize the actuator, correct for the compliance and geometry of the transmission, validate against human biomechanics, and only then close the loop with a controller. The remaining gap — the neural layer — is scoped in Chapter 6: a MuJoCo plant converted from the AARL’s OpenSim models via MyoConverter (the conversion, the SNS-Toolbox coupling, and the assembled two-layer spinal network with its verified antiphase rhythm generator are already in place), wrapped in an SNS-Toolbox synthetic nervous system implementing a two-layer CPG with per-leg rhythm generation and pattern formation, left-right coordination, and Ia, Ib, II, and contact feedback, verified first by tonic-stimulation tests against the literature, then extended with vestibular, ocular, and cerebellar layers, and finally deployed on embedded hardware controlling a treadmill-walking robot. The biomimetic platform that results will permit neuromechanical experiments — neural lesions, sensory deletions, pathway stimulation — that are impossible in living animals, closing the cycle between better robots and better biological understanding, and opening the way toward dynamic whole-body maneuvers.

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APPENDICES

Appendix A: Muscle Force and Joint Torque Calculations

This appendix documents the reference implementation of the force, torque, and compliance equations used in Chapters 3–4, as they exist in the project’s MATLAB code. File names are given so that any number in this dissertation can be traced to the code that produced it.

A.1 BPA Force Model Implementation

A.1.1 Strain Measures

The musculotendon path length l_{LMT} is the sum of straight segments between route via points, transformed into a common frame at each frame crossing. The contractile strain used by the force model subtracts the artificial tendon and both end-cap fittings from the path length before normalization by the resting length,

$$\varepsilon = \frac{l_0 - (l_{LMT} - l_{ten} - 2l_{fitting} - \chi_0)}{l_0}, \quad (A1)$$

where χ_0 is the constant length offset of Section 3.7 (zero in the uncorrected model). [Provenance: `MonoPamDataExplicit.m` `get.Contraction`, lines 210–217; `minimizeFlxPin.m` lines 150–164. End-cap fitting length $l_{fitting} = 25.4$ mm is a constructor default, `MonoPamDataExplicit.m` line 69.]

The maximum free-load contraction at 620 kPa is $\varepsilon_{620} = (l_0 - l_{620})/l_0$, where l_{620} is the BPA length at 620 kPa with no external load (code variable `KMAX`; `MonoPamDataExplicit.m`

line 278). The relative strain is $\epsilon^* = \epsilon / \epsilon_{620}$ (line 281).

A.1.2 Normalized Force Surface

The normalized isometric force surface of Equation (3.4) is implemented in closed form in `Functions/f_festo.m` (lines 1–26),

$$F^* = c_0 \left(e^{-c_1 \epsilon^*} - 1 \right) + P^* e^{-c_2 (\epsilon^*)^2}, \quad F^* \leq 0 \Rightarrow 0, \quad (\text{A2})$$

with $P^* = P / 620$ kPa. The zero-clamp $F^* \leq 0 \Rightarrow 0$ is applied by the callers (e.g. `Functions/festo4.m` line 35), not inside `f_festo.m` itself. The coefficients were fitted with `fitttype` nonlinear least squares and least-absolute-residual robustness (`TestingData/bpaFits.m`, lines 45–60 and 115–140, lower bounds $[0, 0, 0]$) and are reported in Table 4.2. The code additionally holds a 40 mm fit not used in this dissertation ($c_0 = 0.1224$, $c_1 = 10.47$, $c_2 = 2.023$; `f_festo.m` lines 17–20).

At runtime the torque model may alternatively evaluate a spline interpolation of the manufacturer’s datasheet curves (`Functions/festo4.m`, backed by `FestoLookup.mat`), with force clamped to zero for $\epsilon^* > 1$ or $F^* < 0$. Equation (A2) replaced this lookup in the corrected model.

A.1.3 Maximum Force

The resting-length-dependent maximum force of Equation (3.5) is implemented in `Functions/maxBPAfor` (10 mm: line 46; 20 mm: line 57; duplicate for 20 mm in `Functions/Fmax20.m`) with the

fixed 7.5 mm end-contact offset. The fitted 20 mm coefficients carry confidence intervals $a_1 \in [1.4745, 1.5011] \text{ N kPa}^{-1}$ and $a_2 \in [0.0238, 0.0258] \text{ kPa}^{-1} \text{ m}^{-1}$ (Fmax20.m lines 26–27). Total force is the product $F = F^*(\epsilon^*, P^*) F_{620}(l_0)$, resolved into a vector along the force direction (MonoPamDataExplicit.m lines 283–287).

A.2 Muscle Path Length and Moment Arm

Path length accumulates the Euclidean norm of each segment between via points, applying the body-frame transformation whenever the path crosses between the femur and tibia frames (MonoPamDataExplicit.m lines 78–95, total at lines 122–133).

The vector moment arm is implemented as the perpendicular foot of the joint origin onto the line of action,

$$\vec{r} = \vec{p} - \hat{u} (\hat{u} \cdot \vec{p}), \quad (\text{A3})$$

where $\vec{p}$ is the vector from the joint origin to the muscle-path crossing point and $\hat{u}$ is the unit force direction (MonoPamDataExplicit.m lines 156–168; identical implementations in MonoMuscleData.m lines 111–123, minimizeFlxPin.m lines 400–409, and MonoPam_mult.m lines 186–201). This is the same quantity as the cross-product projection of Equation (3.7); the planar scalar moment arm about the joint axis is the magnitude of the in-plane components, computed as $\text{hypot}(\text{mA}(:,1), \text{mA}(:,2))$ (predictKneeFlexor20mm.m line 199). Torque about the joint z-axis is $\vec{M} = \vec{r} \times \vec{F}$, evaluated only where the BPA is in tension (values are discarded for $\epsilon < -0.02$, the compressive region; MonoPamDataExplicit.m line 308).

A.3 Compliance Model Implementation

The compliance-corrected muscle length entering Equation (3.10) is computed as $l_m = l_{LMT} - \chi_0 - l_{ten} - 2 l_{fitting}$ (`minimizeFlxPin.m` line 254). In the two-bracket fit of record, the insertion-bracket stiffness matrix and the projected scalar stiffness along the force direction are

$$\mathbf{K}_{br} = \text{diag}(\chi_1, \chi_2, \chi_1), \quad k_{\hat{u},br} = \left(\hat{u}^T \mathbf{K}_{br}^{-1} \hat{u} \right)^{-1}, \quad (\text{A4})$$

matching Equation (3.11), and the origin-side second bracket contributes an additional scalar compliance c_{b2} in series (Appendix C, Table C6) [Provenance: `minimizeFlxPin2brk.m` lines 387–418]. The earlier single-bracket evaluator used $\text{diag}(\chi_1, \chi_2, \chi_2)$ and is superseded; its array survives only in the legacy file (`minimizeFlxPin.m` lines 266–272).

The artificial tendon is modeled as an axial spring with stiffness

$$k_{ten} = \frac{n A_c E}{L_{ten}}, \quad (\text{A5})$$

with $n = 2$ parallel cable strands ($\times 6$ for the 20 mm assembly), $A_c = 1.51 \times 10^{-6} \text{ m}^2$ (19-strand cable), and $E = 193 \times 10^9 \text{ Pa}$ (`minimizeFlxPin.m` lines 460–471; `MonoPam_mult.m` lines 606–617). Series combination with the projected bracket compliance gives $k_{eq} = \left(k_{\hat{u},br}^{-1} + k_{ten}^{-1} \right)^{-1}$ (lines 273–275).

The scalar compliance displacement δ of Equation (3.25) is found numerically with `fzero` on the bracket $[0, l_m - l_{620}]$,

$$F_{620}(l_0) F^*(\varepsilon^*(\delta), P^*) - k_{eq} \delta = 0, \quad (\text{A6})$$

with $\delta \equiv 0$ when the uncorrected relative strain already exceeds unity (`minimizeFlxPin.m` lines 291–305). The vector bracket deflection is then $\epsilon_{br} = \mathbf{K}_{br}^{-1} \left(\|\vec{F}\| \hat{u} \right)$ (line 324), the cable stretch is $\|\vec{F}\|/k_{ten}$ (line 334), and the attachment point, path, moment arm, and torque are recomputed. For the two-BPA flexor grouping, the coupled balance shares one common force between both muscles and is solved analogously on $[0, F_{620}]$ (`MonoPam_mult.m` lines 490–564).

A.4 Wrapping-Loss Implementation

For the pinned-knee extensor, Equation (3.26) is implemented segment-wise over the wrap arc with radii $R_1 = 12$ mm and $R_2 = 40$ mm and segment boundaries $\theta_1 = 27^\circ$ and $\theta_2 = -23^\circ$ (`minimizeExtX3.m`, nested `Contraction`, lines 327–370). For the biomimetic-knee 20 mm cell, the arc-length factor is the routed bend measure $b(\theta_k)$ (arc length swept by the wrap, built with the `wRap` bend radius in `buildKneeFlexorRoute20mm.m` lines 265–268), giving $\Delta l = \chi_3 b(\theta_k) (1 - \epsilon^*)^2$ (`MonoPam_mult.m` lines 398–441), with a hardcoded-geometry fallback for degenerate routes (lines 428–434). The loss enters only the force-producing strain; the kinematic (prediction) strain excludes it (`minimizeExtX3.m` lines 236–240).

Appendix B: Neuron Model Equations

This appendix documents the neuron and synapse models used by the synthetic nervous systems in this project, both in the AnimatLab walker model and in the SNS-Toolbox implementation of the MuJoCo pipeline (Chapter 6). The authoritative model state is the working AnimatLab project file (`Biped_2xCPG_wSubs/Biped_2xCPG_wSubs.aproj`); after the network reorganization described in Section 3.11, a fresh standalone export (`Biped_2xCPG_wSubs_Standalone.asim`) runs headless and carries the rhythm-generator charts, and the earlier phase-1 instrumented export (`walk new new tester added 2 axis_phase1.asim`) records the run reported in Section 4.4.

B.1 Spiking Neuron Model

Each spiking neuron is a conductance-based leaky integrator with threshold and adaptation,

$$C_m \dot{V} = -g_{\text{leak}} (V - E_{\text{rest}}) + I_{\text{syn}} + I_{\text{app}}, \quad (\text{B1})$$

where V is the membrane potential, C_m the membrane capacitance ($g_{\text{leak}} = C_m / \tau_m$ with membrane time constant τ_m), E_{rest} the resting potential, I_{syn} the summed synaptic current, and I_{app} any applied (tonic stimulation) current. A spike is emitted when V exceeds the firing threshold, after which V resets and an after-hyperpolarization (AHP) of amplitude A_{AHP} with time constant τ_{AHP} is applied; the threshold itself accommodates with the firing history (relative accommodation r_{acc} , time constant τ_{acc}). Optional calcium channels are

defined but disabled in the current model ($G_{\max, \text{Ca}} = 0$).

The parameter values shared by the neurons in the walker model are given in Table B1.

Table B1: Spiking neuron parameters in the AnimatLab bipedal walker model (Biped_2xCPG_wSubs). All neurons share these values except the motor neuron threshold noted below.

Parameter	Symbol	Value
Resting potential	E_{rest}	-60 mV
Membrane time constant	τ_m	5 ms
Initial firing threshold	V_{th}	-55 mV (motor neurons: -59 mV)
Relative accommodation	r_{acc}	0.3
Accommodation time const	τ_{acc}	10 ms
AHP amplitude	A_{AHP}	1 mV
AHP time constant	τ_{AHP}	3 ms

B.2 Non-Spiking (Graded) Chemical Synapses

Pattern-formation and motoneuron-drive connections in the walker use graded chemical synapses, whose conductance is a clipped linear ramp of the presynaptic potential,

$$g_{\text{syn}}(V_{\text{pre}}) = g_{\text{max}} \text{clamp}\left(\frac{V_{\text{pre}} - V_{\text{th}}}{V_{\text{sat}} - V_{\text{th}}}, 0, 1\right), \quad I_{\text{syn}} = g_{\text{syn}}(V_{\text{post}} - E_{\text{syn}}), \quad (\text{B2})$$

where V_{th} and V_{sat} are the synaptic threshold and saturation potentials and E_{syn} the reversal potential. The walker model instantiates three non-spiking synapse types:

- **Nicotinic ACh (excitatory drive):** $E_{\text{syn}} = -10 \text{ mV}$, $g_{\text{max}} = 0.5 \mu\text{S}$, $V_{\text{th}} \rightarrow V_{\text{sat}} = -65 \text{ mV} \rightarrow -20 \text{ mV}$.

- **Hyperpolarizing IPSP (inhibition):** $E_{\text{syn}} = -70 \text{ mV}$, $V_{\text{th}} \rightarrow V_{\text{sat}} = -55 \text{ mV} \rightarrow -40 \text{ mV}$.
- **Depolarizing IPSP:** $E_{\text{syn}} = -55 \text{ mV}$, $V_{\text{th}} \rightarrow V_{\text{sat}} = -60 \text{ mV} \rightarrow -40 \text{ mV}$.

Spiking chemical synapses and electrical synapses (rectifying coupling weight 0.1, non-rectifying 0.2) are also defined in the model and are available for the sensory afferent pathways of the MuJoCo/SNS implementation.

B.3 Relation to the SNS-Toolbox Implementation

These are the standard AnimatLab firing-rate/conductance models. SNS-Toolbox [Nourse et al., 2023] implements corresponding classes of non-spiking and spiking neurons with conductance-based synapses, and the AnimatLab parameters above served as the physically motivated initialization for the first SNS-Toolbox instantiation of the spinal network: a 406-neuron network with per-leg rhythm-generator half-centers, phase-shifted pattern-formation groups, motoneuron pools for all 92 muscle-tendon actuators of the converted model, and Ia, Ib, and II afferent populations. In that implementation the non-spiking membrane and graded-synapse equations take the same forms as Equations (B1) and (B2) with the spiking threshold and accommodation terms removed, and the afferent resting-tone offsets are zeroed so that Ia and Ib signals are pure activity measures; the rhythm-generator layer has been verified to oscillate at a human-like period with the two legs in antiphase under tonic drive (Section 6.2). An explicit parameter-mapping and validation campaign between the two implementations remains future work; the tonic stimulation tests of Section 6.2 correspond to constant I_{app} injections into Equation (B1) at the motoneuron,

pattern-formation, and rhythm-generator layers respectively.

Appendix C: Model Geometry, Bracket Frames, and Compliance Parameters

This appendix documents the numeric geometry and compliance parameters behind the torque models of Chapters 3–4 as they exist in the project’s MATLAB code, with file names given so that any value can be traced to the code that produced it. Route and bracket coordinates are coded in meters and reported here in millimeters; stiffnesses are coded and reported in newtons per meter. The bracket reference points are interpretation choices for the 3D-printed (Onyx FDM) attachment hardware, set from CAD by the author; moving a reference point changes the identified stiffness values, so the points are recorded here alongside the values they produced.

C.1 Static BPA Pathway Data

The static BPA model routes each actuator as a straight-segment polyline through via points stored in a per-test `Location` array of dimensions $M \times 3 \times N$, where N is the number of joint frames in the motion sweep and M the number of via points; the first row in the distal link frame is marked by a cross-point index. There are no binormal or surface-frame quantities: at each frame the force direction is the normalized vector from the cross-point row to the previous row transformed into the crossing frame,

$$\hat{\mathbf{u}} \propto \mathbf{R}^{-1} \mathbf{p}_{C-1} - \mathbf{p}_C, \quad (\text{C1})$$

where $\mathbf{p}_C$ is the cross-point row, $\mathbf{p}_{C-1}$ the previous row, and $\mathbf{R}$ the frame-crossing rotation [Provenance: `Testing_Data/2022_02_Festo/minimizeFlxPin2brk.m` lines 536–547; `minimizeFlxPin.m` lines 381–394; `minimizeExtX3.m` lines 687–700; BPA struct fields, `minimizeFlxPin.m` lines 28–35]. The pinned-knee test routes are archived as `FlxPinBPASet.mat` (5 flexor tests, $M = 2$, knee angle -120° to $+10^\circ$) and `ExtPinBPASet.mat` (9 extensor tests, $M = 7$, knee angle -125° to $+35^\circ$).

Table C1 lists the five pinned-knee flexor tests; their routes are two points, femur-frame origin $\mathbf{p}_1 = [-75.0, 100.0, 32.8]$ mm and tibia-frame insertion $\mathbf{p}_2 = [-50.1, -45.0, 32.3]$ mm (test 1; `FlxPinBPASet.mat`, `kf(1)`, frame 1). Table C3 lists the extensor pinned-knee route of test 1 at $\theta_k = -125^\circ$; rows 1–6 are femur-frame and row 7 is tibia-frame, and as the knee extends, rows 2–6 release onto $\mathbf{p}_1$ (wrap released at $+35^\circ$). Table C4 lists the per-test constants of the nine extensor tests; the validation pool used for the correction terms of Section 3.7 is tests $\{1, 2, 5, 6, 7, 8\}$ [Provenance: `Dig_crossPredict.m` line 46].

Table C1: Pinned-knee flexor test BPA constants, read directly from `FlxPinBPASet.mat` (`kf(1)`–`kf(5)`). Labels are the laboratory shorthand used in the fitting logs (`minimizeFlxPin10mm_2brk.m` line 51); l_0 , tendon, and pressure are the values used by the model. All tests use $\phi 10$ mm BPAs; the 47 cm test’s measured angles carry the $+5.3^\circ$ encoder correction at load time (`minimizeFlxPin2brk.m` lines 66–70), and it was held out of the two-bracket identification as validation.

Test	Label	l_0 (m)	Tendon (m)	P (kPa)	F_m (N)	Use
kf(1)	48 cm	0.485	0	605.7	443.4	training
kf(2)	46 cm	0.457	0	614.2	441.4	training
kf(3)	47 cm	0.479	0	623.2	443.0	held-out validation
kf(4)	40 cm-tendon	0.406	0.035	623.4	436.9	training
kf(5)	41 cm	0.410	0	615.4	437.3	training

The biomimetic flexor route (`buildKneeFlexorContext20mm.m`) is a three-point path —

Table C2: Biomimetic knee, extensor 20 mm BPA route seed (femur frame: p1–p5; tibia frame: p6–p9; buildKneeExtContext20mm.m lines 380–389, Ben-set SolidWorks points). Release schedule of the verified elimination: p5 at -91.1° , p4 at -50.4° , p6 at -47.8° , p7 at -20.2° , p3 at $+6.1^\circ$.

Point	x (mm)	y (mm)	z (mm)	Frame	Role
p1	40.00	35.00	0	femur	BPA origin (attachment)
p2	83.90	-274.76	0	femur	BPA contacts mounting base
p3	66.34	-427.24	0	femur	femoral contact
p4	51.52	-446.74	0	femur	femoral condyle contact
p5	38.26	-453.01	0	femur	femoral condyle contact
p6	69.85	34.33	0	tibia	tibial contact (upper)
p7	73.50	19.13	0	tibia	tibial contact (mid)
p8	71.98	-12.23	0	tibia	tibial contact (lower)
p9	34.06	-66.87	0	tibia	patellar ligament ring (distal, pEnd)

Table C3: Pinned-knee extensor BPA route of test 1 at $\theta_k = -125^\circ$ (ExtPinBPASet.mat, ke(1), frame 1; mm).

Point	x (mm)	y (mm)	z (mm)
p1	30.00	-50.00	0
p2	65.00	-350.00	0
p3	42.40	-408.50	0
p4	28.20	-421.60	0
p5	10.50	-425.20	0
p6	-8.60	-422.80	0
p7	42.50	-75.90	0

Table C4: Pinned-knee extensor test BPA constants, read directly from ExtPinBPASet.mat (ke(1)–ke(9)). Labels are the laboratory shorthand used in the fitting logs (Dig_resultsTable.m line 32); l_0 , tendon, and pressure are the values used by the model — note that the shorthand rounds several lengths (e.g. “42 cm” is $l_0 = 0.415$ m and “43 cm-tendon” is $l_0 = 0.432$ m). All tests use $\phi 10$ mm BPAs. The correction terms of Section 3.7 were identified on the validation pool $\{1, 2, 5, 6, 7, 8\}$; tests 3, 4, and 9 were excluded.

Test	Label	l_0 (m)	Tendon (m)	P (kPa)	F_m (N)	Use
ke(1)	40 cm	0.400	0	624.9	436.3	pool
ke(2)	40 cm-tendon	0.400	0.040	618.4	436.3	pool
ke(3)	42 cm	0.415	0	605.2	437.8	excluded
ke(4)	42 cm-tendon	0.415	0.040	605.2	437.8	excluded
ke(5)	43 cm	0.436	0	621.8	439.7	pool
ke(6)	43 cm-tendon	0.432	0.022	622.9	439.3	pool
ke(7)	46 cm	0.457	0	605.9	441.4	pool
ke(8)	47 cm	0.465	0	605.9	442.0	pool
ke(9)	48 cm	0.480	0	605.9	443.1	excluded

femur-frame origin $\mathbf{p}_1$, a moving wrap point $\mathbf{p}_{\text{wrap}}$ on the tibia, and tibia-frame insertion $\mathbf{p}_{\text{end}}$ — with the wrap point held on a standoff surface of radius b_{paRb} about the joint and its z placed on the projected $\mathbf{p}_1$ – $\mathbf{p}_{\text{end}}$ line; the optimizer seed is $\mathbf{p}_{1,0} = [-50.0, 35.0, 50.0]$ mm and $\mathbf{p}_{2,0} = [-12.24, -8.87, 27.87]$ mm with $l_0 = 415$ mm and 25 mm tendon [Provenance: `buildKneeFlexorContext20mm.m` lines 178–190; `buildKneeFlexorRoute20mm.m` lines 98–186]. The biomimetic extensor seed route is given in Table C2, including the release schedule of the route-elimination rule.

C.2 Joint Transformation Matrices

The knee transform is planar: every joint frame is a rotation about z by the knee angle φ composed with a translation fitted to robot CAD tables of the migrating instantaneous center of rotation. With $\mathbf{R} = \mathbf{R}_z(\varphi)$, the robot (femur→ICR) transform is $\mathbf{T}_{\text{Pam}} = [\mathbf{R}, [f_3(\varphi), f_4(\varphi), 0]^\top; \mathbf{0}^\top \ 1]$ with f_3, f_4 cubic-spline fits (ICR x : 23.30 to 10.47 mm; y : –416.65 to –395.66 mm over $\varphi = 0.17$ to -2.62 rad), and the ICR→tibia transform is the pure translation $[f_{13}(\varphi), f_{14}(\varphi), 0]^\top$ (x : 29.66 to –7.58 mm; y : 25.97 to 22.33 mm), composed as $\mathbf{T}_{t1 \rightarrow f} = \mathbf{T}_{t1 \rightarrow \text{ICR}} \mathbf{T}_{\text{Pam}}^{-1}$ over 100 frames spanning $\varphi \in [-120^\circ, +10^\circ]$ [Provenance: `buildKneeFlexorContext20mm.m` lines 34–80; `buildKneeExtContext20mm.m` lines 30–78; `Functions/ModernRobotics/RpToTrans.m`]. In the archived test sets the per-frame

transform maps tibia to femur/ICR; at $\theta_k = -120^\circ$ (test kf1),

$$\mathbf{T}_k = \begin{bmatrix} -0.5000 & 0.8660 & 0 & 0.0045 \\ -0.8660 & -0.5000 & 0 & -0.3958 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (\text{C2})$$

i.e., $\mathbf{R}_z(-120^\circ)$ with constant translation $[4.5, -395.8, 0]$ mm, rotating to $\mathbf{R}_z(+10^\circ)$ at full extension (FlxPinBPASet.mat, kf(1)).

C.3 Bracket Reference Points and Frames

Table C5 lists the bracket reference points. Each compliant bracket carries an instantaneous body-fixed frame anchored at its reference point, constructed by two rotations (yaw about z , then pitch about the yawed y). With $\mathbf{p}_b$ the vector from the reference point to the bracket-to-BPA attachment target, the yaw and pitch angles are

$$\theta_{br} = \text{atan2}(p_{by}, p_{bx}), \quad \phi_{br} = \text{atan2}(p'_{bz}, p'_{bx}) \quad \text{with} \quad \mathbf{p}'_b = \mathbf{R}_z(\theta_{br})^\top \mathbf{p}_b, \quad (\text{C3})$$

and the bracket-frame pose is

$$\mathbf{R}_{kbr} = \mathbf{R}_z(\theta_{br}) \mathbf{R}_y(\phi_{br})^\top, \quad \mathbf{T}_{kbr} = [\mathbf{R}_{kbr}, \mathbf{p}_{br}; \mathbf{0} \ 1], \quad (\text{C4})$$

so the frame's x axis aims along the nominal load path. The angles are computed per configuration from the bracket geometry, not fixed [Provenance: minimizeFlxPin2brk.m lines

252–298; `minimizeExtX3.m` lines 461–483]. Force is transformed into the bracket frame by **RowVecTrans**($\mathbf{T}_{kbr}^{-1}, \mathbf{F}_k$) and deformed points are mapped back by **RowVecTrans**($\mathbf{T}_{kbr}, \cdot$). The single-rotation (yaw-only) variant used in the earlier one-transform fit survives in `minimizeFlxPinX3.m` lines 200–209.

Table C5: Bracket reference points (mm, as coded; divided by 1000 in code). All are Ben-set CAD interpretation points; frame denotes the frame the vector is expressed in.

Point	x	y	z	Configuration / meaning
Pbri	−27.50	−107.81	−0.54	pinned-knee flexor, insertion bracket (knee-ICR frame)
Pbr2	−52.61	0	75.06	pinned-knee flexor, origin-side 2nd bracket (hip frame)
Pbr	−3.84	−46.44	62.50	pinned-knee extensor, rib midpoint, medial side (hip frame)
Pbr	8.38	20.75	25.10	biomimetic extensor 10 mm, cantilever centroid
Pbr	9.48	−33.38	33.90	biomimetic flexor 20 mm, bolt-pattern centroid
Pbr	−19.00	22.00	27.60	biomimetic flexor 10 mm, cantilever centroid

C.4 Stiffness Arrays and Compliance

Every configuration models bracket compliance as a diagonal Hooke-law spring in the bracket frame, $\mathbf{K}_{br} = \text{diag}(k_1, k_2, k_3)$ newtons per meter. The scalar compliance along the force path and its series combinations are

$$c_b = \hat{\mathbf{u}} \mathbf{K}_{br}^{-1} \hat{\mathbf{u}}^\top, \quad c_{\text{eff}} = c_b + c_{\text{ten}} \text{ (+ } c_{b2} \text{ for the second bracket)}, \quad (\text{C5})$$

exactly as in Equations (3.11)–(3.21); when an artificial tendon is present its axial spring ($\times 2$ cables, effective steel area $1.51 \times 10^{-6} \text{ m}^2$, $E = 193 \text{ GPa}$) adds in series, and the force balance $F_{\text{max}} F^*(\varepsilon^*(\delta), P^*) = k_{\text{eff}} \delta$ is solved per frame by `fzero` on the contraction interval

[Provenance: `minimizeFlxPin2brk.m` lines 387–418 and 439–446; `minimizeFlxPin.m` lines 266–275; `minimizeExtX3.m` lines 575–584; `minimizeExt.m` line 352; `minimizeFlx.m` line 314].

The per-configuration stiffness arrays and their series combinations are given in Table C6.

Table C6: Bracket stiffness arrays by configuration ($K = [k_1, k_2, k_3]$ in bracket-frame axes, N/m; optimizer solves X_1, X_2 assigned as shown). The two-transformation (2trans) and pitch-only (1trans) origin-bracket conventions assign the pair differently; both are carried by the evaluator’s `transMode` argument.

Configuration	K array	Source
Pinned flexor, single bracket (legacy)	$[X_1, X_2, X_2]$	<code>minimizeFlxPin.m</code> line 266
Pinned flexor, insertion bracket	$[X_1, X_2, X_1]$	<code>minimizeFlxPin2brk.m</code> line
Pinned flexor, origin bracket (1trans / 2trans)	$[X_2, X_1, X_2] / [X_1, X_1, X_2]$	<code>minimizeFlxPin2brk.m</code> line
Pinned extensor (with X_3)	$[X_2, X_1, X_2]$	<code>minimizeExtX3.m</code> line 575
Biomimetic extensor	$[X_1, X_2, X_1]$	<code>minimizeExt.m</code> line 352
Biomimetic flexor	$[X_1, X_2, X_2]$	<code>minimizeFlx.m</code> line 314

The wrapping-loss term X_3 (Equations 3.26–3.28) is evaluated as $X_3 R |\Delta\theta| \zeta^2$ per contact arc. On the pinned extensor the two physical contact arcs use $R_1 = 12$ mm (27° to -23°) and $R_2 = 40$ mm, with arc-length normalizers $27.75/(ang_1 - ang_2)$ and $92.46/(ang_2 + 120^\circ)$; on the biomimetic configurations the bend measure is $w_{\text{rap}}|\theta_{\text{turn}}|$ with $w_{\text{rap}} = 25$ mm [Provenance: `minimizeExtX3.m` lines 333–359; `buildKneeFlexorRoute20mm.m` line 268; `MonoPamDataExplicit_balanceX3.m` lines 489–494]. Table C7 records the optimizer parameterization and bounds of the adopted identification campaign: the two-bracket flexor fit solves all three terms, and the extensor refit locks (X_1, X_2) to the adopted flexor-front values — loaded directly from the flexor results file at run time — and solves only the extensor-specific terms (X_0, X_3) , enforcing a single stiffness pair across configurations.

[Provenance: `minimizeFlxPin10mm_2brk.m` lines 82–83 (bounds) and 424–432 (parame-

Table C7: Optimizer parameterization and bounds of the adopted identification. Length offsets are carried in centimeters and stiffnesses on a base-10 logarithmic scale; the initial population of the flexor run was drawn from $X_0 \in [5, 15]$ mm, $X_1 \in [5 \times 10^4, 5 \times 10^5]$ N/m, $X_2 \in [7 \times 10^3, 1.5 \times 10^4]$ N/m.

Identification	Solved terms	Bounds
Two-bracket pinned flexor	X_0, X_1, X_2	$X_0 \in [0, 20]$ mm; $X_1 \in [3 \times 10^4, 10^6]$, $X_2 \in [5 \times 10^3, 2 \times 10^4]$
Pinned extensor refit	X_0, X_3	$X_0 \in [-20, 0]$ mm; $X_3 \in [0, 1]$

terization); minimizeExt10mmX3.m lines 23–26 (lock load) and 67–68 (bounds); Collect.ExtPinX3.sweep.m lines 59–60.]

C.5 Adopted Correction-Term Values

Table C8 records the correction-term values adopted at the time of writing, together with the results file and front row each value was taken from. The initial single-bracket identification reported with the results figures (Table 4.3) was superseded by the two-stage campaign described in Section 4.5: a two-bracket flexor identification over the four training tests of Table C1 (encoder-corrected, cross-validated), followed by an extensor refit that locks (X_1, X_2) to a member of the same flexor Pareto front and solves only (X_0, X_3) on the extensor validation pool. The pinned-flexor data do not uniquely determine (X_1, X_2) — the cross-validation fronts contain a family of stiffness pairs of comparable fit quality — so the adopted flexor row (front row 107) and the locked extensor pair (front row 1) are two operating points on one and the same front, chosen for their respective held-out and cross-configuration performance.

At the adopted flexor values, per-test torque-prediction RMSE against the five pinned-

Table C8: Adopted correction-term values of record. Flexor source: `minimizeFlxPin10_results_20260908_2brkt_2trans_noT3.mat`, pooled two-bracket Pareto front (212 members). Extensor source: `minimizeExt10mmX3_results_20260910_noT3.mat`, driver pick 1. The extensor refit locks (X_1, X_2) to row 1 of the flexor front and never searches them.

Term	Value	Source	Row / pick
Flexor X_0	+8.9 mm (8.86×10^{-3} m)	flexor mat	107
Flexor X_1	5.62×10^4 N/m	flexor mat	107
Flexor X_2	1.85×10^4 N/m	flexor mat	107
Extensor X_0	−10.1 mm (-1.012×10^{-2} m)	extensor mat	1
Extensor X_1 (locked)	4.354×10^4 N/m	flexor mat	1
Extensor X_2 (locked)	1.701×10^4 N/m	flexor mat	1
X_3	0.621	extensor mat	1

flexor tests is 1.94/1.51/2.29/1.63/1.21 N m (48, 46, 47, 40-tendon, 41 cm; FVU 0.14/0.06/0.17/0.04/0.03), the 47 cm test being pure held-out validation. At the adopted extensor values, per-test RMSE on the six-test validation pool spans 0.56–1.88 N m against 0.95–3.62 N m for the uncorrected rigid model, improving on the baseline in every pool test; the three excluded tests (3, 4, 9) are not part of the fit of record. On the biomimetic extensor test ($l_0 = 51.8$ cm) the same values give RMSE 2.20 N m, FVU 2.35, and maximum residual 4.73 N m — a substantial improvement over the baseline (3.03/4.49/5.59) but with FVU above unity, so the corrected model under-predicts that configuration’s measured torque variance and no claim of validated bio-extensor prediction is made. The redesign optimizations load these values directly: `buildKneeFlexorContext20mm.m` reads the flexor front, pick 107, for (X_0, X_1, X_2) at lines 156–174 (its X_3 still taken from the earlier `minimizeExtPin10_results_20260819_2transforms_Z2.mat` fit), and `buildKneeExtContext20mm.m` reads the extensor results file, pick 1, at lines 158–165. Regenerating the results files updates

this table.